Intermediate Macroeconomics

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Preface

These notes accompany the **Macroeconomics** course taught for CASED's Pre-Master/PhD program. The order of the chapters follows the order of the course. There is a quick math refresher in the appendix.

1. **Two-period models.** We introduce basic concepts in the simplest dynamic general-equilibrium economies, in which a country lives for two periods and decides how much to consume today and how much to invest for tomorrow in physical capital, human capital, new technology or foreign assets.

2. **The Solow model.** The horizon is infinite, but saving is a fixed fraction of income. This assumption makes it easy to study the mechanics of capital accumulation, the concepts of steady states, convergence and balanced growth.

3. **The overlapping-generations (OLG) model.** Saving becomes a choice, made by finitely-lived households. Competitive equilibria may be inefficient or multiple and may exhibit poverty traps, and government debt and pensions can matter for welfare.

4. **The Ramsey model.** An infinitely-lived household (or a planner) chooses saving optimally. The equilibrium is efficient and its dynamics are those of a saddle path.

5. **The Real Business Cycle (RBC) model.** The Ramsey model with a labor–leisure choice and random productivity shocks is the workhorse of quantitative macroeconomics.

6. **Endogenous growth.** Long-run growth is explained rather than assumed in the expanding-variety model of Romer (1990), the semi-endogenous model of Jones (1995) and the Schumpeterian quality-ladder model of Aghion and Howitt (1992).

All models are presented in discrete time. I try my best to cover as much stuff as possible while leaving the materials intuitive and accessible. Be advised that the notes may contain errors. If you spot one, please email me.

Chapter 1

Two-Period Models

1.1 Optimal consumption and saving

Before turning to growth models, we study the most basic problem of consumption and saving. This section introduces the vocabulary that we use throughout the notes, such as the marginal rate of substitution, the Euler equation and the elasticity of intertemporal substitution.

1.1.1 Setup

We consider a household that lives for two periods, 0 and 1, and receives exogenous income y_0 and y_1 in these periods. The household can save or borrow an amount s at the gross interest rate $R \equiv 1 + r > 0$. Preferences are

$$U(c_0, c_1) = u(c_0) + \beta\, u(c_1), \qquad 0 < \beta < 1,$$

where the period utility u is strictly increasing ($u' > 0$), strictly concave ($u'' < 0$) and satisfies the Inada condition $u'(0) = +\infty$. The discount factor β measures the degree of *patience*, and the smaller β is, the less the household cares about the future. The budget constraints in the two periods are

$$\begin{aligned} c_0 + s &= y_0, \\ c_1 &= y_1 + Rs. \end{aligned}$$

Eliminating s gives the **intertemporal (lifetime) budget constraint**

$$c_0 + \frac{c_1}{R} = y_0 + \frac{y_1}{R} \equiv W, \tag{1.1}$$

which says that the present value of consumption equals the present value of income, W ("lifetime wealth"). The relative price of c_1 in terms of c_0 is $1/R$ because in order to obtain one more unit of consumption today, the household gives up $1/R$ units that can be consumed tomorrow.

1.1.2 Euler equation

To solve the household's optimization problem, we can write the Lagrangian as

$$\mathcal{L} = u(c_0) + \beta u(c_1) + \lambda\Big(W - c_0 - \frac{c_1}{R}\Big).$$

The first-order conditions (FOCs) are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_0} = 0: \quad & u'(c_0) = \lambda, \\ \frac{\partial \mathcal{L}}{\partial c_1} = 0: \quad & \beta u'(c_1) = \frac{\lambda}{R}. \end{aligned}$$

The multiplier λ is the *shadow value of wealth*. If lifetime wealth W rises by one unit, maximal utility rises by λ units. Eliminating λ gives the **Euler equation**

$$u'(c_0) = \beta R\, u'(c_1) \tag{1.2}$$

or equivalently

$$\frac{u'(c_0)}{\beta u'(c_1)} = R. \tag{1.3}$$

Meaning of the Euler equation

Consider the thought experiment in which the household reduces c_0 by one (small) unit and saves it.

- *Cost*: utility today falls by $u'(c_0)$.
- *Benefit*: tomorrow the household has R additional units of consumption, each worth $u'(c_1)$ in period-1 utility, which is worth β in today's utility (present value). Thus, the total benefit is $\beta R u'(c_1)$.

If the cost were smaller than the benefit, saving one more unit would raise utility, so the plan could not be optimal. If the cost were larger, saving one unit less would raise utility, and the plan would again fail to be optimal. The household is therefore *indifferent* at the margin only at the optimum (1.2).
The second form of the condition, (1.3), says that the *marginal rate of substitution* between c_0 and c_1 (the slope of the indifference curve) equals the *relative price* R (the slope of the budget line).

1.1.3 Elasticity of intertemporal substitution

We now adopt a specific functional form for utility. The workhorse instantaneous utility function in macroeconomics is the **constant relative risk**

aversion (CRRA) function[1]

$$u(c) = \begin{cases} \dfrac{c^{1-\sigma} - 1}{1-\sigma} & \text{if } \sigma > 0, \\ \ln c & \text{if } \sigma = 1. \end{cases} \tag{1.4}$$

Since $u'(c) = c^{-\sigma}$ and $u''(c) = -\sigma c^{-\sigma-1}$, the Arrow–Pratt coefficient of relative risk aversion, which measures the curvature of u, is

$$-\frac{c\,u''(c)}{u'(c)} = \sigma.$$

The constant σ is the elasticity of marginal utility with respect to consumption. Under uncertainty, it is the coefficient of relative risk aversion and tells us how much the household dislikes consumption volatility. The larger σ is, the more the household dislikes volatility, and as $\sigma \to \infty$ it becomes infinitely averse to fluctuations and insists on perfectly smooth consumption. In a deterministic intertemporal problem its inverse measures the willingness to substitute consumption over time.

With CRRA utility, the Euler equation becomes

$$c_0^{-\sigma} = \beta R\, c_1^{-\sigma},$$

which we can rewrite as

$$\frac{c_1}{c_0} = (\beta R)^{1/\sigma} \qquad \Longleftrightarrow \qquad \ln \frac{c_1}{c_0} = \frac{1}{\sigma} \ln \beta + \frac{1}{\sigma} \ln R. \tag{1.5}$$

Definition 1.1 (Elasticity)**.** For a differentiable relation $y = h(x)$ with $x, y > 0$, the elasticity of y with respect to x is

$$\varepsilon_{y,x} \equiv \frac{\mathrm{d}y/y}{\mathrm{d}x/x} = \frac{\mathrm{d}\ln y}{\mathrm{d}\ln x} = \frac{x\,h'(x)}{h(x)}.$$

It is the percentage change in y caused by a one-percent change in x.

Definition 1.2 (Elasticity of intertemporal substitution)**.** The **elasticity of intertemporal substitution** (EIS) is the elasticity of the consumption ratio c_1/c_0 with respect to its relative price R, holding the household on its Euler equation:

$$\mathrm{EIS} \equiv \frac{\mathrm{d}\ln(c_1/c_0)}{\mathrm{d}\ln R}.$$

For a general u, differentiating $\ln u'(c_0) - \ln u'(c_1) = \ln\beta + \ln R$ shows that, locally around $c_0 = c_1 = c$,

$$\mathrm{EIS}(c) = -\frac{u'(c)}{c\,u''(c)}.$$

[1] The "-1" in the numerator is harmless and ensures that the limit as $\sigma \to 1$ is $\ln c$, by l'Hôpital's rule.

From (1.5), EIS $= 1/\sigma$ for CRRA utility. A household with a high σ (low EIS) dislikes uneven consumption paths and barely reacts to the interest rate. As $\sigma \to \infty$, the EIS tends to zero and the consumption ratio no longer reacts to the interest rate. The intuition comes from $u'(c) = c^{-\sigma}$. With a high σ, marginal utility rises steeply when consumption falls and drops steeply when it rises, so uneven paths are costly and the consumer strongly wants to smooth consumption across time (low EIS). A low σ, in contrast, means that the consumer is happy to let consumption swing (high EIS) and is therefore willing to postpone today's consumption in exchange for future payoff.

To obtain a closed-form solution, we substitute $c_1 = (\beta R)^{1/\sigma} c_0$ into (1.1) and find

$$c_0\left(1 + \beta^{1/\sigma} R^{\frac{1}{\sigma}-1}\right) = W$$

which gives

$$c_0 = \frac{W}{1 + \beta^{1/\sigma} R^{\frac{1-\sigma}{\sigma}}},$$
$$c_1 = (\beta R)^{1/\sigma} c_0.$$

Consumption today is thus a fraction of lifetime wealth. With log utility ($\sigma = 1$), we have $c_0 = W/(1+\beta)$, so the household consumes the fraction $1/(1+\beta)$ of its wealth today regardless of R.

1.1.4 Substitution and income effects of the interest rate

How does a higher interest rate affect consumption today and, therefore, saving $s = y_0 - c_0$?

We first rewrite the lifetime budget constraint (1.1) so that the interest rate appears as a relative price,

$$c_0 + p\, c_1 = y_0 + p\, y_1 \equiv W, \qquad p \equiv \frac{1}{R}.$$

Read in this way, the two-period problem is an ordinary two-good consumer problem. The two "goods" are consumption today, c_0, whose price is normalized to one, and consumption tomorrow, c_1, whose price is $p = 1/R$, since the household must save $1/R$ units today to obtain one unit of c_1. A rise in R is therefore a *fall* in the price of future consumption.

It is useful to distinguish between the Marshallian and the Hicksian demand functions.

- The **Marshallian** (ordinary) demand $c_0(p, W)$ solves max $u(c_0) + \beta u(c_1)$ subject to $c_0 + pc_1 = W$, with wealth W held fixed.

- The **Hicksian** (compensated) demand $c_0^h(p, \overline{U})$ solves min $c_0 + pc_1$ subject to $u(c_0) + \beta u(c_1) \geq \overline{U}$, with *utility* held fixed at $\overline{U}$. It tells us how the household would rearrange its consumption if the price changed but it were compensated, by just enough wealth, to stay on its original indifference curve.

Let $e(p, \overline{U})$ be the minimized expenditure in the second problem. If $\overline{U}$ is the utility the household actually attains with wealth W, then $e(p, \overline{U}) = W$, and the two demands coincide:

$$c_0^h(p, \overline{U}) = c_0\big(p,\, e(p, \overline{U})\big).$$

This identity holds for every p, so we can differentiate both sides with respect to p:

$$\frac{\partial c_0^h}{\partial p} = \frac{\partial c_0}{\partial p} + \frac{\partial c_0}{\partial W}\,\frac{\partial e}{\partial p}.$$

By Shephard's lemma (or the envelope theorem applied to the expenditure minimization problem), $\partial e/\partial p = c_1^h = c_1$. In words, when the price of c_1 rises by $\mathrm{d}p$, the cost of keeping the same bundle rises by $c_1\,\mathrm{d}p$. Rearranging gives the **Slutsky equation**

$$\underbrace{\frac{\partial c_0}{\partial p}}_{\text{total, } W \text{ fixed}} = \underbrace{\frac{\partial c_0^h}{\partial p}}_{\text{substitution}} - \underbrace{c_1\,\frac{\partial c_0}{\partial W}}_{\text{income } (W \text{ fixed})}. \tag{1.6}$$

The first term is the **substitution effect**: the change in c_0 along the original indifference curve. Because the household is compensated, only the relative price matters, and since there are just two goods a higher price of c_1 always shifts demand towards c_0, so that $\partial c_0^h/\partial p > 0$. The second term is the income effect for a consumer whose *money income* W is fixed. A higher p makes the planned bundle more expensive by $c_1\,\mathrm{d}p$, which is equivalent to a loss of wealth of that size, and with c_0 a normal good ($\partial c_0/\partial W > 0$) this lowers c_0. As we now show, this is *not* the relevant income effect for a household that owns an endowment.

The income effect with an endowment. In our economy, however, wealth is not fixed. Lifetime wealth $W = y_0 + py_1$ rises with p, at the rate $\partial W/\partial p = y_1$, because the household *sells* future income y_1 at the price p (by borrowing against it). The *total* effect of the price on c_0 is therefore

$$\begin{aligned}\frac{\mathrm{d}c_0}{\mathrm{d}p} &= \frac{\partial c_0}{\partial p} + \frac{\partial c_0}{\partial W}\,\frac{\partial W}{\partial p} \\ &= \frac{\partial c_0^h}{\partial p} - c_1\,\frac{\partial c_0}{\partial W} + y_1\,\frac{\partial c_0}{\partial W} \\ &= \frac{\partial c_0^h}{\partial p} - (c_1 - y_1)\,\frac{\partial c_0}{\partial W}.\end{aligned}$$

What matters for the income effect is not the whole bundle c_1 but the household's *net demand* for future consumption, $c_1 - y_1$, that is, the amount it buys net of what it already owns.

As a consequence, a price increase makes a *net buyer* of the good poorer and a *net seller* richer. The period budget constraints imply $c_1 - y_1 = Rs$, so the net demand for future consumption has the sign of saving. We can therefore distinguish three cases.

- A **net lender** ($s > 0$) is a net buyer of future consumption, $c_1 > y_1$.
- A **net borrower** ($s < 0$) is a net seller of future consumption, $c_1 < y_1$, because it sells part of its future income to consume more today.
- A household with $s = 0$ consumes exactly its endowment, and price changes have no first-order income effect on it.

Since $p = 1/R$, we have $\mathrm{d}p/\mathrm{d}R = -1/R^2$, and the chain rule gives

$$\frac{\mathrm{d}c_0}{\mathrm{d}R} = -\frac{1}{R^2}\,\frac{\mathrm{d}c_0}{\mathrm{d}p}.$$

Multiplying both terms by $-1/R^2$ and using $c_1 - y_1 = Rs$, we obtain the decomposition of the effect of the interest rate on current consumption:

$$\frac{\mathrm{d}c_0}{\mathrm{d}R} = \underbrace{-\frac{1}{R^2}\frac{\partial c_0^h}{\partial p}}_{\text{substitution effect } (<0)} + \underbrace{\frac{s}{R}\frac{\partial c_0}{\partial W}}_{\substack{\text{income effect:}\\ \text{sign of } s}}. \tag{1.7}$$

The substitution effect always lowers current consumption. The sign of the income effect, by contrast, depends on whether the household is a net lender or a net borrower.

Substitution and income effects of a higher interest rate

1. **Substitution effect** ($c_0 \downarrow$, $s \uparrow$). Waiting is better rewarded, since each unit saved today buys more consumption tomorrow. Holding utility constant, the household substitutes away from the good that has become relatively expensive, c_0, towards c_1. This is true for every household.
2. **Income effect, net lender** ($s > 0$, $c_0 \uparrow$, $s \downarrow$). A lender receives the interest rate, so a higher R makes it richer because the same saving now finances more future consumption. Since both goods are normal, it consumes more in both periods, including today. The income effect works *against* the substitution effect, and the response of saving is ambiguous.

3. **Income effect, net borrower** ($s < 0$, $c_0 \downarrow$, $s \uparrow$). A borrower pays the interest rate, so a higher R makes it poorer because its debt is more expensive to repay out of future income. It consumes less in both periods. The income effect *reinforces* the substitution effect, and a borrower unambiguously reduces current consumption and borrows less.

Remark 1.1 (Two components of the income effect). It is sometimes useful to split the income effect into the two terms of the derivation above,

$$\frac{s}{R}\frac{\partial c_0}{\partial W} = \frac{c_1}{R^2}\frac{\partial c_0}{\partial W} - \frac{y_1}{R^2}\frac{\partial c_0}{\partial W}.$$

The first term is the gain from the fall in the price of the planned future consumption c_1. The second term, the **human-wealth effect**, is the loss from the fall in the present value y_1/R of future (labor) income. A net lender is one for whom the first dominates, a net borrower one for whom the second dominates. For a household with no future income ($y_1 = 0$), such as the young of the OLG model, only the first term is present. Such a household is necessarily a lender, and its income effect is always positive.

With CRRA utility (1.4), we can compute both effects explicitly. We define

$$D \equiv 1 + \beta^{1/\sigma} R^{\frac{1-\sigma}{\sigma}} = 1 + \beta^{1/\sigma} p^{\frac{\sigma-1}{\sigma}},$$

so that the Marshallian demand is $c_0 = W/D$ and $c_1 = (\beta R)^{1/\sigma} c_0 = \beta^{1/\sigma} p^{-1/\sigma} c_0$.

- *Marginal propensity to consume out of wealth*, $\partial c_0/\partial W = 1/D$.
- *Price effect at fixed wealth.* Since $\partial D/\partial p = \beta^{1/\sigma}\,\frac{\sigma-1}{\sigma}\,p^{-1/\sigma}$, we obtain

$$\frac{\partial c_0}{\partial p} = -\frac{W}{D^2}\frac{\partial D}{\partial p} = -\frac{c_0}{D}\,\beta^{1/\sigma} p^{-1/\sigma}\,\frac{\sigma-1}{\sigma} = -\frac{\sigma-1}{\sigma}\,\frac{c_1}{D}.$$

- *Substitution effect*. From the Slutsky equation (1.6), we have

$$\frac{\partial c_0^h}{\partial p} = \frac{\partial c_0}{\partial p} + c_1\frac{\partial c_0}{\partial W} = -\frac{\sigma-1}{\sigma}\,\frac{c_1}{D} + \frac{c_1}{D} = \frac{1}{\sigma}\,\frac{c_1}{D} > 0.$$

  As expected, the substitution effect is proportional to the EIS $1/\sigma$, which measures how easily the household substitutes consumption across periods when utility is held fixed.

Substituting into (1.7) gives

$$\frac{\mathrm{d}c_0}{\mathrm{d}R} = \underbrace{-\frac{c_1}{\sigma R^2 D}}_{\text{substitution}} + \underbrace{\frac{s}{RD}}_{\text{income}} = \frac{Rs - \frac{1}{\sigma}c_1}{R^2 D} = \frac{c_1\left(1-\frac{1}{\sigma}\right) - y_1}{R^2 D}, \tag{1.8}$$

where the last step uses $Rs = c_1 - y_1$. Since $s = y_0 - c_0$ with y_0 fixed,

$$\frac{\mathrm{d}s}{\mathrm{d}R} = -\frac{\mathrm{d}c_0}{\mathrm{d}R} = \frac{\frac{1}{\sigma}c_1 - Rs}{R^2 D}.$$

Saving rises with the interest rate if and only if the substitution term c_1/σ exceeds the income term $Rs = c_1 - y_1$.

Who saves more when the interest rate rises? The sign of (1.8) depends on the position of the household as a lender or a borrower, and, for a lender, on the EIS.

1. **Net borrowers ($s < 0$).** Both terms are negative, so a higher R lowers c_0 and raises saving (that is, reduces borrowing), whatever the value of σ.

2. **Households with $s = 0$.** The income effect vanishes and only the substitution effect remains, so c_0 falls and saving rises.

3. **Net lenders ($s > 0$).** The two effects have opposite signs, and saving rises if and only if $c_1/\sigma > c_1 - y_1$.

   - *High EIS ($\sigma < 1$).* In this case $c_1/\sigma > c_1 \geq c_1 - y_1$, so the substitution effect dominates and saving rises.
   - *Log utility ($\sigma = 1$).* The condition becomes $y_1 > 0$. A lender with future income saves more. A lender with no future income ($y_1 = 0$) has $c_0 = y_0/(1+\beta)$ and $s = \beta y_0/(1+\beta)$, which do not depend on R because the two effects cancel exactly. This is why log utility is so convenient in the OLG model, where the young have no future labor income.
   - *Low EIS ($\sigma > 1$).* Saving rises if and only if $y_1 > c_1(\sigma - 1)/\sigma$. A lender with little future income and a strong desire for consumption smoothing saves *less* when the interest rate rises. The income effect dominates in this case, because the household can reach its target of future consumption with a smaller sacrifice today.

Example 1.1 (A graphical decomposition: lender versus borrower)**.** Let $u(c) = \ln c$ and $\beta = 1$, and let the interest rate double from $R = 1$ to $R = 2$. With log utility and $\beta = 1$ the household always consumes half of its wealth, $c_0 = W/2$, and chooses $c_1 = Rc_0$.

A lender. Let the endowment be $(y_0, y_1) = (1.5, 0.5)$. At $R = 1$, $W = 2$ and the household chooses $c_0 = c_1 = 1$, saving $s = 0.5$ (point A in Figure 1.1). When R rises the budget line pivots around the endowment point E and becomes steeper.

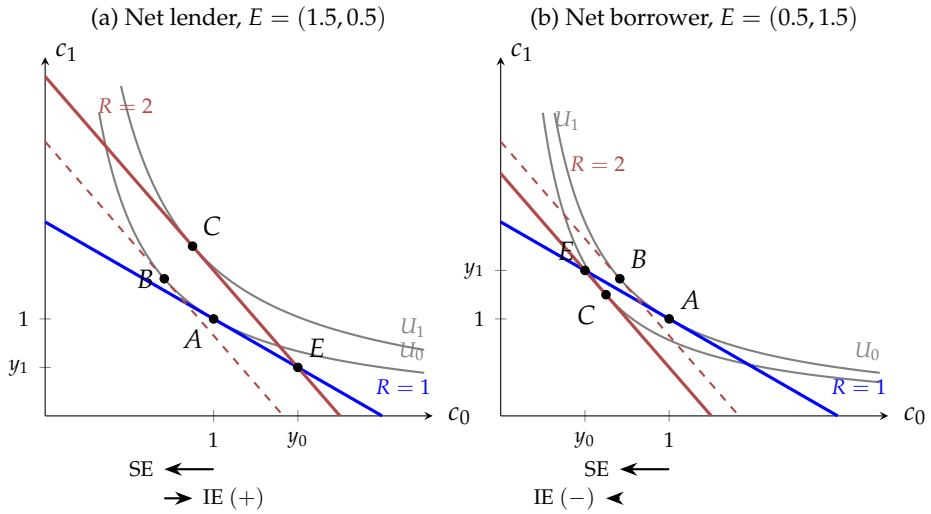


Figure 1.1: Slutsky decomposition of a rise in the interest rate from $R = 1$ (blue) to $R = 2$ (red) with log utility and $\beta = 1$ (Example 1.1). The budget line pivots around the endowment E. In both panels the substitution effect (SE) moves the household along its original indifference curve U_0 from A to the compensated bundle B (on the dashed line). The income effect (IE) moves it from B to C: towards more current consumption for the lender, who is made richer (a), and towards less for the borrower, who is made poorer (b).

- *Substitution effect.* Keep utility at its initial level $\ln c_0 + \ln c_1 = 0$, i.e. $c_0 c_1 = 1$, and let the household face the new price. The Euler equation gives $c_1 = 2c_0$, so the compensated bundle is $B = (1/\sqrt{2}, \sqrt{2}) \approx (0.707, 1.414)$ and current consumption falls by 0.293.
- *Income effect.* New wealth is $W' = 1.5 + 0.5/2 = 1.75$, so the actual choice is $C = (0.875, 1.75)$, on a higher indifference curve than A. Moving from B to C raises current consumption by 0.168, because the lender is richer.

In total, c_0 falls from 1 to 0.875 and saving rises from 0.5 to 0.625. The substitution effect dominates, as case 3 predicts for log utility with $y_1 > 0$.

A borrower. Now let the endowment be $(y_0, y_1) = (0.5, 1.5)$. At $R = 1$ the household again chooses $c_0 = c_1 = 1$, but it now borrows, with $s = -0.5$. The substitution effect is the same as before (c_0 falls to 0.707), because the initial bundle and indifference curve are the same. But new wealth is $W' = 0.5 + 1.5/2 = 1.25$, so $c_0 = 0.625$. The income effect *lowers* current consumption by a further 0.082, and the household ends on a *lower* indifference curve. Both effects push in the same direction, and borrowing falls from 0.5 to 0.125.

While the consumption–saving decision is the bedrock of many frameworks in modern macroeconomics, we now digress and study several two-period models that illustrate how economists build models. We start with a model of human capital and then turn to models of technology adoption and of foreign debt.

1.2 Human capital and growth

1.2.1 Economy without education

We consider a two-period *closed* economy in which output is produced with labor only, so that the production function is simply

$$Y_t = F(L_t)$$

where $F' > 0$, $F'' < 0$ and $F(0) = 0$. Population is constant and normalized to one, $L_0 = L_1 = 1$. Without any way of transferring resources across periods (there is no capital and the economy is closed), the country simply consumes its output in each period,

$$c_0 = F(1), \qquad c_1 = F(1).$$

Consumption is therefore flat, the growth rate of output and consumption is zero, and lifetime utility is $(1 + \beta)\, u(F(1))$.

1.2.2 Investing in human capital

We now suppose that the country has access to a labor-augmenting technology. It can spend $G \geq 0$ units of period-0 output on education and training to acquire such a technology. Education raises the human capital of each period-1 worker from its "natural" level 1 to $h = H(G)$, so that *effective* labor in period 1 is $L_1 H(G) = H(G)$. The resource constraints become

$$\begin{aligned} c_0 + G &= F(1), \\ c_1 &= F\big(H(G)\big). \end{aligned}$$

In period 0, output is split between consumption and education. In period 1, output depends on the quality of the workforce and is entirely consumed. The country solves

$$\max_G \ \Phi(G) \equiv u\big(F(1) - G\big) + \beta\, u\big(F(H(G))\big) \qquad \text{s.t.} \qquad 0 \leq G \leq F(1),$$

under the following assumptions.

- $H(0) = 1$. Without training, human capital stays at its natural level and we are back to the economy without education.

- $H' > 0$ and $H'' \leq 0$. Training raises human capital, with non-increasing returns.
- $u' > 0$, $u'' < 0$ and $u'(0) = +\infty$.

Education as an investment

Education is an investment like any other. It costs consumption today (one unit of G is one unit of c_0 lost) and pays off tomorrow, because better-trained workers produce more. The gross return on education is the increase in tomorrow's output from one more unit of G,

$$\frac{\mathrm{d}c_1}{\mathrm{d}G} = F'\big(H(G)\big)\, H'(G),$$

the product of the marginal product of effective labor and the marginal productivity of training. It plays exactly the role of the gross interest rate R in Section 1.1, or of $F'(k_1) + 1 - \delta$ in the models with physical capital below.

Differentiating Φ by the chain rule, we obtain

$$\Phi'(G) = -u'\big(F(1) - G\big) + \beta\, u'\big(F(H(G))\big)\, F'\big(H(G)\big)\, H'(G).$$

An interior optimum $G^* \in (0, F(1))$ satisfies $\Phi'(G^*) = 0$, that is,

$$u'(c_0^*) = \beta\, u'(c_1^*)\, F'\big(H(G^*)\big)\, H'(G^*) \tag{1.9}$$

This is the Euler equation of the economy, with the return on education in place of R. The left-hand side is the *marginal cost* of education, namely the utility value of the unit of consumption sacrificed today. The right-hand side is the *marginal benefit*. One more unit of G raises human capital by $H'(G)$, each unit of human capital raises output by $F'(H(G))$, and each unit of output is worth $u'(c_1)$ tomorrow and $\beta u'(c_1)$ today.

Uniqueness. We now let G increase from 0 to $F(1)$ and inspect (1.9).

- The marginal cost is *increasing*, because $c_0 = F(1) - G$ falls and u' is decreasing. It equals $u'(F(1))$ at $G = 0$ and tends to $+\infty$ as $G \to F(1)$, because $c_0 \to 0$ and $u'(0) = +\infty$.
- The marginal benefit is *decreasing*. As $F(H(G))$ rises, $u'(F(H(G)))$ falls. As $H(G)$ rises, $F'(H(G))$ falls because $F'' < 0$. Finally, $H'(G)$ is non-increasing because $H'' \leq 0$. The marginal benefit is the product of three positive non-increasing functions, at least one of which is strictly decreasing.

Hence the two sides of (1.9) cross *at most once*, and by the Intermediate Value Theorem they cross exactly once whenever the marginal benefit starts above the marginal cost. Equivalently, Φ' is strictly decreasing, so Φ is strictly concave, the first-order condition is sufficient, and the optimum is unique. Whether the two curves cross at all depends on how productive the *first* unit of education is, which is the subject of the next subsection.

1.2.3 When does a country invest in education?

Case 1: $H'(0) = +\infty$**.** If the first unit of training is infinitely productive, the marginal benefit is $+\infty$ at $G = 0$ while the marginal cost $u'(F(1))$ is finite. Since the marginal benefit decreases and the marginal cost rises to $+\infty$, there is a *unique* interior $G^* \in (0, F(1))$ that solves (1.9), and the country always educates.

Case 2: $H'(0) < +\infty$**.** Suppose that the training technology is linear,

$$H(G) = aG + 1, \qquad a > 0,$$

so that $H'(G) = a$ for all G. The parameter a measures the *quality* of the education system. At $G = 0$, both sides of (1.9) are finite,

$$\begin{aligned} \text{marginal cost at } G = 0 &= u'(F(1)), \\ \text{marginal benefit at } G = 0 &= \beta\, u'(F(1))\, F'(1)\, a. \end{aligned}$$

Since the marginal cost increases and the marginal benefit decreases in G, we must distinguish between the following cases.

- If $\beta F'(1) a \leq 1$, the marginal benefit is below the marginal cost already at $G = 0$, and therefore for all $G > 0$. Then $\Phi'(G) < 0$ on $(0, F(1))$, and the optimum is the **corner** $G^* = 0$.
- If $\beta F'(1) a > 1$, the curves cross once and there is a unique $G^* > 0$.

When does a country invest in education?

With $H(G) = aG + 1$,

$$G^* > 0 \iff a > \frac{1}{\beta F'(1)} \iff \underbrace{a\, F'(1)}_{\substack{\text{gross return of the} \\ \text{first unit of } G}} > \underbrace{\frac{1}{\beta}}_{\substack{\text{required} \\ \text{return}}}.$$

The country educates if and only if the education system is of high enough **quality** (a high), labor is **productive** ($F'(1)$ high), or the country is **patient** (β high). Otherwise, the optimum is the corner $G^* = 0$ and the economy behaves as without education.

We can also read the corner solution off the Kuhn–Tucker conditions (Appendix A). With the constraint $G \geq 0$ and multiplier $\mu \geq 0$, the optimum satisfies $\Phi'(G^*) + \mu = 0$ and $\mu G^* = 0$. At the corner $G^* = 0$, the multiplier is

$$\mu = -\Phi'(0) = u'(F(1)) \left[1 - \beta F'(1) a\right],$$

which is non-negative exactly when $\beta F'(1) a \leq 1$. The multiplier measures how much the country would gain if it were allowed to "un-educate", i.e. to set G slightly below zero.

1.2.4 Is growth improved by education?

Proposition 1.1. *Whenever the country invests in education ($G^* > 0$), the following hold.*

1. *Welfare is higher than without education,*

$$u(c_0^*) + \beta u(c_1^*) > (1 + \beta)\, u(F(1)).$$

2. *Consumption grows,*

$$\frac{c_1^*}{c_0^*} = \frac{F(H(G^*))}{F(1) - G^*} > 1,$$

whereas without education $c_1 / c_0 = 1$.

Proof. (1) $G = 0$ is feasible and gives utility $(1 + \beta) u(F(1))$. Since Φ is strictly concave, its maximizer is unique, so the optimal plan $G^* > 0$ yields strictly more utility than $G = 0$. (2) $G^* > 0$ and H increasing imply $H(G^*) > H(0) = 1$, so $c_1^* = F(H(G^*)) > F(1) > F(1) - G^* = c_0^*$. □

Does education raise total consumption? Education raises welfare whenever it is chosen, but does it raise the *sum* of consumption over the two periods? This is a different question, because a patient country might be willing to sacrifice a lot today for a small gain tomorrow. With $H(G) = aG + 1$, we need

$$\begin{aligned} c_0^* + c_1^* > 2F(1) \quad &\Longleftrightarrow \quad F(1) - G^* + F(aG^* + 1) > 2F(1) \\ &\Longleftrightarrow \quad F(aG^* + 1) - F(1) > G^*. \end{aligned}$$

A strictly concave function lies below its tangent lines, so that for any $x \neq y$,

$$F(y) < F(x) + F'(x)(y - x).$$

Taking $x = aG^* + 1$ and $y = 1$, we obtain

$$\begin{aligned} F(1) &< F(aG^* + 1) - F'(aG^* + 1)\, aG^* \\ \Longrightarrow \quad & F(aG^* + 1) - F(1) > F'(aG^* + 1)\, aG^*. \end{aligned}$$

Therefore a *sufficient* condition for $c_0^* + c_1^* > 2F(1)$ is

$$F'(aG^* + 1)\, a \geq 1, \tag{1.10}$$

which says that the *last* unit spent on training returns at least one unit of output tomorrow.

Example 1.2 (Log utility and Cobb–Douglas technology)**.** Let $u(c) = \ln c$, $F(L) = AL^{\alpha}$ with $0 < \alpha < 1$, and $H(G) = aG + 1$. Then $F(1) = A$, $F'(L) = \alpha A L^{\alpha-1}$, and (1.9) becomes

$$\frac{1}{A - G^*} = \beta \, \frac{1}{A(aG^* + 1)^{\alpha}} \, \alpha A (aG^* + 1)^{\alpha - 1} \, a = \frac{\beta \alpha a}{aG^* + 1}.$$

We note that the level A cancels on the right. With log utility, $u'(c_1)F'(\cdot) = F'(\cdot)/F(\cdot)$ is the *proportional* return to effective labor, which equals $\alpha/(aG + 1)$. Cross-multiplying gives $aG^* + 1 = \beta\alpha a(A - G^*)$, so that

$$G^* = \frac{\beta \alpha a A - 1}{a(1 + \beta\alpha)}.$$

Since $F'(1) = \alpha A$, the threshold $a > 1/(\beta F'(1))$ reads $\beta\alpha a A > 1$, which is exactly the condition for $G^* > 0$. The optimal consumptions are

$$\begin{aligned} c_0^* &= A - G^* = \frac{1 + aA}{a(1 + \beta\alpha)}, \\ aG^* + 1 &= \frac{\beta\alpha(1 + aA)}{1 + \beta\alpha}, \\ c_1^* &= A\,(aG^* + 1)^{\alpha} = A \left[\frac{\beta\alpha(1 + aA)}{1 + \beta\alpha} \right]^{\alpha}. \end{aligned}$$

The sufficient condition (1.10) becomes

$$\alpha A a \, (aG^* + 1)^{\alpha - 1} \geq 1 \quad \Longleftrightarrow \quad \alpha a A \;\geq\; \left(\frac{\beta\alpha}{1 + \beta\alpha} \right)^{1-\alpha} (1 + aA)^{1-\alpha}.$$

The left-hand side is linear in the product aA while the right-hand side grows like $(aA)^{1-\alpha}$, so the inequality holds when aA is large enough.

Numerical illustration. With $A = 1$, $\alpha = 0.5$, $\beta = 0.9$ and $a = 5$, we have $\beta\alpha a A = 2.25 > 1$, so the country educates. Optimal spending is $G^* = 1.25/7.25 \approx 0.172$, with $c_0^* \approx 0.828$ and $c_1^* = (1.862)^{0.5} \approx 1.365$. Consumption grows by about 65%, and total consumption 2.19 exceeds $2F(1) = 2$, as the sufficient condition predicts since $\alpha a A (aG^* + 1)^{\alpha - 1} = 2.5/1.365 \approx 1.83 \geq 1$.

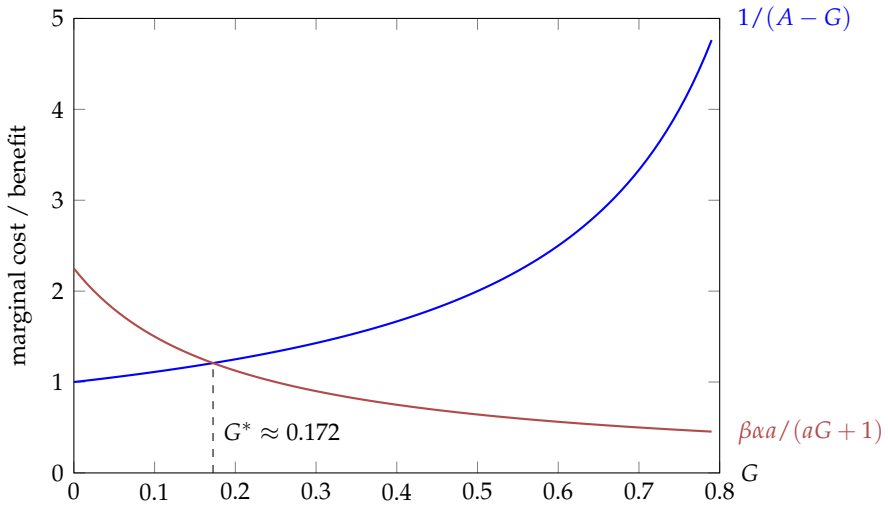


Figure 1.2: Optimal education in Example 1.2 with $A = 1$, $\alpha = 0.5$, $\beta = 0.9$, $a = 5$. The marginal cost is increasing and the marginal benefit decreasing, so they cross once, at $G^* = (\beta\alpha aA - 1)/(a(1 + \beta\alpha)) = 1.25/7.25 \approx 0.172$.

Why the product aA matters

a is the efficiency of the education sector (human capital per unit spent) and A is the productivity of the economy (output per unit of effective labor). Education pays when the *combination* is high. A good school system is wasted in an unproductive economy, and a productive economy gains little from a poor school system. Rich countries (high A) therefore find it optimal to invest more in education and grow faster, a first hint that growth and development can be self-reinforcing.

1.3 New technology and growth

1.3.1 Economy without new technology

We now consider a two-period closed economy in which capital is the only input,

$$y_t = f(k_t),$$

where f is increasing and concave with $f(0) = 0$. The depreciation rate is 1, so capital lasts one period. Given $k_0 > 0$, the economy solves

$$\max_{k_1} \{u(c_0) + \beta u(c_1)\}$$

subject to

$$\begin{aligned} c_0 + k_1 &= f(k_0), \\ c_1 &= f(k_1). \end{aligned}$$

where c_0 and c_1 are consumption today and tomorrow. In period 0, output $f(k_0)$ is split between consumption and next period's capital. In period 1, the last period, nothing is saved and all output is consumed. After substituting the constraints, we obtain the FOC for k_1,

$$u'\big(f(k_0) - k_1\big) = \beta\, u'\big(f(k_1)\big)\, f'(k_1). \tag{1.11}$$

The LHS increases in k_1 (from $u'(f(k_0))$ to $+\infty$) and the RHS decreases (both $u'(f(k_1))$ and $f'(k_1)$ fall). Because $f(0) = 0$ and $u'(0) = +\infty$, the RHS is $+\infty$ at $k_1 = 0$ (as long as $f'(0) > 0$), so there is a unique interior solution for k_1, say $\hat{k}_1$.

1.3.2 Investing in a new technology

We now suppose that there is a sector that produces a **new technology** (NT) with capital, $y_e = \phi(k_e)$, where ϕ is strictly increasing. If the country adopts the NT, total factor productivity in period 1 becomes $A(y_e)$. Since ϕ is strictly increasing, we may write TFP directly as a function of k_e, $A(k_e)$. The constraints are

$$\begin{aligned} c_0 + S_0 &= f(k_0), \\ S_0 &= k_1 + \lambda k_e, \\ c_1 &= A(k_e)\, f(k_1). \end{aligned}$$

Saving S_0 is used to build ordinary capital k_1 and NT capital k_e. One unit of NT capital costs $\lambda > 1$ units of the consumption good simply because new technologies are expensive. We assume $A(0) = 1$ (no NT means no TFP gain), $A' > 0$ and A concave. The problem can be written as

$$\max_{k_1, k_e} \; L = u\big(f(k_0) - k_1 - \lambda k_e\big) + \beta u\big(A(k_e) f(k_1)\big)$$

subject to

$$\begin{aligned} k_1 \geq 0,\; k_e \geq 0, \\ k_1 + \lambda k_e \leq f(k_0) \end{aligned}$$

At an interior optimum the two FOCs are

$$\frac{\partial L}{\partial k_1}: \quad u'(c_0^*) = \beta u'(c_1^*)\, A(k_e^*)\, f'(k_1^*), \tag{1.12}$$

$$\frac{\partial L}{\partial k_e}: \quad \lambda\, u'(c_0^*) = \beta u'(c_1^*)\, A'(k_e^*)\, f(k_1^*). \tag{1.13}$$

Each asset, k_1 and k_e, has its own Euler equation. In (1.12), one more unit of k_1 costs $u'(c_0)$ but yields $Af'(k_1)$ more units of output tomorrow. In (1.13), one more unit of k_e costs λ units of consumption, or $\lambda u'(c_0)$ units of utility, but raises tomorrow's output by $A'(k_e)f(k_1)$, because a higher TFP multiplies *all* of tomorrow's production.

Dividing (1.13) by (1.12), we obtain

$$\lambda \frac{f'(k_1^*)}{f(k_1^*)} = \frac{A'(k_e^*)}{A(k_e^*)} \qquad \Longleftrightarrow \qquad \frac{f'(k_1^*)}{f(k_1^*)} = \frac{1}{\lambda}\frac{A'(k_e^*)}{A(k_e^*)}. \tag{1.14}$$

1.3.3 Is growth improved by the new technology?

To answer this question, we let $f(k) = \gamma k$ with $\gamma > 0$, $A(k_e) = ak_e + 1$ with $a > 0$, and $u(c) = \ln c$. The objective

$$\ln(\gamma k_0 - k_1 - \lambda k_e) + \beta \ln(ak_e + 1) + \beta \ln \gamma + \beta \ln k_1$$

is a sum of concave functions, so the FOCs are sufficient. Since $f'/f = 1/k_1$ and $A'/A = a/(ak_e + 1)$, condition (1.14) gives

$$\frac{\lambda}{k_1^*} = \frac{a}{ak_e^* + 1} \quad \Longrightarrow \quad ak_e^* + 1 = \frac{ak_1^*}{\lambda} \quad \Longrightarrow \quad \lambda k_e^* = k_1^* - \frac{\lambda}{a}.$$

With log utility, we have $u'(c_1)Af'(k_1) = A\gamma/(A\gamma k_1) = 1/k_1$, so that (1.12) reads

$$\frac{1}{\gamma k_0 - k_1^* - \lambda k_e^*} = \frac{\beta}{k_1^*}.$$

Substituting $\lambda k_e^* = k_1^* - \lambda/a$ into the denominator gives $\gamma k_0 - k_1^* - \lambda k_e^* = \gamma k_0 - 2k_1^* + \lambda/a$, so the second equation becomes

$$k_1^* = \beta\Big(\gamma k_0 - 2k_1^* + \frac{\lambda}{a}\Big) \quad \Longleftrightarrow \quad (1+2\beta)k_1^* = \beta\Big(\gamma k_0 + \frac{\lambda}{a}\Big).$$

Using the first equation once more for k_e^* yields

$$k_1^* = \frac{\beta}{1+2\beta}\left(\gamma k_0 + \frac{\lambda}{a}\right),$$
$$k_e^* = \frac{k_1^*}{\lambda} - \frac{1}{a} = \frac{1}{1+2\beta}\left[\frac{\beta\gamma}{\lambda}k_0 - \frac{1+\beta}{a}\right].$$

When does a country adopt the new technology?

$$k_e^* > 0 \quad \Longleftrightarrow \quad a\beta\gamma k_0 > \lambda(1+\beta).$$

The country invests in the new technology if and only if it is **rich enough** (k_0 high), the **quality** of the NT is high (a high), the economy is

productive (γ high), it is **patient** (β high), or the NT is **cheap** (λ low). If the inequality fails, the optimum is the corner $k_e^* = 0$ and the economy behaves as in the benchmark.

We now compare output with and without the NT. Without NT, the benchmark (1.11) with $f(k) = \gamma k$ and log utility gives

$$\frac{1}{\gamma k_0 - \hat{k}_1} = \frac{\beta}{\hat{k}_1},$$

so that

$$\hat{k}_1 = \frac{\beta\gamma}{1+\beta} k_0, \qquad \hat{y}_1 = \gamma \hat{k}_1 = \frac{\beta\gamma^2}{1+\beta} k_0.$$

With the NT (in the interior case), we have

$$y_1^* = (ak_e^* + 1)\gamma k_1^* = \frac{\lambda\gamma}{a}(ak_e^* + 1)^2, \qquad ak_e^* + 1 = \frac{\beta}{1+2\beta}\left(\frac{a\gamma k_0}{\lambda} + 1\right),$$

where we used $k_1^* = \lambda(ak_e^* + 1)/a$. Defining $x \equiv a\gamma k_0/\lambda$, we obtain

$$\begin{aligned} & y_1^* > \hat{y}_1 \\ \iff \quad & \frac{\lambda\gamma}{a}\frac{\beta^2}{(1+2\beta)^2}(x+1)^2 > \frac{\beta\gamma^2 k_0}{1+\beta} \\ \iff \quad & \beta(1+\beta)(x+1)^2 > (1+2\beta)^2 x. \end{aligned}$$

Expanding, this is the quadratic inequality

$$Q(x) \equiv \beta(1+\beta)x^2 - (1+2\beta+2\beta^2)x + \beta(1+\beta) > 0.$$

The product of the roots of Q is 1, and we can check directly that $x = (1+\beta)/\beta$ is a root,

$$Q\left(\tfrac{1+\beta}{\beta}\right) = \tfrac{1+\beta}{\beta}\left[(1+\beta)^2 - (1+2\beta+2\beta^2) + \beta^2\right] = 0.$$

The roots are therefore $\beta/(1+\beta) < 1$ and $(1+\beta)/\beta > 1$, and $Q(x) > 0$ whenever $x > (1+\beta)/\beta$. This last inequality is precisely the adoption condition $a\beta\gamma k_0 > \lambda(1+\beta)$.

Proposition 1.2. *In the example, whenever the country chooses to invest in the new technology ($k_e^* > 0$), period-1 output and hence the growth rate y_1/y_0 are strictly higher than without the NT.*

Intuition

A country adopts the NT only when it pays off, and when it does it uses the NT to multiply the productivity of its whole capital stock. Poor countries (k_0 small) or countries facing a low-quality or expensive technology stay with the old technology, and their growth is lower. This is a simple mechanism for *divergence*: richer countries adopt better technologies and grow faster.

1.4 Open economy and debt

1.4.1 Economy without debt

We consider an open developing country that lives for two periods. Without access to foreign borrowing, it solves

$$\max \{u(c_0) + \beta u(c_1)\}$$

subject to

$$\begin{aligned} c_0 + k_1 &= F(k_0) + (1-\delta)k_0, \\ c_1 &= F(k_1) + (1-\delta)k_1 \end{aligned}$$

where $\delta \in [0,1]$ is the depreciation rate and $k_0 > 0$ is given. In period 0, output plus undepreciated capital, $F(k_0) + (1-\delta)k_0$, is split between consumption and next period's capital k_1. In period 1, the country consumes its output together with the undepreciated capital. After substituting the constraints, we obtain the FOC for k_1,

$$u'\big(F(k_0) + (1-\delta)k_0 - k_1^*\big) = \beta\, u'\big(F(k_1^*) + (1-\delta)k_1^*\big)\,\big(F'(k_1^*) + 1 - \delta\big). \tag{1.15}$$

The gross return on one unit of capital is $F'(k_1) + 1 - \delta$, the marginal product plus the undepreciated part. As before, the LHS increases from $u'(F(k_0) + (1-\delta)k_0)$ to $+\infty$ and the RHS decreases, so the optimum k_1^* is unique.

We assume CRRA utility, as in (1.4), and a linear technology,

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0, \qquad F(k) = Ak.$$

Let $R \equiv A + 1 - \delta$ be the gross return on capital and $W \equiv Rk_0$ period-0 wealth. The constraints become $c_0 = W - k_1$ and $c_1 = Rk_1$, and the Euler equation (1.15) is $c_0^{-\sigma} = \beta R c_1^{-\sigma}$, that is,

$$\frac{c_1}{c_0} = (\beta R)^{\frac{1}{\sigma}}.$$

Substituting $c_1 = Rk_1$ and $c_0 = W - k_1$, we obtain $Rk_1 = (\beta R)^{\frac{1}{\sigma}}(W - k_1)$, or

$$k_1\Big[R + (\beta R)^{\frac{1}{\sigma}}\Big] = (\beta R)^{\frac{1}{\sigma}} W.$$

Dividing numerator and denominator by R and defining

$$D \equiv \beta^{\frac{1}{\sigma}} R^{\frac{1-\sigma}{\sigma}} = \beta^{\frac{1}{\sigma}} (A + 1 - \delta)^{\frac{1-\sigma}{\sigma}},$$

we obtain

$$\begin{aligned} k_1^* &= \frac{D}{1+D}\, W = k_0\, \frac{\beta^{\frac{1}{\sigma}} (A + 1 - \delta)^{\frac{1}{\sigma}}}{1 + \beta^{\frac{1}{\sigma}} (A + 1 - \delta)^{\frac{1-\sigma}{\sigma}}}, \\ c_0^* &= \frac{W}{1+D}, \\ c_1^* &= (A + 1 - \delta)\, k_1^*. \end{aligned} \tag{1.16}$$

The saving rate out of wealth is $D/(1 + D)$.

1.4.2 Growth and the role of debt

We now suppose that in period -1 the country borrows an amount $E > 0$ from abroad at the world interest rate r. The loan is used as additional capital in period 0, and $(1 + r)E$ is repaid at the end of period 0. The constraints become

$$\begin{aligned} c_0 + k_1 - (1 - \delta)(k_0 + E) &= A(k_0 + E) - (1 + r)E, \\ c_1 &= Ak_1 + (1 - \delta)k_1. \end{aligned}$$

The only change is in period-0 wealth, which becomes

$$\widetilde{W} \equiv (A + 1 - \delta)(k_0 + E) - (1 + r)E = (A + 1 - \delta)k_0 + (A - r - \delta)E.$$

Since the problem from period 0 onward is identical to the benchmark with W replaced by $\widetilde{W}$, the solution (1.16) applies and gives

$$\begin{aligned} \tilde{k}_1 &= \frac{D}{1+D}\Big[(A + 1 - \delta)k_0 + (A - r - \delta)E\Big], \\ \tilde{c}_0 &= \frac{(A + 1 - \delta)k_0 + (A - r - \delta)E}{1+D}, \\ \tilde{c}_1 &= (A + 1 - \delta)\tilde{k}_1. \end{aligned}$$

We can now see the effect of debt on the capital stock.

Debt and growth

$$\frac{\partial \tilde{k}_1}{\partial E} = \frac{D}{1+D}\,(A - r - \delta).$$

Borrowing raises investment, period-1 output and welfare if and only if

$$A - \delta > r.$$

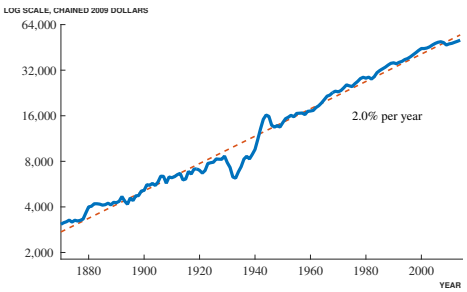
Economic meaning of $A > r + \delta$

One borrowed unit installed as capital produces A units of output and loses δ to depreciation, so its net return is $A - \delta$. The country must pay interest r on it. Borrowing to invest therefore creates value only if the *net marginal product of capital* exceeds the *cost of funds*, $A - \delta > r$. When $A < r + \delta$, each borrowed unit destroys $r + \delta - A$ units of wealth, and debt reduces growth and welfare. This is a stylized story of how poorly-used foreign debt can impoverish a developing country.

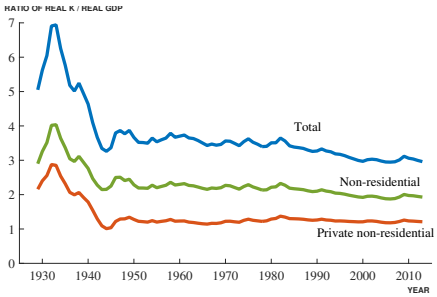
Chapter 2

The Solow Growth Model

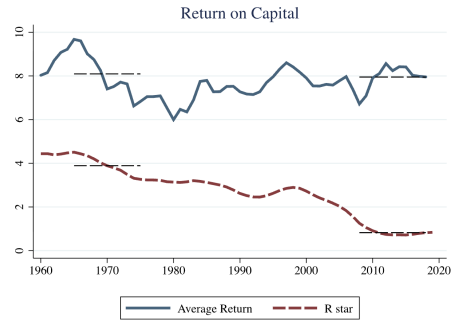
Kaldor (1961) summarized the long-run behaviour of advanced economies in a list of "stylized facts", which Figure 2.1 illustrates with US data.



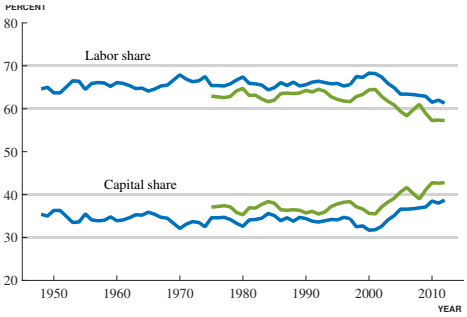
(a) GDP per capita growth (US)



(b) Capital/GDP ratio US



(c) Return on capital and natural rate of interests (R star)



(d) Labor share and capital share in US

Figure 2.1: Facts of growth from Eggertsson et al. (2021) and Jones (2015).

1. Output per worker grows at a roughly constant rate.
2. Capital per worker grows at roughly the same rate.

3. The capital–output ratio K/Y is roughly constant.
4. The real return to capital is roughly constant.
5. The shares of capital and labor in national income are constant.

In addition, income per capita differs enormously across countries, and so do growth rates. The Solow model is designed to explain the first set of facts and to organize our thinking about the second.

2.1 The environment

The model of Solow (1956) and Swan (1956) is the starting point of growth theory. Its key simplification is that households save a constant fraction of their income, so that all the action comes from the accumulation of capital and from technology.

2.1.1 Technology

Time is discrete, $t = 0, 1, 2, \ldots$, and there is one good, produced from capital K_t and labor L_t according to

$$Y_t = F(K_t, L_t).$$

Definition 2.1 (Neoclassical production function)**.** $F : \mathbb{R}^2_+ \to \mathbb{R}_+$ is *neoclassical* if it satisfies the following properties.

1. **Constant returns to scale** (CRS), $F(\lambda K, \lambda L) = \lambda F(K, L)$ for all $\lambda > 0$.
2. **Positive and diminishing marginal products**, $F_K > 0$, $F_L > 0$, $F_{KK} < 0$ and $F_{LL} < 0$.
3. The **Inada conditions**,
$$\lim_{K\to 0} F_K = \lim_{L\to 0} F_L = +\infty, \qquad \lim_{K\to\infty} F_K = \lim_{L\to\infty} F_L = 0.$$
4. **Essential inputs**, $F(0, L) = F(K, 0) = 0$.

The intensive form. CRS allows us to write everything per worker. Taking $\lambda = 1/L$, we have

$$\frac{Y}{L} = F\Big(\frac{K}{L}, 1\Big) \qquad \text{so we write output per worker as} \quad y = f(k),$$

and thus

$$k \equiv \frac{K}{L}, \;\; y \equiv \frac{Y}{L}, \;\; f(k) \equiv F(k, 1).$$

The properties of F translate into $f(0) = 0$, $f' > 0$, $f'' < 0$, $f'(0) = +\infty$, $f'(\infty) = 0$. Since $F(K, L) = Lf(K/L)$, the marginal products are [1]

$$F_K = f'(k), \qquad F_L = f(k) - kf'(k). \tag{2.1}$$

Factor prices and shares. If factor markets are competitive, firms pay each factor its marginal product. The rental rate of capital is $R^k_t = f'(k_t)$ and the wage is $w_t = f(k_t) - k_t f'(k_t)$. By Euler's theorem for homogeneous functions,

$$R^k_t K_t + w_t L_t = L_t\big[k_t f'(k_t) + f(k_t) - k_t f'(k_t)\big] = L_t f(k_t) = Y_t,$$

so factor payments exhaust output and profits are zero. The capital share is

$$\alpha_K(k) \equiv \frac{R^k K}{Y} = \frac{kf'(k)}{f(k)} = \frac{\mathrm{d}\ln f(k)}{\mathrm{d}\ln k},$$

the *elasticity of output per worker with respect to capital per worker*.

Example 2.1 (Cobb–Douglas)**.** The function $F(K, L) = AK^\alpha L^{1-\alpha}$ with $0 < \alpha < 1$ gives $f(k) = Ak^\alpha$, $f'(k) = \alpha A k^{\alpha-1}$, $R^k = \alpha A k^{\alpha-1}$, $w = (1-\alpha)Ak^\alpha$, and $\alpha_K(k) = \alpha$ for all k. Factor shares are therefore constant, in line with Kaldor's fifth fact. The *elasticity of substitution* between capital and labor,

$$\sigma_{KL} \equiv \frac{\mathrm{d}\ln(K/L)}{\mathrm{d}\ln(F_L/F_K)},$$

the percentage change in the capital–labor ratio when the relative factor price w/R^k rises by one percent, equals one for Cobb–Douglas, because $w/R^k = \frac{1-\alpha}{\alpha}k$ implies $\mathrm{d}\ln k/\mathrm{d}\ln(w/R^k) = 1$.

2.1.2 Equilibrium

The economy is closed and there is no government, so output is either consumed or invested. The goods market clears when

$$Y_t = C_t + I_t,$$

where the left-hand side is aggregate supply and the right-hand side is aggregate demand. The defining assumption of the model is that a constant fraction $s \in (0, 1)$ of output is saved, and that saving equals investment,

$$I_t = S_t = sY_t, \qquad C_t = (1-s)Y_t.$$

[1] To derive the second, differentiate $Lf(K/L)$ with respect to L, which gives $f(k) + Lf'(k)\cdot(-K/L^2) = f(k) - kf'(k)$.

Capital depreciates at rate $\delta \in [0,1]$ and the labor force grows at a deterministic rate $n > -1$,

$$K_{t+1} = (1-\delta)K_t + I_t,$$
$$L_{t+1} = (1+n)L_t.$$

Investment made in period t becomes productive capital in period $t+1$ ("time to build" of one period).

2.2 The dynamic equation

Combining the accumulation equation with $I_t = sY_t$ and dividing by L_t, we obtain

$$\frac{K_{t+1}}{L_t} = s\frac{Y_t}{L_t} + (1-\delta)\frac{K_t}{L_t}.$$

The left-hand side equals $\frac{K_{t+1}}{L_{t+1}}\frac{L_{t+1}}{L_t} = (1+n)k_{t+1}$, and hence

$$k_{t+1} = g(k_t) \equiv \frac{sf(k_t) + (1-\delta)k_t}{1+n} \tag{2.2}$$

This is a **first-order nonlinear difference equation**: given k_0, it determines the entire path $\{k_t\}$. Subtracting k_t from both sides gives

$$\Delta k_t \equiv k_{t+1} - k_t = \frac{sf(k_t) - (n+\delta)k_t}{1+n}. \tag{2.3}$$

Actual versus break-even investment

The term $sf(k)$ is *actual investment* per worker. The term $(n+\delta)k$ is *break-even investment*, the investment needed just to keep k constant by replacing depreciated capital (δk) and equipping new workers (nk). Capital per worker rises when actual investment exceeds break-even investment and falls otherwise. The division by $1+n$ appears because the capital installed today is shared among tomorrow's (larger) labor force.

2.3 Steady state

Definition 2.2 (Steady state)**.** A steady state is a k^* such that $g(k^*) = k^*$, i.e. $\Delta k = 0$. Equivalently,

$$sf(k^*) = (n+\delta)k^*. \tag{2.4}$$

The point $k = 0$ is always a (trivial) steady state, and we are interested in positive steady states.

Proposition 2.1 (Existence, uniqueness and global stability)**.** *Suppose f is neoclassical, $s \in (0,1)$ and $n + \delta > 0$. Then the following hold.*

1. *There is a unique steady state $k^* > 0$.*
2. *For every $k_0 > 0$, the path $k_{t+1} = g(k_t)$ converges monotonically to k^*. If $k_0 < k^*$, then k_t increases to k^*, and if $k_0 > k^*$, then k_t decreases to k^*.*

Proof. Existence and uniqueness. For $k > 0$, we define $\psi(k) \equiv sf(k)/k$, so that a positive steady state solves $\psi(k) = n + \delta$. Because f is strictly concave with $f(0) = 0$, the average product $f(k)/k$ is strictly decreasing,

$$\frac{\mathrm{d}}{\mathrm{d}k}\frac{f(k)}{k} = \frac{kf'(k) - f(k)}{k^2} < 0,$$

since $f(k) = f(k) - f(0) > kf'(k)$ (the chord lies above the tangent for a strictly concave function). By l'Hôpital's rule and the Inada conditions,

$$\lim_{k\to 0}\frac{f(k)}{k} = \lim_{k\to 0} f'(k) = +\infty, \qquad \lim_{k\to\infty}\frac{f(k)}{k} = \lim_{k\to\infty} f'(k) = 0.$$

Hence ψ falls continuously from $+\infty$ to 0 and crosses $n + \delta > 0$ exactly once.

Monotonicity of the map. Since $g'(k) = [sf'(k) + 1 - \delta]/(1 + n) > 0$, the map g is strictly increasing. Hence if $k_t < k^*$, then $k_{t+1} = g(k_t) < g(k^*) = k^*$, and the path never overshoots.

Direction of motion. From (2.3), $\Delta k_t > 0$ if and only if $\psi(k_t) > n + \delta$, that is, if and only if $k_t < k^*$. For $k_0 < k^*$, the sequence is therefore increasing and bounded above by k^*, so it converges to some limit $\bar{k} \le k^*$. By continuity of g, $\bar{k} = g(\bar{k})$, and since $\bar{k} \ge k_0 > 0$, uniqueness gives $\bar{k} = k^*$. The case $k_0 > k^*$ is symmetric. □

Why does the economy converge?

The engine of convergence is *diminishing returns to capital*. When k is low, an extra unit of capital is very productive, so saving $sf(k)$ is large relative to the capital stock and k grows fast. As k rises, the average product $f(k)/k$ falls and saving eventually just covers break-even investment. Nothing in the model can keep k growing forever, so in the long run *income per worker stops growing*.

2.4 An example with Cobb–Douglas technology

With $f(k) = Ak^\alpha$, the steady-state condition (2.4) becomes $sAk^{*\alpha} = (n + \delta)k^*$, so that

$$k^* = \left(\frac{sA}{n+\delta}\right)^{\frac{1}{1-\alpha}}, \qquad y^* = A^{\frac{1}{1-\alpha}}\left(\frac{s}{n+\delta}\right)^{\frac{\alpha}{1-\alpha}}, \qquad c^* = (1-s)y^*.$$

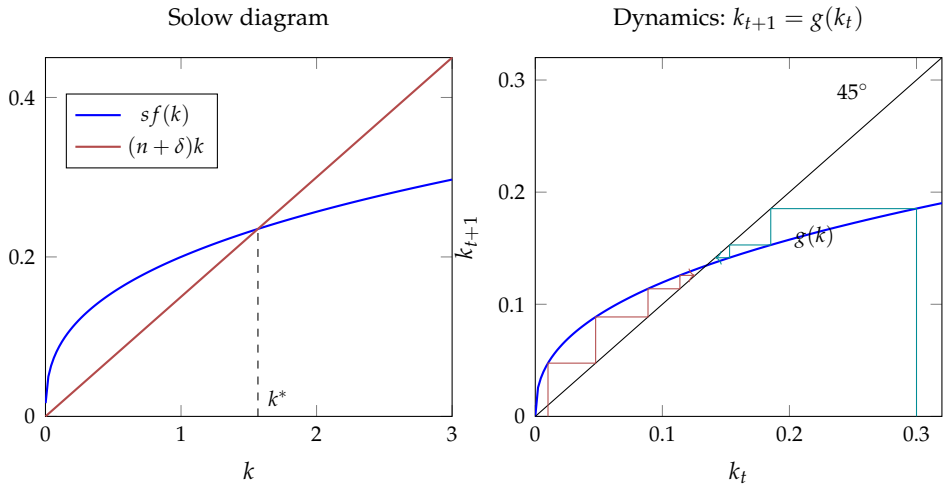


Figure 2.2: Left, the Solow diagram for $f(k) = k^{0.36}$, $s = 0.2$, $n = 0.05$ and $\delta = 0.1$. The steady state is where actual and break-even investment cross, at $k^* = (0.2/0.15)^{1/0.64} \approx 1.5675$. Right, the 45°-line diagram for $k_{t+1} = 0.3\,k_t^{0.4}$ (the case $\delta = 1$, $n = 0$, $s = 0.3$, $\alpha = 0.4$). The staircases show monotone convergence to $k^* = 0.3^{1/0.6} \approx 0.134$ from below and from above.

Elasticities of steady-state income

Taking logs, we obtain

$$\ln y^* = \frac{1}{1-\alpha}\ln A + \frac{\alpha}{1-\alpha}\ln s - \frac{\alpha}{1-\alpha}\ln(n+\delta),$$

and hence

$$\frac{\partial \ln y^*}{\partial \ln s} = \frac{\alpha}{1-\alpha}.$$

With a capital share $\alpha = 1/3$, this elasticity is $1/2$, so doubling the saving rate raises steady-state income per worker by a factor $2^{1/2} \approx 1.41$, or by about 41%. The effect of saving on income is *modest* because of diminishing returns. More saving raises k, but each additional unit of k adds less output. The effect of TFP is larger than one-for-one, $\partial \ln y^*/\partial \ln A = 1/(1-\alpha) = 1.5$, because higher TFP raises output directly and also raises saving and hence capital.

Example 2.2. Let $\alpha = 0.36$, $n = 0.05$, $\delta = 0.10$, $s = 0.20$ and $A = 1$. Then $k^* = (0.2/0.15)^{1/0.64} \approx 1.5675$, $y^* \approx 1.1756$ and $c^* \approx 0.9405$. Iterating (2.2)

from $k_0 = 0.3$ gives

$$\begin{aligned} k_t: \quad & 0.3000,\ 0.3806,\ 0.4608,\ 0.5391,\ 0.6145,\\ & 0.6866,\ 0.7549,\ 0.8192,\ \ldots \to 1.5675, \end{aligned}$$

and from $k_0 = 3$ gives 3.0000, 2.8543, 2.7244, 2.6084, $\ldots \to 1.5675$. Convergence is monotone from both sides, as Proposition 2.1 predicts.

If capital fully depreciates ($\delta = 1$), a common assumption when a model period is long, the law of motion becomes

$$k_{t+1} = \frac{sA}{1+n} k_t^{\alpha}.$$

Taking logs, we obtain

$$\ln k_{t+1} = \ln \frac{sA}{1+n} + \alpha \ln k_t,$$

which is a *linear* first-order difference equation in $x_t \equiv \ln k_t$. Its fixed point is $x^* = \ln(sA/(1+n))/(1-\alpha)$. Subtracting $x^* = \ln \frac{sA}{1+n} + \alpha x^*$, we find

$$x_{t+1} - x^* = \alpha(x_t - x^*) \quad \Longrightarrow \quad \ln k_t - \ln k^* = \alpha^t \big(\ln k_0 - \ln k^* \big).$$

The log-gap to the steady state shrinks by the factor α each period (Appendix A.4).

2.5 Comparative dynamics and convergence

Suppose that the economy is in its steady state and that s rises permanently to $s' > s$. The $sf(k)$ curve shifts up, and the new steady state $k^{*\prime}$ is higher. At the moment of the change, k is predetermined and output does not jump. Consumption therefore *falls* on impact, from $(1-s)f(k^*)$ to $(1-s')f(k^*)$, after which k, y and $c = (1-s')f(k)$ rise monotonically to their new steady-state values. The new steady-state consumption exceeds the old one only if the economy remains below the golden rule (Section 2.7). Otherwise, consumption ends lower. The growth rate of y jumps up and then declines back to zero, as Figure 2.3 shows. A higher saving rate has a **level effect**, not a **growth effect**, on income per worker.

A rise in n or δ rotates the break-even line $(n+\delta)k$ upwards. Capital must then be spread over more workers or replaced faster, so k^* and y^* fall ("capital dilution").

We now turn to growth rates. Dividing (2.3) by k_t gives

$$g_{k,t} \equiv \frac{k_{t+1} - k_t}{k_t} = \frac{1}{1+n} \left[s \frac{f(k_t)}{k_t} - (n+\delta) \right]. \tag{2.5}$$

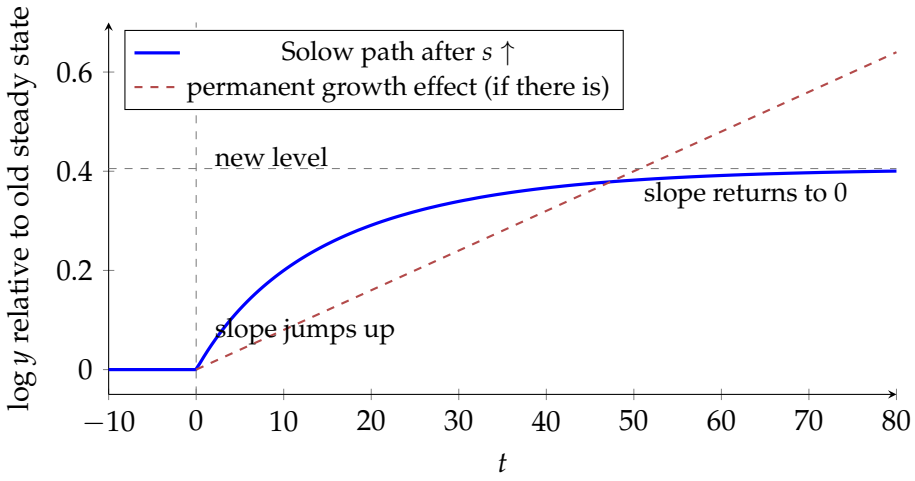


Figure 2.3: Level versus growth effect of a permanent rise in the saving rate at $t = 0$.

Because $f(k)/k$ is decreasing, *the growth rate of capital per worker falls as k rises*. If two economies share the same parameters (s, n, δ, f) but differ in initial capital, they converge to the same k^*, and the poorer one grows faster along the way. This is **absolute convergence** *conditional* on parameters. If parameters differ, each economy converges to its own steady state, and the prediction becomes one of **conditional convergence**, in which an economy grows faster the further it is *below its own* steady state. We validate these predictions with the data in Section 2.6.

2.6 Empirical evidence

The Solow model makes sharp predictions about the *levels* of income per worker. Taking logs of the Cobb–Douglas steady state, we obtain

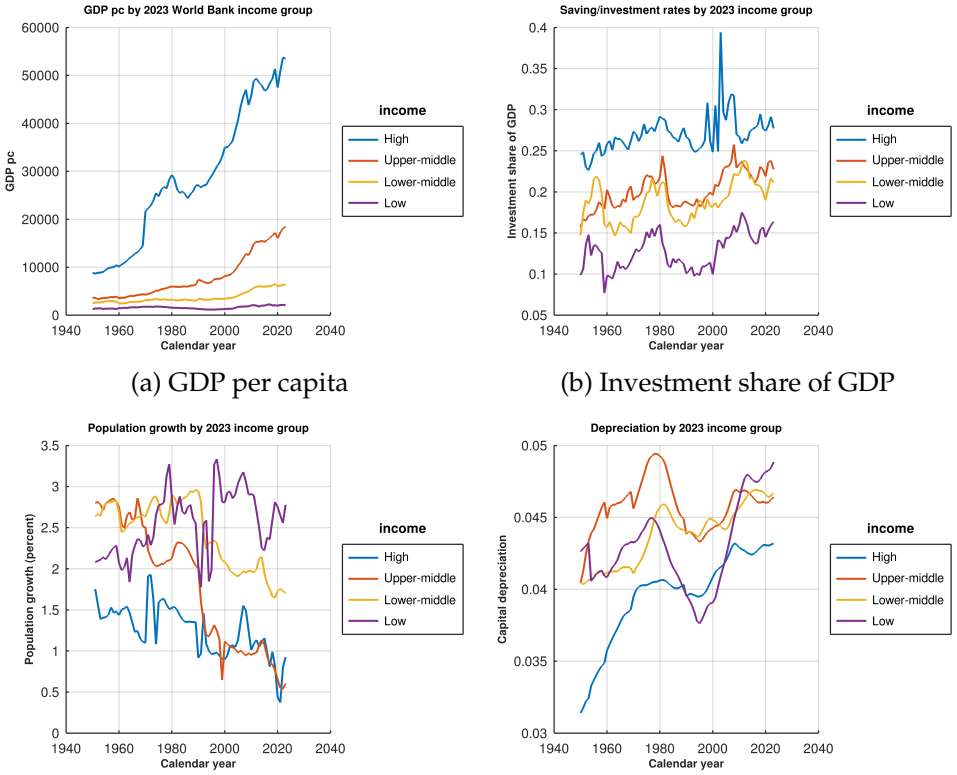
$$\ln y^* = \frac{1}{1-\alpha} \ln A + \frac{\alpha}{1-\alpha} \ln s - \frac{\alpha}{1-\alpha} \ln(n + \delta).$$

According to this expression, countries that save and invest more should be richer, and countries with faster population growth or faster depreciation should be poorer. The model also makes a prediction about *growth rates*: an economy grows faster the further it is below its own steady state. In this section we confront these predictions with cross-country data.

2.6.1 Income, saving and population growth across income groups

Figure 2.4 groups countries according to the World Bank's 2023 income classification and plots, for each group, the average of the four objects that appear in the steady-state formula, namely income per capita, the investment (saving) rate s, the population growth rate n and the depreciation rate δ.

hol



(a) GDP per capita

(b) Investment share of GDP

(c) Population growth (percent)

(d) Depreciation rate of capital

Figure 2.4: Income per capita and the determinants of the Solow steady state, by 2023 World Bank income group, 1950–2023.

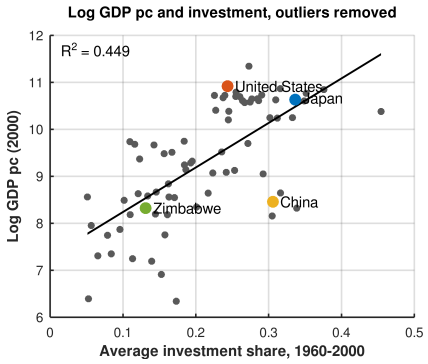
1. **Income differences are large and persistent** (panel a). Income per capita in the high-income group is many times that of the low-income group, and the gap in levels has widened over seventy years. Only the upper-middle-income group has clearly accelerated since 2000.

2. **Richer countries invest more** (panel b). Throughout the period the ranking of investment rates coincides with the ranking of income, with roughly 25–30% of GDP in high-income countries against 10–

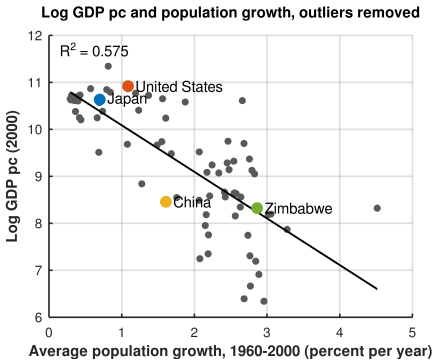
16% in low-income countries. This is exactly the ordering predicted by $\partial \ln y^* / \partial \ln s > 0$.

3. **Poorer countries have faster population growth** (panel c). Low-income countries' populations have grown by 2–3% per year, high-income countries' by about 1–1.5%. Again this matches the prediction $\partial \ln y^* / \partial \ln n < 0$, since capital has to be spread over a faster-growing number of workers ("capital dilution").

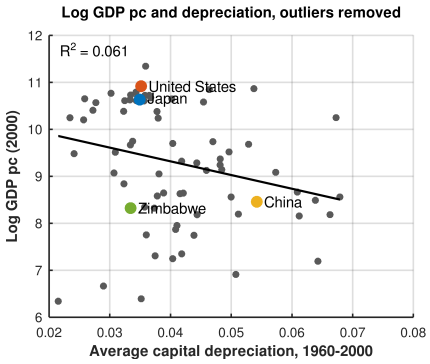
4. Depreciation rates (panel d) are somewhat lower in high-income countries, but the differences across groups are small. They are therefore unlikely to account for much of the income gap.



(a) Investment share



(b) Population growth



(c) Depreciation

Figure 2.5: Log GDP per capita in 2000 against average investment shares, population growth and depreciation rates over 1960–2000, with OLS fits.

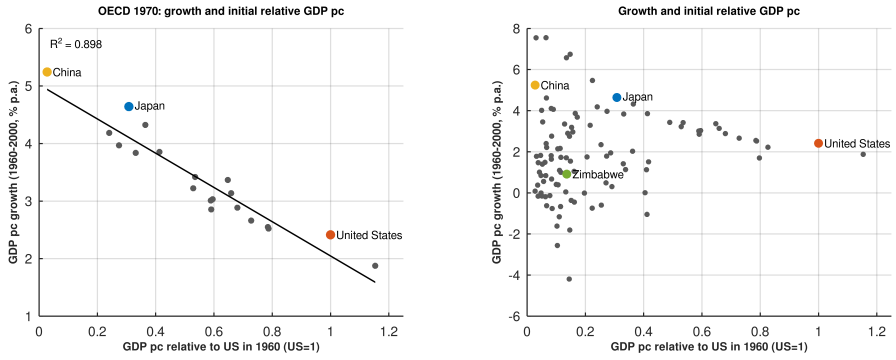
2.6.2 Income levels and their determinants across countries

Group averages can hide a lot of heterogeneity. Figure 2.5 therefore plots each country separately, showing the logarithm of GDP per capita in 2000

against the average investment share, the average population growth rate and the average depreciation rate over 1960–2000 (a few extreme outliers are removed). The black lines are OLS fits.

The signs are those predicted by the model. Investment alone "explains" about 45% of the cross-country variance of log income ($R^2 = 0.449$), population growth alone about 57% ($R^2 = 0.575$), while depreciation explains very little ($R^2 = 0.061$). We first note that the slopes are *steep*. Along the fitted line in panel a, moving from an investment share of 0.1 to 0.3 raises log income by about 1.9, which multiplies income per capita by a factor of $e^{1.9} \approx 7$. The Solow model with $\alpha = 1/3$ predicts an elasticity of income with respect to s of only $\alpha/(1-\alpha) = 1/2$, so that tripling s should raise income by a factor of only $3^{1/2} \approx 1.7$. The scatter plots also reveal interesting outliers. China invests as much as Japan but was far poorer in 2000, because its TFP A and its stock of human capital were much lower and it was still far from its steady state. We return to the first point in Section 2.10.

2.6.3 Convergence



(a) OECD members of 1970 (b) All countries

Figure 2.6: Average growth of GDP per capita over 1960–2000 against GDP per capita relative to the US in 1960. China, Japan, the US and Zimbabwe are highlighted for reference.

Equation (2.5) implies that, *holding the steady state fixed*, a poorer economy grows faster. Figure 2.6 tests this idea in two samples. Panel a considers the countries that were members of the OECD in 1970. These economies have similar saving rates, population growth, institutions and access to technology, so they plausibly share a similar steady state. The relationship between growth over 1960–2000 and income relative to the US in 1960 is strongly negative ($R^2 = 0.898$). Japan, which started at about 30% of US income, grew at more than 4.5% per year, while the US grew at about 2.5%.

Panel b uses the whole world, and here we see no relation at all. Many poor countries grew slowly or even shrank, while a few (China among them) grew very fast.

Absolute versus conditional convergence

The two panels do not contradict the model. They are exactly what the Solow model predicts once we recall that convergence is *conditional*. Among the OECD economies, which have similar (s, n, δ, A), initial income is a good proxy for the distance to the common steady state, and poorer countries do grow faster. In other words, **absolute convergence** holds within this club. In the world as a whole, poor countries are poor partly *because* their steady states are low (low saving, fast population growth, low TFP). Being poor then does not mean being far below one's own steady state, and there is no reason to expect fast growth. The proper test of the model is whether poor countries grow faster *after controlling for the determinants of the steady state*. We run this test in Section 2.10, and the answer turns out to be yes.

2.7 The golden rule and dynamic efficiency

Steady-state consumption per worker is

$$c^* = (1 - s) f(k^*) = f(k^*) - s f(k^*) = f(k^*) - (n + \delta) k^*.$$

Different saving rates give different k^* and hence different c^*, which raises the question of which steady state yields the most consumption.

Definition 2.3 (Golden rule)**.** The golden-rule capital stock k_{GR} maximizes steady-state consumption $c(k) = f(k) - (n + \delta)k$ and therefore satisfies

$$f'(k_{GR}) = n + \delta.$$

The golden-rule saving rate $s_{GR} = (n + \delta) k_{GR} / f(k_{GR})$ is the one that makes k_{GR} the steady state.

Intuition

In a steady state, one more unit of capital per worker produces $f'(k)$ more output but must be maintained, which requires $(n + \delta)$ units of extra investment every period. Consumption is maximized when the marginal product equals this "maintenance cost". We note that

$$s_{GR} = \frac{(n + \delta) k_{GR}}{f(k_{GR})} = \frac{f'(k_{GR}) k_{GR}}{f(k_{GR})} = \alpha_K(k_{GR}),$$

so that **at the golden rule, the saving rate equals the capital share**. With Cobb–Douglas technology, $s_{GR} = \alpha$ and $k_{GR} = (\alpha A/(n+\delta))^{1/(1-\alpha)}$. In the numerical example above, $k_{GR} = (0.36/0.15)^{1/0.64} \approx 3.93 > k^* = 1.57$, so that economy saves too little to maximize steady-state consumption.

Proposition 2.2 (Dynamic inefficiency)**.** *If $k^* > k_{GR}$ (with Cobb–Douglas, $s > \alpha$), the economy is **dynamically inefficient**, in the sense that lowering the saving rate to s_{GR} raises consumption in* every *period.*

Proof. We start at $k_0 = k^* > k_{GR}$ and set $s = s_{GR}$ from $t = 0$ onward. By Proposition 2.1, the capital stock decreases monotonically to k_{GR}, so $k_t \geq k_{GR}$ for all t. Consumption is then

$$c_t = (1 - s_{GR}) f(k_t) \geq (1 - s_{GR}) f(k_{GR}) = c_{GR}.$$

Moreover, $c_{GR} > c^*$ by definition of the golden rule (strict concavity of $c(k)$ and $k^* \neq k_{GR}$). Hence $c_t > c^*$ for all $t \geq 0$. □

If $k^* < k_{GR}$, the economy is **dynamically efficient**. Raising s would increase consumption in the long run, but only at the cost of lower consumption during the transition. Whether that is desirable depends on how we weigh present against future generations. The Solow model cannot answer this question because it has no preferences, and we return to it with the Ramsey model in Chapter 4. Empirically, most economies appear to be dynamically efficient, since the gross return to capital exceeds gross investment (Abel et al., 1989).

2.8 The speed of convergence

We now ask how fast the economy approaches its steady state. A first-order Taylor expansion of g around k^* (Appendix A.4) gives

$$k_{t+1} - k^* \approx g'(k^*)\,(k_t - k^*), \qquad g'(k^*) = \frac{s f'(k^*) + 1 - \delta}{1 + n}.$$

Using the steady-state condition $s = (n+\delta)k^*/f(k^*)$, we have

$$s f'(k^*) = (n+\delta)\frac{k^* f'(k^*)}{f(k^*)} = (n+\delta)\alpha_K^*, \qquad \alpha_K^* \equiv \alpha_K(k^*).$$

Therefore

$$g'(k^*) = \frac{1 - \delta + \alpha_K^*(n+\delta)}{1+n} = 1 - \frac{(1-\alpha_K^*)(n+\delta)}{1+n}.$$

We define the **speed of convergence** as $\lambda \equiv 1 - g'(k^*)$, so that

$$\lambda = \frac{(1-\alpha_K^*)(n+\delta)}{1+n}, \qquad k_t - k^* \approx (1-\lambda)^t (k_0 - k^*).$$

A fraction λ of the remaining gap is closed each period. The **half-life**, the number of periods needed to close half of the gap, solves $(1-\lambda)^{t_{1/2}} = 1/2$ and is given by

$$t_{1/2} = \frac{\ln 2}{-\ln(1-\lambda)} \approx \frac{0.69}{\lambda}.$$

Intuition

Convergence is faster when depreciation and population growth are high (break-even investment is large, so deviations are corrected quickly) and when the capital share is low (returns diminish fast). In the limit $\alpha_K \to 1$, diminishing returns disappear and convergence stops, which is the case of the AK model of Chapter 6. In the numerical example ($\alpha = 0.36$, $n = 0.05$, $\delta = 0.1$), we obtain $\lambda = 0.64 \times 0.15/1.05 \approx 0.091$ and $t_{1/2} \approx 7.2$ periods.

2.9 Technological progress and balanced growth

2.9.1 Labor-augmenting technology

Without technical progress, income per worker is constant in the long run, which contradicts Kaldor's first fact. We therefore introduce *labor-augmenting* (Harrod-neutral) technology,

$$Y_t = F(K_t, A_t L_t), \qquad A_{t+1} = (1+g) A_t.$$

The product $A_t L_t$ is labor measured in *efficiency units*. With Cobb–Douglas technology, $K^\alpha (AL)^{1-\alpha} = A^{1-\alpha} K^\alpha L^{1-\alpha}$, so labor-augmenting progress at rate g is the same as TFP growing at gross rate $(1+g)^{1-\alpha}$. We use this form because a theorem due to Uzawa shows that labor-augmenting progress is the only form compatible with a balanced growth path for general CRS technologies.

We define variables per effective worker, $\tilde{k}_t \equiv K_t/(A_t L_t)$ and $\tilde{y}_t \equiv Y_t/(A_t L_t) = f(\tilde{k}_t)$. Dividing $K_{t+1} = sY_t + (1-\delta)K_t$ by $A_t L_t$ and using $A_{t+1}L_{t+1} = (1+g)(1+n)A_t L_t$, we obtain

$$\tilde{k}_{t+1} = \frac{s f(\tilde{k}_t) + (1-\delta)\tilde{k}_t}{(1+n)(1+g)}, \qquad \Delta \tilde{k}_t = \frac{s f(\tilde{k}_t) - (n + g + ng + \delta)\tilde{k}_t}{(1+n)(1+g)}.$$

This has exactly the structure of (2.2), with $(1+n)$ replaced by $(1+n)(1+g)$ and n by $n + g + ng \approx n + g$. Proposition 2.1 therefore applies, and $\tilde{k}_t$

converges monotonically to a unique $\tilde{k}^*$ that satisfies

$$sf(\tilde{k}^*) = (n + g + ng + \delta)\tilde{k}^*.$$

With Cobb–Douglas technology,

$$\tilde{k}^* = \left(\frac{s}{n + g + ng + \delta} \right)^{\frac{1}{1-\alpha}}.$$

2.9.2 The balanced growth path

Definition 2.4 (Balanced growth path)**.** A balanced growth path (BGP) is a path along which all variables grow at constant (possibly different) rates.

Once $\tilde{k}_t = \tilde{k}^*$, the economy has the following properties.

- Capital and income per worker, $k_t = K_t/L_t = A_t\tilde{k}^*$ and $y_t = A_t f(\tilde{k}^*)$, grow at rate g.
- Aggregate capital K_t and output Y_t grow at gross rate $(1 + n)(1 + g)$.
- The capital–output ratio $K_t/Y_t = \tilde{k}^*/f(\tilde{k}^*)$ is constant.
- The rental rate $R^k = f'(\tilde{k}^*)$ is constant, and the wage $w_t = A_t[f(\tilde{k}^*) - \tilde{k}^* f'(\tilde{k}^*)]$ grows at rate g.
- Factor shares are constant.

All five Kaldor facts therefore hold. The growth rate of income per worker, g, is however *exogenous*, so the model explains everything except the one number we care about most. This is the motivation for endogenous growth theory (Chapter 6).

We can verify this result with the production function. The relation $y_t = k_t^\alpha A_t^{1-\alpha}$ implies

$$1 + g_y = (1 + g_k)^\alpha (1 + g)^{1-\alpha}.$$

On the BGP, $g_k = g_y$, so $(1 + g_y)^{1-\alpha} = (1 + g)^{1-\alpha}$ and $g_y = g$.

The golden rule in efficiency units is $f'(\tilde{k}_{GR}) = n + g + ng + \delta$, and again $s_{GR} = \alpha$ in the Cobb–Douglas case. The speed of convergence becomes

$$\lambda = \frac{(1-\alpha)(n + g + ng + \delta)}{(1 + n)(1 + g)} \approx (1 - \alpha)(n + g + \delta).$$

With $\alpha = 1/3$, $n = 0.01$, $g = 0.02$ and $\delta = 0.05$ (annual data), we obtain

$$\lambda = \frac{(2/3)(0.0802)}{1.0302} \approx 0.052,$$

and $t_{1/2} \approx 13$ years. Empirical estimates are closer to $\lambda \approx 0.02$, a half-life of about 35 years, so the textbook Solow model converges too fast.

2.10 Taking the Solow model to the data

The pictures of Section 2.6 are suggestive, but they look at one determinant at a time and do not tell us whether the *magnitudes* are those implied by the model. Mankiw et al. (1992) (MRW) proposed a simple and influential way to estimate the Solow model directly. We reproduce their main results using their original data set (75 countries, 1960–1985), following the presentation in Favero (2001, ch. 1).

2.10.1 From the steady state to a regression

We assume that countries are on their balanced growth paths and that technology grows at the common rate g, so that $A_t = A_0(1+g)^t$. The Cobb–Douglas steady state in efficiency units is $\tilde{y}^* = (\tilde{k}^*)^{\alpha}$, with

$$\tilde{k}^* = \left(\frac{s}{n + g + \delta} \right)^{\frac{1}{1-\alpha}},$$

so income per worker $y_t = A_t \tilde{y}^*$ satisfies

$$\ln y_t = \ln A_0 + t \ln(1+g) + \frac{\alpha}{1-\alpha} \ln s - \frac{\alpha}{1-\alpha} \ln(n + g + \delta),$$

where we use the approximation $n + g + ng + \delta \approx n + g + \delta$. Countries differ in s and n, which we observe, and in the initial level of technology A_0, which we do not. MRW write $\ln A_0 = a + \varepsilon_i$, where a is a constant and ε_i is a country-specific shock (resource endowments, climate, institutions). For country i in a given year this gives the regression

$$\ln y_i = c + b_1 \ln s_i + b_2 \ln(n_i + g + \delta) + \varepsilon_i, \qquad b_1 = \frac{\alpha}{1-\alpha}, \quad b_2 = -\frac{\alpha}{1-\alpha}. \tag{2.6}$$

Following MRW, y_i is real GDP per working-age person in 1985, s_i is the average share of real investment in GDP over 1960–1985, n_i is the average growth rate of the working-age population over the same period, and $g + \delta = 0.05$ for all countries.

The model makes the following testable predictions about (2.6).

1. The coefficient on saving is positive and that on population growth is negative.
2. The two coefficients are equal in absolute value, $b_1 + b_2 = 0$, because only the ratio $s/(n+g+\delta)$ matters for the steady state.
3. Their size reveals the capital share. With $\alpha \approx 1/3$, the share of capital income in national accounts, we should find $b_1 = -b_2 \approx 0.5$.

OLS estimation of (2.6) is valid if ε_i is uncorrelated with s_i and n_i. This is the key identifying assumption, and we will come back to it.

2.10.2 Results

We report the estimates in Table 2.1.

Table 2.1: Estimation of the textbook Solow model. Dependent variable: log GDP per working-age person in 1985. Standard errors in parentheses. 75 countries.

	Unrestricted	Restricted
$\ln s_i$	1.33 (0.17)	
$\ln(n_i + g + \delta)$	-2.01 (0.53)	
$\ln s_i - \ln(n_i + g + \delta)$		1.44 (0.14)
R^2	0.60	0.60
Standard error of regression	0.61	0.61
Test of $b_1 + b_2 = 0$	$F(1, 72) = 1.26$ $(p = 0.27)$	
Test of $b_1 = -b_2 = 0.5$	$F(2, 72) = 23.48$ $(p < 0.001)$	
Implied α		0.59 (0.02)

What the Solow model gets right. The data confirm the first two predictions. Saving enters with a positive coefficient and population growth with a negative one, and both are highly significant. The restriction $b_1 + b_2 = 0$ is not rejected ($p = 0.27$), so we can impose it and regress $\ln y_i$ on the single variable $\ln\big(s_i/(n_i + g + \delta)\big)$, which leaves the fit essentially unchanged. It is remarkable that, in such a simple model, saving and population growth alone account for 60% of the cross-country variation in income per worker.

What it gets wrong. The third prediction fails. Since $b = \alpha/(1-\alpha)$, the restricted coefficient 1.44 implies

$$\hat{\alpha} = \frac{\hat{b}}{1 + \hat{b}} = \frac{1.44}{2.44} \approx 0.59,$$

with a standard error of about 0.02 (by the delta method, $\mathrm{se}(\hat{\alpha}) = \mathrm{se}(\hat{b})/(1+\hat{b})^2$). A capital share of 0.6 is far above the value of about $1/3$ measured in national accounts, and the hypothesis $b_1 = -b_2 = 0.5$ is overwhelmingly rejected. In other words, saving and population growth affect income *in the direction* predicted by Solow, but *much more strongly* than a model with $\alpha = 1/3$ can explain. This is the formal counterpart of the steep slopes we saw in Figure 2.5.

2.10.3 Omitted human capital

We now ask why OLS would overstate the effect of saving. Suppose that the true model contains a third variable, the rate of investment in *human*

capital $s_{H,i}$,

$$\ln y_i = c + \beta_1 \ln s_i + \beta_2 \ln(n_i + g + \delta) + \beta_3 \ln s_{H,i} + u_i,$$

and that we omit it. Then $\beta_3 \ln s_{H,i}$ ends up in the error term of (2.6), and the identifying assumption fails if $s_{H,i}$ is correlated with the included regressors. This is the typical *omitted-variable bias*. Let γ_1 and γ_2 be the coefficients of the auxiliary regression of the omitted variable on the included ones,

$$\ln s_{H,i} = d + \gamma_1 \ln s_i + \gamma_2 \ln(n_i + g + \delta) + v_i.$$

Substituting into the true model shows that the short regression (2.6) estimates

$$b_1 = \beta_1 + \beta_3 \gamma_1, \qquad b_2 = \beta_2 + \beta_3 \gamma_2.$$

In the data, countries that invest a lot in physical capital also invest a lot in schooling ($\gamma_1 > 0$), and countries with fast population growth invest less in schooling ($\gamma_2 < 0$). Since schooling raises income ($\beta_3 > 0$), the coefficient on saving is biased *upwards* and the coefficient on population growth is biased *downwards* (more negative). Both biases go in the direction we observe. Using the percentage of the working-age population enrolled in secondary school as a proxy for $s_{H,i}$, the auxiliary regression gives $\hat{\gamma}_1 = 0.84$ and $\hat{\gamma}_2 = -0.71$. Combined with the estimates of the long regression reported in Table 2.2 below ($\hat{\beta}_1 = 0.71$, $\hat{\beta}_2 = -1.50$, $\hat{\beta}_3 = 0.73$), the formula reproduces the short-regression estimates exactly,

$$0.71 + 0.73 \times 0.84 \approx 1.33, \qquad -1.50 + 0.73 \times (-0.71) \approx -2.01.$$

This suggests a natural remedy, which is to add human capital to the model. This is what the Mankiw–Romer–Weil model does.

2.11 The Mankiw–Romer–Weil model

2.11.1 Setup

Motivated by the empirical failure of the textbook model, Mankiw et al. (1992) add human capital H as a second accumulable factor. The aim is to explain the large income differences across countries with a plausible capital share, and at the same time to slow down convergence. Output is

$$Y_t = K_t^{\alpha_K} H_t^{\alpha_H} (A_t L_t)^{1-\alpha_K-\alpha_H}, \qquad \alpha_K + \alpha_H < 1.$$

Fractions s_K and s_H of output are invested in physical and human capital, and for simplicity both depreciate at rate δ,

$$K_{t+1} = s_K Y_t + (1-\delta) K_t, \qquad H_{t+1} = s_H Y_t + (1-\delta) H_t.$$

With $k_t \equiv K_t/(A_t L_t)$, $h_t \equiv H_t/(A_t L_t)$ and $y_t = k_t^{\alpha_K} h_t^{\alpha_H}$, dividing by $A_t L_t$ gives the system

$$(1+n)(1+g)k_{t+1} = s_K y_t + (1-\delta)k_t,$$
$$(1+n)(1+g)h_{t+1} = s_H y_t + (1-\delta)h_t.$$

2.11.2 Steady state

Let $\phi \equiv n + g + ng + \delta$. In the steady state, $s_K y^* = \phi k^*$ and $s_H y^* = \phi h^*$, and dividing the two conditions gives $h^*/k^* = s_H/s_K$. Substituting $h^* = (s_H/s_K)k^*$ into $s_K k^{*\alpha_K} h^{*\alpha_H} = \phi k^*$, we obtain

$$k^* = \left[\left(\frac{s_K}{\phi}\right)^{1-\alpha_H}\left(\frac{s_H}{\phi}\right)^{\alpha_H}\right]^{\frac{1}{1-\alpha_K-\alpha_H}}, \qquad h^* = \left[\left(\frac{s_K}{\phi}\right)^{\alpha_K}\left(\frac{s_H}{\phi}\right)^{1-\alpha_K}\right]^{\frac{1}{1-\alpha_K-\alpha_H}},$$

and

$$\ln y^* = \frac{\alpha_K}{1-\alpha_K-\alpha_H}\ln s_K + \frac{\alpha_H}{1-\alpha_K-\alpha_H}\ln s_H - \frac{\alpha_K+\alpha_H}{1-\alpha_K-\alpha_H}\ln \phi.$$

With $\alpha_K = \alpha_H = 1/3$, the elasticity of y^* with respect to s_K is 1 (instead of $1/2$ in the basic model), so moderate differences in saving and schooling can explain large income differences.

2.11.3 Convergence

We write the system in logs, $x_t = \ln k_t$ and $z_t = \ln h_t$, and linearize it around the steady state. With $\Gamma \equiv (1+n)(1+g)$, we have

$$\frac{\partial x_{t+1}}{\partial x_t} = \frac{\alpha_K s_K y^* + (1-\delta)k^*}{\Gamma k^*} = \frac{1-\delta+\alpha_K\phi}{\Gamma}, \qquad \frac{\partial x_{t+1}}{\partial z_t} = \frac{\alpha_H \phi}{\Gamma},$$

and symmetrically for z_{t+1}. The Jacobian is

$$J = \frac{1-\delta}{\Gamma} I + \frac{\phi}{\Gamma}\begin{pmatrix} \alpha_K & \alpha_H \\ \alpha_K & \alpha_H \end{pmatrix}.$$

The second matrix has rank one, with eigenvalues 0 and $\alpha_K + \alpha_H$. Therefore the eigenvalues of J are

$$\mu_1 = \frac{1-\delta}{\Gamma}, \qquad \mu_2 = \frac{1-\delta+(\alpha_K+\alpha_H)\phi}{\Gamma}.$$

Both lie in $(0,1)$ because $\Gamma - (1-\delta) = \phi$, so the steady state is a *stable node*. The slower root μ_2 governs convergence,

$$\lambda = 1 - \mu_2 = \frac{(1-\alpha_K-\alpha_H)\phi}{\Gamma} \approx (1-\alpha_K-\alpha_H)(n+g+\delta).$$

With $\alpha_K = \alpha_H = 1/3$ and the same n, g, δ as before, $\lambda \approx 0.026$ and $t_{1/2} \approx 26$ years, much closer to the data. The *broad* capital share $\alpha_K + \alpha_H$ is what matters for the speed of convergence.

2.11.4 Empirical performance

Income levels. Exactly as for the textbook model, we write $\ln A_0 = a + \varepsilon_i$ and convert the steady state into income per worker, $\ln y_t = \ln A_t + \ln y^*$, which gives

$$\ln y_i = c + \frac{\alpha_K}{1-\alpha_K-\alpha_H}\ln s_{K,i} + \frac{\alpha_H}{1-\alpha_K-\alpha_H}\ln s_{H,i} - \frac{\alpha_K+\alpha_H}{1-\alpha_K-\alpha_H}\ln(n_i+g+\delta) + \varepsilon_i.$$

The model again implies that the coefficients sum to zero, and with $\alpha_K = \alpha_H = 1/3$ it predicts coefficients of 1, 1 and -2. Table 2.2 reports the estimates, with $s_{H,i}$ measured by the percentage of the working-age population in secondary school.

Table 2.2: Estimation of the augmented Solow (MRW) model. Dependent variable: log GDP per working-age person in 1985. Standard errors in parentheses. 75 countries.

	Unrestricted	Restricted
$\ln s_{K,i}$	0.71 (0.15)	
$\ln s_{H,i}$	0.73 (0.10)	
$\ln(n_i+g+\delta)$	-1.50 (0.40)	
$\ln s_{K,i} - \ln(n_i+g+\delta)$		0.72 (0.14)
$\ln s_{H,i} - \ln(n_i+g+\delta)$		0.73 (0.09)
R^2	0.78	0.78
Standard error of regression	0.45	0.45
Test of zero sum of coefficients	$F(1,71) = 0.02$ ($p = 0.90$)	
Implied α_K		0.29
Implied α_H		0.30

The results improve markedly on those of Table 2.1.

- **Fit.** The share of the cross-country variance of income explained rises from 60% to 78%.
- **Restriction.** The zero-sum restriction is comfortably accepted ($p = 0.90$).
- **Magnitudes.** The coefficient on physical capital investment falls from 1.33 to 0.71, as the omitted-variable argument predicted. Solving $\beta_K = \alpha_K/(1-\alpha_K-\alpha_H)$ and $\beta_H = \alpha_H/(1-\alpha_K-\alpha_H)$, we find that the restricted estimates $\hat\beta_K = 0.72$ and $\hat\beta_H = 0.73$ imply

$$\hat\alpha_K = \frac{\hat\beta_K}{1+\hat\beta_K+\hat\beta_H} \approx 0.29, \qquad \hat\alpha_H = \frac{\hat\beta_H}{1+\hat\beta_K+\hat\beta_H} \approx 0.30.$$

The capital share is now close to its national-accounts value of $1/3$, and the implied share of human capital is of a similar size. Since $\alpha_K + \alpha_H \approx 0.6$, it is the *broad* capital share, physical plus human, that the textbook regression was picking up when it delivered $\hat{\alpha} \approx 0.59$.

Conditional convergence. The model also tells us how to test convergence properly. From the linearization of the previous subsection, the log-gap of income per effective worker shrinks at rate λ,

$$\ln \tilde{y}_T - \ln \tilde{y}^* = (1-\lambda)^T (\ln \tilde{y}_0 - \ln \tilde{y}^*).$$

Since $\ln y_t = \ln A_0 + t \ln(1+g) + \ln \tilde{y}_t$, the growth of income per worker between 0 and T is

$$\ln y_T - \ln y_0 = T \ln(1+g) + \left[1 - (1-\lambda)^T\right]\left(\ln \tilde{y}^* + \ln A_0\right) - \left[1 - (1-\lambda)^T\right] \ln y_0.$$

Growth depends negatively on initial income, but only once we control for the determinants of the steady state $\tilde{y}^*$, namely s_K, s_H and n. The coefficient on $\ln y_0$ is $b_0 = -[1-(1-\lambda)^T]$, so the implied speed of convergence is

$$\lambda = 1 - (1 + b_0)^{1/T}.$$

Table 2.3 reports the regressions for $T = 25$ (1960–1985).

Table 2.3: Convergence regressions. Dependent variable: $\ln y_{i,1985} - \ln y_{i,1960}$. Standard errors in parentheses. 75 countries.

	Unconditional	Textbook Solow	Augmented Solow
$\ln y_{i,1960}$	-0.002 (0.055)	-0.227 (0.057)	-0.364 (0.067)
$\ln s_{K,i}$		0.65 (0.10)	0.55 (0.10)
$\ln(n_i + g + \delta)$		-0.45 (0.31)	-0.54 (0.29)
$\ln s_{H,i}$			0.27 (0.08)
R^2	0.00	0.39	0.47
Implied λ (per year)	0.000	0.010	0.018

Without controls (first column) there is no convergence at all, which is the formal version of Figure 2.6b. Once we control for saving and population growth, the coefficient on initial income becomes negative and highly significant. Countries do converge, but each to its *own* steady state. Adding human capital strengthens the result and raises the implied speed of convergence to about 1.8% per year, a half-life of roughly 38 years. This is close to the speed predicted by the MRW model with a broad capital share of about $2/3$ ($\lambda \approx 0.026$ above), and much slower than the 5% predicted by the textbook model with $\alpha = 1/3$. Human capital thus resolves both puzzles at once, since it explains why saving and population growth have such large effects on income levels, and why convergence is so slow.

Chapter 3

The Overlapping-Generations Model

In the Solow model, saving is a fixed fraction of income. We now let households *choose* how much to save but give them finite lives. Each person lives for two periods, works when young and lives off savings when old. Because a new generation of workers is born every period, the economy can last forever even though each worker's lifetime is finite. This is the overlapping-generations (OLG) model of Diamond (1965), which builds on Samuelson (1958), and a comprehensive treatment can be found in de la Croix and Michel (2002). The OLG framework is celebrated for its tractability and for the ease with which it can be extended to explain various development traps.

3.1 The environment

3.1.1 Demography

Time is discrete and infinite, $t = 1, 2, \ldots$, and a new generation is born in each period. Each individual lives for two periods, being *young* in the period of birth and *old* in the next. Generation t is the cohort born at t, and it has N_t members, with

$$N_t = (1+n)N_{t-1}, \qquad n > -1.$$

At each date t, N_t young and $N_{t-1} = N_t/(1+n)$ old are alive. In period 1 there is also an *initial old* generation (generation 0), who live only in that period and make no economic decisions.

	$t=1$	$t=2$	$t=3$	$\cdots$	t	$t+1$
Gen. 0	old					
Gen. 1	young	old				
Gen. 2		young	old			
$\vdots$				$\ddots$		
Gen. t					young	old

3.1.2 Endowments, preferences and technology

- **Labor.** Each young person supplies one unit of labor inelastically, and the old do not work.
- **Capital.** The initial old own the initial capital stock K_1. Capital depreciates at rate $\delta \in [0,1]$.
- **Preferences.** A member of generation $t \geq 1$ consumes c_t when young and d_{t+1} when old and has lifetime utility

  $$U_t = u(c_t) + \beta u(d_{t+1}),$$

  with $u' > 0$, $u'' < 0$, $u'(0) = +\infty$ and $\beta \in (0,1]$ (because a period is a generation, β is the discount factor over roughly 30 years). There is no bequest motive and no utility from leisure. The initial old care only about d_1.
- **Technology.** Competitive firms use a neoclassical technology with constant returns to scale, $Y_t = F(K_t, N_t)$. Per young worker, output is $y_t = f(k_t)$, where $k_t \equiv K_t/N_t$.

Output plus undepreciated capital is either consumed by the young and the old or carried into the next period as capital,

$$N_t c_t + N_{t-1} d_t + K_{t+1} = F(K_t, N_t) + (1-\delta)K_t.$$

Dividing by N_t and using $N_{t-1}/N_t = 1/(1+n)$ and $K_{t+1}/N_t = (1+n)k_{t+1}$, we obtain

$$c_t + \frac{d_t}{1+n} + (1+n)k_{t+1} = f(k_t) + (1-\delta)k_t. \tag{3.1}$$

An allocation is *feasible* if it is non-negative and satisfies (3.1) at every date. A feasible allocation is *Pareto optimal* if no other feasible allocation makes some generation better off without making any other generation worse off.

3.2 Decision problems

3.2.1 Firms

Firms rent capital at rate R_t^k, hire labor at wage w_t, and maximize $F(K_t, N_t) - R_t^k K_t - w_t N_t$. Using (2.1), we have

$$R_t^k = f'(k_t), \qquad w_t = f(k_t) - k_t f'(k_t) \equiv w(k_t). \tag{3.2}$$

We note that $w'(k) = -kf''(k) > 0$, so a higher capital stock raises wages. The gross return on one unit of saving carried from t to $t+1$ is

$$R_{t+1} \equiv 1 + r_{t+1} = 1 + R_{t+1}^k - \delta = 1 + f'(k_{t+1}) - \delta.$$

3.2.2 Households

The initial old. They simply consume their wealth, $d_1 = R_1 K_1 / N_0$.

Generation t. A young person earns w_t, consumes c_t and saves s_t. When old, the person consumes the gross return on these savings,

$$c_t + s_t = w_t, \qquad d_{t+1} = R_{t+1} s_t.$$

Combining the two constraints gives the lifetime budget constraint $c_t + d_{t+1}/R_{t+1} = w_t$, and the Lagrangian is

$$\mathcal{L} = u(c_t) + \beta u(d_{t+1}) + \lambda_t \Big[w_t - c_t - \frac{d_{t+1}}{R_{t+1}} \Big],$$

with FOCs $u'(c_t) = \lambda_t$ and $\beta u'(d_{t+1}) = \lambda_t / R_{t+1}$. Eliminating λ_t gives the OLG **Euler equation**

$$u'(c_t) = \beta R_{t+1} u'(d_{t+1}) \qquad \Longleftrightarrow \qquad u'(w_t - s_t) = \beta R_{t+1} u'(R_{t+1} s_t). \tag{3.3}$$

It has exactly the interpretation of Chapter 1. Consuming one unit less when young costs $u'(c)$, and saving that unit yields R_{t+1} units in old age, which are worth $\beta R_{t+1} u'(d)$ today.

3.2.3 The saving function

Equation (3.3) defines saving implicitly as a function of the wage and the interest factor:

$$s_t = s(w_t, R_{t+1}).$$

We define $\Psi(s, w, R) \equiv u'(w - s) - \beta R u'(Rs) = 0$, whose partial derivatives are

$$\Psi_s = -u''(c) - \beta R^2 u''(d) > 0, \qquad \Psi_w = u''(c) < 0,$$
$$\Psi_R = -\beta u'(d) - \beta R s\, u''(d) = -\beta u'(d) \big[1 - \sigma(d) \big],$$

where $\sigma(c) \equiv -cu''(c)/u'(c)$ is the elasticity of marginal utility (the inverse of the EIS). By the implicit function theorem (Appendix A), we obtain

$$s_w = -\frac{\Psi_w}{\Psi_s} = \frac{u''(c)}{u''(c) + \beta R^2 u''(d)} \in (0,1),$$
$$s_R = -\frac{\Psi_R}{\Psi_s} = \frac{\beta u'(d)\left[1-\sigma(d)\right]}{\Psi_s}.$$

Wages and interest rates

- Since $0 < s_w < 1$, the young consume part of a one-unit wage increase and save the rest, so that *both* c and d rise (both goods are normal).
- The sign of s_R is ambiguous. A higher return makes old-age consumption cheaper, and the *substitution effect* leads the young to save more. It also makes the saver richer, and the *income effect* leads the young to consume more today and save less. The income effect is positive here because the young have no future labor income and are therefore net lenders, $s > 0$ (see Section 1.1.4). If $\sigma < 1$ (EIS > 1), the substitution effect dominates and $s_R > 0$. If $\sigma > 1$, we have $s_R < 0$. With log utility ($\sigma = 1$), the two effects cancel and $s_R = 0$.

Example 3.1 (CRRA saving function)**.** With $u(c) = (c^{1-\sigma} - 1)/(1-\sigma)$, the Euler equation gives $d/c = (\beta R)^{1/\sigma}$, that is, $Rs = (\beta R)^{1/\sigma}(w - s)$, so that

$$s(w,R) = \frac{w}{1 + \beta^{-1/\sigma} R^{\frac{\sigma-1}{\sigma}}}.$$

The saving rate s/w increases with R if $\sigma < 1$ and decreases if $\sigma > 1$. With log utility, $s = \frac{\beta}{1+\beta} w$.

3.3 Competitive equilibrium

Definition 3.1 (Competitive equilibrium)**.** Given K_1, a competitive equilibrium is a sequence of prices $\{R_t^k, w_t\}_{t\geq 1}$ and allocations $\{d_1\}$, $\{c_t, d_{t+1}, s_t\}_{t\geq 1}$, $\{K_t\}_{t\geq 1}$ such that the following conditions hold.

1. Given prices, each generation solves its utility maximization problem.
2. Given prices, firms maximize profits, so that (3.2) holds.
3. Markets clear. Labor N_t is fully employed, the capital stock at $t+1$ equals the saving of the young at t, $K_{t+1} = N_t s_t$, and the goods market clears.

Walras' law. Adding the budget constraints of the young and the old alive at t, we have

$$N_t(c_t + s_t) + N_{t-1}d_t = N_t w_t + N_{t-1}R_t s_{t-1}.$$

Using $N_t s_t = K_{t+1}$, $N_{t-1}s_{t-1} = K_t$ and $w_t N_t + R_t^k K_t = F(K_t, N_t)$, we obtain exactly the resource constraint

$$N_t c_t + N_{t-1}d_t + K_{t+1} = F(K_t, N_t) + (1-\delta)K_t.$$

If households are on their budget constraints and factor markets clear, the goods market clears automatically.

3.3.1 The law of motion of capital

Capital-market clearing $K_{t+1} = N_t s_t$, divided by $N_{t+1} = (1+n)N_t$, gives

$$(1+n)k_{t+1} = s\Big(w(k_t),\ 1 + f'(k_{t+1}) - \delta\Big) \tag{3.4}$$

This is an *implicit* first-order difference equation, in which k_{t+1} appears on both sides because tomorrow's interest rate depends on tomorrow's capital stock, which depends on today's saving.

- If $s_R \geq 0$ (e.g. $\sigma \leq 1$), the right-hand side is non-increasing in k_{t+1} (a higher k_{t+1} lowers R_{t+1}) while the left-hand side is increasing, so for each k_t there is at most one k_{t+1} and the equilibrium path is *unique*.
- If $s_R < 0$ (low EIS), the right-hand side increases with k_{t+1} and several values of k_{t+1} may be consistent with the same k_t, and the equilibrium may be *indeterminate*.

Definition 3.2 (Steady state)**.** A steady state is $k^{ss} > 0$ with $(1+n)k^{ss} = s\big(w(k^{ss}), 1 + f'(k^{ss}) - \delta\big)$.

Unlike in the Solow model, nothing guarantees a unique positive steady state, or even the existence of one, for general u and f. The log–Cobb–Douglas case, however, is well behaved.

3.4 Log utility and Cobb–Douglas technology

Let $u = \ln$ and $f(k) = Ak^\alpha$. Then $s_t = \frac{\beta}{1+\beta} w_t$ and $w_t = (1-\alpha)Ak_t^\alpha$, so (3.4) becomes *explicit*,

$$k_{t+1} = \frac{\beta(1-\alpha)A}{(1+\beta)(1+n)}\, k_t^\alpha \equiv \Phi(k_t). \tag{3.5}$$

The map Φ is increasing and strictly concave with $\Phi'(0) = +\infty$ and $\Phi'(\infty) = 0$. Exactly as in Proposition 2.1, there is a unique positive steady state and it is globally stable. We note that δ does not appear in (3.5). With log utility, saving does not respond to the return, and depreciation only affects how much the old consume.

Steady state. Setting $k_{t+1} = k_t = k^{ss}$, we obtain

$$k^{ss} = \left(\frac{\beta(1-\alpha)A}{(1+\beta)(1+n)} \right)^{\frac{1}{1-\alpha}}, \qquad R^{k,ss} = \alpha A (k^{ss})^{\alpha-1} = \frac{\alpha(1+\beta)(1+n)}{\beta(1-\alpha)},$$

$$w^{ss} = (1-\alpha)A(k^{ss})^{\alpha}, \qquad c^{ss} = \frac{w^{ss}}{1+\beta}, \qquad d^{ss} = (1 + R^{k,ss} - \delta)\frac{\beta w^{ss}}{1+\beta}.$$

To derive $R^{k,ss}$, we use $(k^{ss})^{\alpha-1} = \frac{(1+\beta)(1+n)}{\beta(1-\alpha)A}$.

Comparative statics. The steady state k^{ss} increases with TFP A and patience β (more saving), and decreases with population growth n (capital dilution). The effect of α is ambiguous, because a higher capital share lowers the wage share $(1-\alpha)$, and hence saving, but raises the curvature exponent $1/(1-\alpha)$.

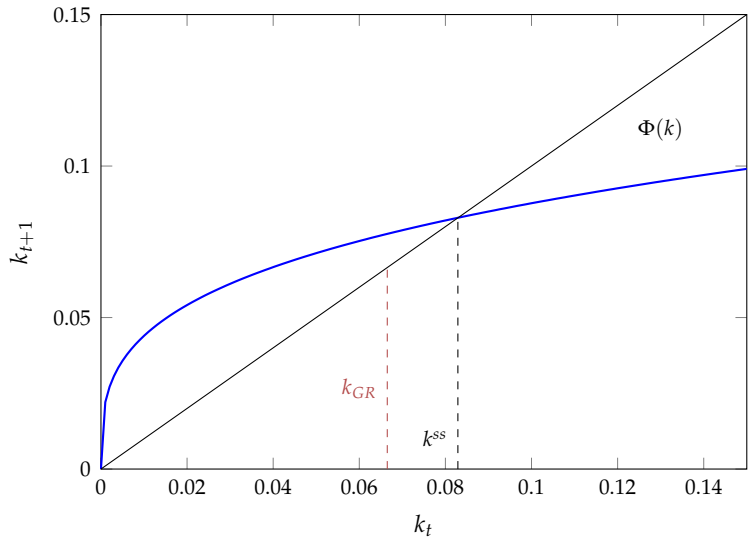


Figure 3.1: The OLG law of motion (3.5) for $A = 1$, $\alpha = 0.3$, $\beta = 1$ and $n = 1$, which gives $k_{t+1} = 0.175\, k_t^{0.3}$. The steady state $k^{ss} \approx 0.0829$ lies above the golden rule $k_{GR} \approx 0.0665$ computed with $\delta = 1$, so this economy over-accumulates capital.

3.5 Dynamic efficiency and over-accumulation

3.5.1 The golden rule in the OLG model

In a steady state, (3.1) says that total consumption per young person is

$$c + \frac{d}{1+n} = f(k) - (n+\delta)k.$$

It is maximized at the golden rule

$$f'(k_{GR}) = n + \delta \qquad \Longleftrightarrow \qquad R = 1 + n.$$

Proposition 3.1 (Dynamic inefficiency)**.** *If a competitive steady state has $k^{ss} > k_{GR}$, equivalently $r^{ss} \equiv R^{k,ss} - \delta < n$, it is not Pareto optimal.*

Proof. Suppose that the economy is in the steady state from date T onward, and that a planner changes the allocation from date T by setting $k_{T+1} = k_{T+2} = \cdots = k_{GR}$. Let $C_t \equiv c_t + d_t/(1+n)$ be total consumption per young person. By (3.1), we have the following.

- At date T, because $k_{GR} < k^{ss}$,

$$\begin{aligned} C_T &= f(k^{ss}) + (1-\delta)k^{ss} - (1+n)k_{GR} \\ &> f(k^{ss}) + (1-\delta)k^{ss} - (1+n)k^{ss} = f(k^{ss}) - (n+\delta)k^{ss}. \end{aligned}$$

- At dates $t > T$, by definition of the golden rule,

$$C_t = f(k_{GR}) - (n+\delta)k_{GR} > f(k^{ss}) - (n+\delta)k^{ss}.$$

Total consumption is therefore strictly higher than in the competitive steady state in *every* period from T on. The planner can divide the surplus in each period between the young and the old so that every person alive receives more of both goods than before. Every generation, including the one that is old at T, is then strictly better off. □

Example 3.2. Let $\delta = n = A = 1$, $\alpha = 0.3$ and $\beta = 1$. Since a generation is long, full depreciation and a doubling of the population are not absurd. Then $k_{GR} = (0.3/2)^{1/0.7} = 0.15^{10/7} \approx 0.0665$ and $k^{ss} = (0.7/4)^{1/0.7} = 0.175^{10/7} \approx 0.0829 > k_{GR}$. Equivalently, $R^{k,ss} = \frac{0.3\times 2\times 2}{0.7} \approx 1.714$, so $r^{ss} = R^{k,ss} - \delta \approx 0.714 < n = 1$. Steady-state consumption per young person is 0.3080 in the competitive equilibrium and 0.3105 at the golden rule. The competitive economy has *too much capital*. For log–Cobb–Douglas economies the general condition is

$$k^{ss} > k_{GR} \quad \Longleftrightarrow \quad \beta(1-\alpha)(n+\delta) > \alpha(1+\beta)(1+n).$$

Why can a competitive economy over-accumulate?

There is no externality and no market power, yet the First Welfare Theorem fails. The failure comes from the following features of the OLG economy.

- **The only store of value is capital.** The young *must* save to consume in old age, and in this economy the only way to save is to accumulate capital, however low its return.
- **An infinity of agents.** The planner can do something no market can, namely transfer one unit from each young person to the old in every period, forever. Since there are $1+n$ young people per old person, each old person receives $1+n$ units for every unit given when young. This "chain" never has to be repaid, because there is always a new generation. If the economy's return on capital $R = 1+r$ is below $1+n$, the chain beats capital as a way to provide for old age.

In the Ramsey model, such over-accumulation is impossible. A single infinitely-lived household would never hold capital whose return is below the growth rate of the economy, because it would be better off eating it, and the transversality condition rules this out.

If instead $k^{ss} < k_{GR}$ ($r^{ss} > n$), the steady state is dynamically efficient, because raising capital to the golden rule would require some generation to consume less.

3.6 Government debt

3.6.1 Set-up

The government issues one-period bonds that pay the same gross return as capital. Let B_t be the debt outstanding at the beginning of period t (held by the old), and suppose the government keeps debt per young person constant, $B_t/N_t = b$. Each young person pays a lump-sum tax τ_t. The government budget constraint is

$$\underbrace{R_t B_t}_{\text{repay old debt with interest}} = \underbrace{B_{t+1}}_{\text{new borrowing}} + \underbrace{N_t \tau_t}_{\text{taxes}}.$$

Dividing by N_t and using $B_{t+1}/N_t = (1+n)b$, we obtain

$$\tau_t = (R_t - 1 - n)b = (r_t - n)\, b.$$

If the interest rate exceeds the growth rate ($r > n$), the debt must be serviced with positive taxes. If $r < n$, new borrowing more than covers interest and

the government can *transfer* resources to the young ($\tau < 0$) while keeping debt per person constant.

3.6.2 Equilibrium

The young now face $c_t + s_t = w_t - \tau_t$, so $s_t = s(w_t - \tau_t, R_{t+1})$. Saving is held either as capital or as bonds, $N_t s_t = K_{t+1} + B_{t+1}$, which gives

$$(1+n)(k_{t+1} + b) = s\big(w(k_t) - (r_t - n)b,\; R_{t+1}\big).$$

Debt affects capital through two channels. It absorbs part of private saving (*crowding out*), and the taxes (or transfers) needed to service it change disposable income.

Log–Cobb–Douglas. In this case $s_t = \frac{\beta}{1+\beta}(w_t - \tau_t)$, so that

$$k_{t+1} = \frac{\beta}{(1+\beta)(1+n)}\Big[(1-\alpha)Ak_t^{\alpha} - \big(\alpha A k_t^{\alpha-1} - \delta - n\big)b\Big] - b.$$

Proposition 3.2. *In the log–Cobb–Douglas economy, $\mathrm{d}k^{ss}/\mathrm{d}b < 0$ at any locally stable steady state, so government debt crowds out capital.*

Proof. We write the steady-state condition as

$$\Gamma(k, b) \equiv \frac{\beta}{1+\beta}\big[w(k) - (f'(k) - \delta - n)b\big] - (1+n)(k+b) = 0.$$

Then

$$\begin{aligned}\Gamma_b &= -\frac{\beta(f'(k) - \delta - n)}{1+\beta} - (1+n) = -\frac{(1+n)(1+\beta) + \beta(R - 1 - n)}{1+\beta} \\ &= -\frac{1 + n + \beta R}{1+\beta} < 0.\end{aligned}$$

Local stability means that the slope of the map $k_t \mapsto k_{t+1}$ at the steady state is less than one, which here is $\Gamma_k < 0$. By the implicit function theorem, $\mathrm{d}k^{ss}/\mathrm{d}b = -\Gamma_b/\Gamma_k < 0$. □

Debt can cure dynamic inefficiency

If the debt-free steady state is dynamically inefficient, the government can choose b so that the steady state with debt is exactly the golden rule. At $k = k_{GR}$, we have $r = n$ and therefore $\tau = 0$, so debt is rolled over forever *without any taxes*. The required debt level solves

$(1+n)(k_{GR}+b) = \frac{\beta}{1+\beta} w(k_{GR})$, which gives

$$b_{GR} = \frac{\beta w(k_{GR})}{(1+\beta)(1+n)} - k_{GR},$$

which is positive precisely when $k^{ss} > k_{GR}$. In the example above, $w(k_{GR}) \approx 0.3104$ and $b_{GR} \approx 0.0111$.

Intuition

Public debt provides the young with an alternative store of value. Part of their saving goes into bonds instead of capital, which reduces over-accumulation. When $r < n$, debt is like the chain of transfers in Proposition 3.1. It is a Ponzi scheme, but a sustainable one, because the economy grows faster than the debt accumulates interest. When $r > n$ (the economy is dynamically efficient), debt must be serviced with taxes and simply crowds out capital, making future generations poorer.

3.7 Pay-as-you-go social security

3.7.1 Set-up

In a pay-as-you-go (PAYG) pension system, each young person pays a contribution p and the proceeds are paid out immediately to the current old. Since there are $1+n$ young per old person, each old person receives $(1+n)p$. The budget constraints are

$$c_t + s_t = w_t - p, \qquad d_{t+1} = R_{t+1} s_t + (1+n)p.$$

Eliminating s_t gives the lifetime budget constraint

$$c_t + \frac{d_{t+1}}{R_{t+1}} = w_t - p + \frac{(1+n)p}{R_{t+1}} = w_t - p\,\frac{R_{t+1} - (1+n)}{R_{t+1}}.$$

At given prices, the PAYG system raises the lifetime wealth of a contributor if and only if $1+n > R_{t+1}$, that is, if its implicit return, the growth factor of the population, exceeds the return on capital.

3.7.2 Saving and capital

With log utility, the Euler equation $d_{t+1} = \beta R_{t+1} c_t$ and the two budget constraints give

$$R_{t+1} s_t + (1+n)p = \beta R_{t+1}(w_t - p - s_t).$$

Collecting the terms in s_t yields $(1+\beta)R_{t+1}s_t = \beta R_{t+1}(w_t - p) - (1+n)p$, so that

$$s_t = \frac{\beta}{1+\beta}(w_t - p) - \frac{(1+n)p}{(1+\beta)R_{t+1}}.$$

Saving falls for two reasons. Disposable income when young is lower, and the pension reduces the need to save for old age. Capital-market clearing gives the law of motion

$$(1+n)k_{t+1} = \frac{\beta}{1+\beta}\big(w(k_t) - p\big) - \frac{(1+n)p}{(1+\beta)R(k_{t+1})}, \qquad R(k) \equiv 1 + f'(k) - \delta, \tag{3.6}$$

which is implicit in k_{t+1}.

Proposition 3.3. *At any locally stable steady state of* (3.6), $\mathrm{d}k^{ss}/\mathrm{d}p < 0$.

Proof. We write (3.6) as $H(k_{t+1}, p) = \frac{\beta}{1+\beta}w(k_t)$, with

$$H(k', p) \equiv (1+n)k' + \frac{\beta p}{1+\beta} + \frac{(1+n)p}{(1+\beta)R(k')}.$$

Since $R'(k) = f''(k) < 0$, we have

$$H_{k'} = (1+n) - \frac{(1+n)pR'(k')}{(1+\beta)R(k')^2} > 0,$$

so (3.6) defines an increasing map $k_{t+1} = \Phi(k_t)$ with slope $\Phi' = \frac{\beta w'(k_t)/(1+\beta)}{H_{k'}}$. In a steady state, we define $\Gamma(k, p) \equiv \frac{\beta}{1+\beta}w(k) - H(k, p) = 0$. Local stability, $\Phi'(k^{ss}) < 1$, is equivalent to $\Gamma_k = \frac{\beta w'(k)}{1+\beta} - H_{k'} < 0$. Moreover,

$$\Gamma_p = -\frac{\beta}{1+\beta} - \frac{1+n}{(1+\beta)R(k)} < 0.$$

By the implicit function theorem, $\mathrm{d}k^{ss}/\mathrm{d}p = -\Gamma_p/\Gamma_k < 0$. □

PAYG and welfare

- If the economy is dynamically inefficient ($R < 1+n$), introducing a PAYG system makes the current old better off (they receive a pension they never paid for) and, by lowering capital towards the golden rule, can make every future generation better off as well. In the example of Section 3.5, the contribution that implements the golden rule (where $R = 1+n$) solves (3.6) at $k = k_{GR}$ and equals $p_{GR} \approx 0.0222$.
- If the economy is dynamically efficient ($R > 1+n$), PAYG makes the initial old better off but future generations worse off. Each generation pays p and receives the return $1+n < R$, and capital

is crowded out.

- A **fully funded** system, in which contributions are invested in capital and returned with interest R, is neutral, because it simply replaces private saving one for one (as long as contributions do not exceed desired saving).

	PAYG	**Fully funded**
Who finances pensions	current young	own past contributions
Implicit return	$1 + n$ (population growth)	$R = 1 + r$ (capital)
Effect on capital	reduces saving and k	none
Under dynamic inefficiency	can be Pareto improving	no effect
Exposure to ageing ($n \downarrow$)	high	low

3.8 Extensions: Development traps

The basic Diamond model has a unique, globally stable steady state (with log utility and Cobb–Douglas technology) and no long-run growth of income per worker. We now study four extensions. In the first three, *poverty traps* arise from the behaviour of poor households, who under-invest in their children's education, cannot save because of subsistence needs, or are too impatient. In the last extension, a production externality generates either growth or a trap. To focus on the mechanisms, we set $\delta = 1$ (a period is a generation) unless stated otherwise.

3.8.1 Human capital and a poverty trap

This extension follows Ceroni (2001). Physical capital plays no role, the only reproducible factor is human capital, and education is financed by parents.

Set-up. Population is constant and every adult has one child. An adult of generation t has human capital h_t. Output is produced linearly from human capital, $Y_t = H_t$, so the wage per efficiency unit is one and the adult's income is h_t. The adult divides this income between own consumption c_t and spending $e_t \geq 0$ on the education of the child,

$$c_t + e_t = h_t.$$

The child's human capital is

$$h_{t+1} = \ln(\gamma e_t + \nu), \qquad \gamma > 0,\ \nu > 1. \tag{3.7}$$

- Without education, a child still has human capital $\underline{\mu} \equiv \ln \nu > 0$.
- The marginal return to education, $\partial h_{t+1}/\partial e_t = \gamma/(\gamma e_t + \nu)$, is *finite* at $e_t = 0$, where it equals γ/ν, and it is diminishing.

Parents are altruistic in a "warm-glow" way, in the sense that they value the human capital of their child,

$$U_t = \ln c_t + \omega\, h_{t+1}, \qquad \omega > 0,$$

where ω is the weight on the child.

The optimal education choice. After substituting the budget constraint and (3.7), the parent solves

$$\max_{0 \le e_t \le h_t} \ln(h_t - e_t) + \omega \ln(\gamma e_t + \nu).$$

With multiplier $\kappa \ge 0$ on $e_t \ge 0$, the Kuhn–Tucker conditions are

$$-\frac{1}{h_t - e_t} + \frac{\omega\gamma}{\gamma e_t + \nu} + \kappa = 0, \qquad \kappa \ge 0, \qquad \kappa e_t = 0.$$

The first term is the marginal cost of education, that is, the marginal utility of the consumption given up. The second term is its marginal benefit, the extra human capital of the child valued at ω.

- **Interior solution** ($\kappa = 0$). The condition $\gamma e_t + \nu = \omega\gamma(h_t - e_t)$ gives

  $$e_t = \frac{\omega}{1+\omega} h_t - \frac{\nu}{\gamma(1+\omega)},$$

  which is positive if and only if $h_t > \underline{h} \equiv \dfrac{\nu}{\omega\gamma}$.

- **Corner solution** ($e_t = 0$). In this case $\kappa = \frac{1}{h_t} - \frac{\omega\gamma}{\nu} \ge 0$, that is, $h_t \le \underline{h}$. At $e_t = 0$, the marginal benefit $\omega\gamma/\nu$ does not exceed the marginal cost $1/h_t$.

Why poor parents do not educate

At zero education, the return to the first unit spent on schooling is bounded (γ/ν), whereas the marginal utility of consumption, $1/h_t$, is very high for a poor parent. Only parents whose income exceeds $\underline{h}$ find it worthwhile to give up consumption for their child's education. Edu-

cation behaves like a luxury good.

Dynamics. We substitute the education choice into (3.7). For $h_t \le \underline{h}$, we have $h_{t+1} = \ln \nu = \underline{\mu}$. For $h_t > \underline{h}$, we have

$$\gamma e_t + \nu = \frac{\gamma\omega h_t - \nu + \nu(1+\omega)}{1+\omega} = \frac{\omega(\gamma h_t + \nu)}{1+\omega}.$$

Hence

$$h_{t+1} = \phi(h_t) \equiv \begin{cases} \underline{\mu} = \ln \nu & h_t \le \underline{h}, \\ \ln \dfrac{\omega(\gamma h_t + \nu)}{1+\omega} & h_t > \underline{h}. \end{cases} \tag{3.8}$$

The map is continuous, since $\omega(\gamma \underline{h} + \nu)/(1+\omega) = \nu$ at $h_t = \underline{h}$. It is flat below $\underline{h}$, and increasing and strictly concave above it, with slope $\phi'(h) = \gamma/(\gamma h + \nu)$.

Conditions for multiple steady states. Multiple steady states require the following conditions.

1. **A low steady state on the flat branch.** The point $h = \underline{\mu}$ is a steady state if it lies on the flat branch, $\underline{\mu} < \underline{h}$, that is,

$$\ln \nu < \frac{\nu}{\omega\gamma} \quad \Longleftrightarrow \quad \gamma < \frac{\nu}{\omega \ln \nu}.$$

   It is stable, since $\phi' = 0$ there.

2. **A steep upper branch at the kink.** Let $G(h) \equiv \phi(h) - h$ on $(\underline{h}, \infty)$. Under condition 1, $G(\underline{h}) = \underline{\mu} - \underline{h} < 0$, and G is strictly concave with $G(h) \to -\infty$. Hence G crosses zero (twice) if and only if its maximum is positive. The maximum is interior, at the point h_m where $\phi'(h_m) = 1$, only if $\phi'(\underline{h}^+) > 1$, which requires

$$\phi'(\underline{h}^+) = \frac{\gamma}{\nu/\omega + \nu} = \frac{\omega\gamma}{\nu(1+\omega)} > 1 \quad \Longleftrightarrow \quad \gamma > \frac{\nu(1+\omega)}{\omega}.$$

3. **The upper branch reaches the** 45° **line.** The condition $\phi'(h_m) = 1$ gives $\gamma h_m + \nu = \gamma$, so $h_m = 1 - \nu/\gamma$, and we need

$$G(h_m) = \ln \frac{\omega\gamma}{1+\omega} - 1 + \frac{\nu}{\gamma} > 0.$$

Conditions 1–2 together require $\frac{\nu(1+\omega)}{\omega} < \gamma < \frac{\nu}{\omega \ln \nu}$, but they are *not sufficient*, and condition 3 is also needed.

Proposition 3.4 (Human-capital poverty trap)**.** *Under conditions 1–3, ϕ has three fixed points $\underline{\mu} < h_U < h_H$ with $\underline{h} < h_U < h_m < h_H$. The steady states $\underline{\mu}$ and h_H are stable and h_U is unstable. If $h_0 < h_U$, then $h_t \to \underline{\mu}$, and if $h_0 > h_U$, then $h_t \to h_H$.*

Proof. On $(\underline{h}, \infty)$, G is strictly concave, negative at $\underline{h}$, positive at h_m and tends to $-\infty$, so it has exactly two zeros $h_U < h_m < h_H$. At h_U, $\phi'(h_U) > \phi'(h_m) = 1$, so this steady state is unstable. At h_H, $\phi'(h_H) < 1$, so this steady state is stable. Since ϕ is non-decreasing, paths are monotone. For $h_0 \in (\underline{h}, h_U)$, $G < 0$ and h_t falls until it is below $\underline{h}$, after which $h_{t+1} = \underline{\mu}$. For $h_0 > h_U$, the path converges monotonically to h_H, as in Proposition 2.1. □

Example 3.3. Let $\nu = 1.3$, $\omega = 1$ and $\gamma = 4.5$. Then $\underline{\mu} = \ln 1.3 \approx 0.262$ and $\underline{h} = 1.3/4.5 \approx 0.289$. Condition 1 holds because $4.5 < 1.3/\ln 1.3 \approx 4.95$, condition 2 holds because $4.5 > 2.6$, and condition 3 holds because $\ln 2.25 + 1.3/4.5 \approx 1.10 > 1$. The steady states are $\underline{\mu} \approx 0.262$, $h_U \approx 0.328$ and $h_H \approx 1.227$. With $\gamma = 3.5$ instead, conditions 1 and 2 still hold but condition 3 fails, since $\ln 1.75 + 1.3/3.5 \approx 0.93$. The upper branch then stays below the 45° line and $\underline{\mu}$ is the only steady state.

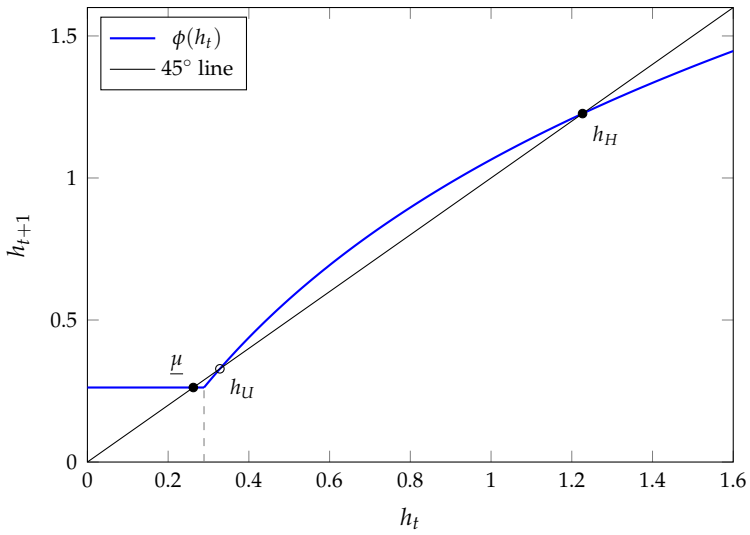


Figure 3.2: The human-capital map (3.8) for $\nu = 1.3$, $\omega = 1$ and $\gamma = 4.5$. Dynasties starting below h_U fall into the low steady state $\underline{\mu}$, where parents do not educate their children, while those above h_U converge to h_H.

Intuition

Poverty is self-perpetuating. Poor parents do not educate their children because the first unit of education is not worth its consumption

cost to them, so their children are poor too. A one-time policy that raises h above h_U (compulsory schooling, an education transfer) moves a dynasty permanently into the high steady state, whereas a small or temporary subsidy that leaves h below h_U has no lasting effect.

3.8.2 Subsistence consumption and a poverty trap

The next two extensions follow Azariadis and Stachurski (2005).

Set-up. Young people need to consume at least $\bar{c} > 0$ to survive. Preferences are of the Stone–Geary form

$$U_t = \ln(c_t - \bar{c}) + \beta \ln d_{t+1}.$$

Saving. If $w_t \leq \bar{c}$, the young consume their whole wage and save nothing. If $w_t > \bar{c}$, the FOC

$$\frac{1}{w_t - s_t - \bar{c}} = \frac{\beta}{s_t}$$

gives

$$s_t = \frac{\beta}{1+\beta}(w_t - \bar{c}) \qquad \Longrightarrow \qquad \frac{s_t}{w_t} = \frac{\beta}{1+\beta}\Big(1 - \frac{\bar{c}}{w_t}\Big).$$

Why the poor save a smaller fraction

Only income above subsistence is available for intertemporal allocation. The local EIS of the young is $-u'(c)/(cu''(c)) = (c-\bar{c})/c$, which tends to zero as $c \to \bar{c}$. Households close to subsistence are extremely unwilling to shift consumption to the future. Hence the *saving rate* increases with the wage.

Dynamics. With $f(k) = Ak^\alpha$ and $\delta = 1$, the law of motion is

$$k_{t+1} = \frac{\beta}{(1+\beta)(1+n)} \max\Big\{(1-\alpha)Ak_t^\alpha - \bar{c},\ 0\Big\}.$$

The map is zero for $k_t \leq k_{\min} \equiv (\bar{c}/((1-\alpha)A))^{1/\alpha}$ and increasing and concave above. If $\bar{c}$ is not too large, it crosses the 45° line twice, at $k_U < k_S$.

Proposition 3.5 (Subsistence poverty trap)**.** *Suppose the map crosses the* 45° *line at* $0 < k_U < k_S$*. Then* $k = 0$ *and* k_S *are locally stable steady states and* k_U *is unstable. If* $k_0 < k_U$*, then* $k_t \to 0$*, and if* $k_0 > k_U$*, then* $k_t \to k_S$*.*

Example 3.4. Let $A = 5$, $\alpha = 0.3$, $\beta = 0.8$, $n = 0$ and $\bar{c} = 1.6$, so that

$$k_{t+1} = \frac{0.8}{1.8} \max\{3.5k_t^{0.3} - 1.6, 0\}.$$

Then $k_{\min} \approx 0.074$, $k_U \approx 0.128$ and $k_S \approx 0.666$. The slope of the map is about 1.97 at k_U, which is unstable, and about 0.62 at k_S, which is stable. Starting from $k_0 = 0.12$, the economy collapses, whereas starting from $k_0 = 0.14$ it converges to k_S.

3.8.3 An impatience trap

Set-up. Suppose that patience depends on income, so that poor households discount the future more heavily. This may reflect a higher mortality risk, a stronger focus on immediate needs, or less exposure to the future-oriented attitudes that come with education. In the simplest version,

$$\beta(w) = \begin{cases} \beta_L & w < \hat{w}, \\ \beta_H & w \geq \hat{w}, \end{cases} \qquad \beta_L < \beta_H.$$

Dynamics. With log utility, $s_t = \frac{\beta(w_t)}{1+\beta(w_t)} w_t$, and the law of motion is

$$k_{t+1} = \frac{\beta(w(k_t))}{1 + \beta(w(k_t))} \cdot \frac{(1-\alpha)A}{1+n} k_t^{\alpha}.$$

It follows the "impatient" branch for $k_t < \hat{k} \equiv (\hat{w}/((1-\alpha)A))^{1/\alpha}$ and the "patient" branch for $k_t \geq \hat{k}$. Each branch has its own steady state,

$$k_j^* = \left(\frac{\beta_j (1-\alpha) A}{(1+\beta_j)(1+n)} \right)^{\frac{1}{1-\alpha}}, \qquad j \in \{L, H\}.$$

If $k_L^* < \hat{k} < k_H^*$, *both* are steady states of the economy.

Proposition 3.6 (Impatience trap). *If $k_L^* < \hat{k} < k_H^*$, an economy with $k_0 < \hat{k}$ converges to the low steady state k_L^*, and one with $k_0 \geq \hat{k}$ converges to k_H^*.*

Example 3.5. Let $A = 5$, $\alpha = 0.3$, $n = 0$, $\beta_L = 0.3$, $\beta_H = 0.9$ and $\hat{w} = 3.8$. Then $k_L^* \approx 0.737$, $k_H^* \approx 2.059$ and $\hat{k} = (3.8/3.5)^{1/0.3} \approx 1.315$, so the condition holds. The wage at k_L^* is about $3.19 < \hat{w}$. In the low steady state, households are poor *because* they are impatient, and impatient *because* they are poor.

Intuition

Low income leads to low patience, which leads to a low saving rate, low capital and low wages, so the loop is self-reinforcing. A one-time transfer that lifts k above $\hat{k}$ moves the economy permanently to the high steady state, whereas the same transfer spread thinly over time may have no permanent effect. With a smooth, S-shaped $\beta(w)$ the same logic produces three steady states, the middle one unstable.

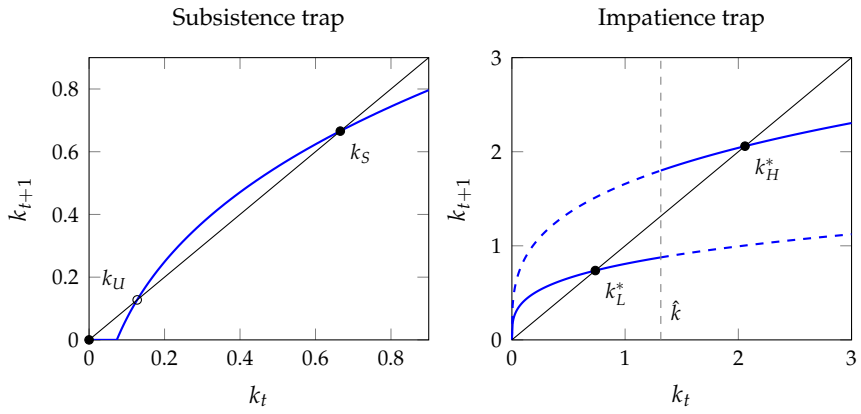


Figure 3.3: Poverty traps in the OLG model. The left panel shows subsistence consumption ($A = 5$, $\alpha = 0.3$, $\beta = 0.8$, $\bar{c} = 1.6$), where 0 and k_S are stable and k_U is the threshold. The right panel shows wealth-dependent patience ($\beta_L = 0.3$, $\beta_H = 0.9$, $\hat{w} = 3.8$), where the map jumps at $\hat{k}$ and both k_L^* and k_H^* are stable. Dashed curves show the shadow dynamics that would apply under the other patience regime (β_H below $\hat{k}$, β_L above $\hat{k}$).

3.8.4 A positive production externality

Set-up. Following Arrow (1962) and Romer (1986), we suppose that firms learn by doing, so that the productivity of every firm increases with the economy-wide capital intensity. Firm i produces

$$Y_{it} = a\,\bar{k}_t^{\,\varepsilon}\,K_{it}^{\alpha}N_{it}^{1-\alpha}, \qquad \varepsilon > 0,$$

where $\bar{k}_t$ is aggregate capital per worker. Each firm is small and takes $\bar{k}_t$ as given, so the effect of its own investment on everybody's productivity is an *externality*.

Private versus social returns. Profit maximization gives the *private* factor prices

$$R_t^k = \alpha a \bar{k}_t^{\varepsilon} k_t^{\alpha-1}, \qquad w_t = (1-\alpha) a \bar{k}_t^{\varepsilon} k_t^{\alpha}.$$

In equilibrium all firms choose the same capital intensity, so $k_t = \bar{k}_t$ and

$$y_t = a k_t^{\alpha+\varepsilon}, \qquad R_t^k = \alpha a k_t^{\alpha+\varepsilon-1}, \qquad w_t = (1-\alpha) a k_t^{\alpha+\varepsilon}.$$

The *social* marginal product of capital is $\mathrm{d}y/\mathrm{d}k = (\alpha+\varepsilon) a k^{\alpha+\varepsilon-1}$, which is larger than the private return R^k. Savers are therefore paid only part of what their saving contributes.

Dynamics. With log utility and $\delta = 1$, we have $(1+n)k_{t+1} = \frac{\beta}{1+\beta} w_t$, so that

$$k_{t+1} = \Lambda\, k_t^{\alpha+\varepsilon}, \qquad \Lambda \equiv \frac{\beta(1-\alpha)a}{(1+\beta)(1+n)}.$$

The behaviour depends on the *social* returns to capital, $\alpha+\varepsilon$.

- $\alpha+\varepsilon < 1$. The map is concave, and there is a unique, globally stable steady state $k^* = \Lambda^{1/(1-\alpha-\varepsilon)}$ with no long-run growth. The externality raises the level of income but not its growth rate.
- $\alpha+\varepsilon = 1$. The map is linear, $k_{t+1} = \Lambda k_t$. Capital and income per worker grow at the constant rate $g = \Lambda - 1$ from the first period on, provided $\Lambda > 1$. Growth is endogenous and depends on preferences and technology, since more patient ($\beta \uparrow$), more productive ($a \uparrow$) or slower-growing ($n \downarrow$) economies grow faster. There are no transitional dynamics.
- $\alpha+\varepsilon > 1$. The map is convex, and the positive steady state $\bar{k} = \Lambda^{-1/(\alpha+\varepsilon-1)}$ is unstable. Economies starting below $\bar{k}$ converge to zero while those above grow without bound, so increasing social returns create a poverty trap.

Intuition

The externality makes the *social* technology $y = a k^{\alpha+\varepsilon}$ less concave than the private one. When it exactly offsets diminishing returns ($\alpha+\varepsilon = 1$), the economy behaves like an AK model. No force stops accumulation, and the growth rate is pinned down by how much the young save. The equilibrium is generally inefficient, because no one is rewarded for the productivity gains that their saving generates for others. This "knowledge spillover" theme reappears at the centre of the Romer model in Chapter 6.

Chapter 4

The Ramsey Model

In the Solow model, households save a constant fraction of income regardless of the return to saving, which is hard to justify because people respond to incentives. In the OLG model, households optimize, but their horizon is only two periods. The Ramsey model has an infinite horizon and admits two interpretations.

- A **benevolent social planner** chooses the path of consumption and capital to maximize the discounted utility of a representative household.
- A **dynasty**, in which parents care about the utility of their children, who care about theirs, and so on. Operative altruism turns a sequence of finitely-lived generations into one infinitely-lived decision maker.

We show that, under perfect competition, the competitive equilibrium coincides with the planner's solution. The model's steady state is of the same kind as Solow's, but now the saving rate is *derived*, and the economy is always *dynamically efficient*.

4.1 Warm-up: the cake-eating problem

Before letting capital produce output, we consider the simplest infinite-horizon problem. A household owns a cake of size $k_0 > 0$. Whatever is not eaten today is left for tomorrow, but the cake does not grow,

$$k_{t+1} = k_t - c_t, \qquad c_t \geq 0, \quad k_{t+1} \geq 0. \tag{4.1}$$

The household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad (4.1),\ k_0 \text{ given.}$$

The *state* variable is the remaining cake k_t, and the *control* is consumption c_t.

Lagrangian and Euler equation. We attach to the period-t constraint the multiplier $\beta^t \lambda_t$, so that λ_t measures the value of one unit of cake at date t in units of date-t utility. This is a *current-value* multiplier, and we discuss the alternative *present-value* form in Section 4.3. The Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ u(c_t) + \lambda_t \big[k_t - c_t - k_{t+1} \big] \Big\}.$$

The choice c_t appears only in period t, while the choice k_{t+1} appears both in the period-t constraint (as a use of cake) and in the period-$(t+1)$ constraint (as cake available). The FOCs are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_t} = 0: &\quad u'(c_t) = \lambda_t, \\ \frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0: &\quad -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} = 0 \implies \lambda_t = \beta \lambda_{t+1}. \end{aligned}$$

Eliminating the multipliers gives the Euler equation

$$u'(c_t) = \beta\, u'(c_{t+1}). \tag{4.2}$$

Reading the cake-eating Euler equation

Eating one unit less today costs $u'(c_t)$. The unit is kept as cake and, since the cake does not grow, it delivers exactly one unit tomorrow, worth $\beta u'(c_{t+1})$ today. At the optimum, the two are equal. Because $\beta < 1$, (4.2) requires $u'(c_{t+1}) = u'(c_t)/\beta > u'(c_t)$, that is, $c_{t+1} < c_t$. An impatient household eats more early and consumption declines over time.

The Euler equation is not enough. The Euler equation (4.2) links c_t and c_{t+1}, but it does not tell us where consumption *starts*. Given any initial consumption $c_0 > 0$, the Euler equation determines c_1, then c_2, and so on, and (4.1) determines the remaining cake. Every choice of c_0 therefore generates a whole path that satisfies all the first-order conditions. Only one of these paths is optimal, and we need one more condition to find it. In a finite-horizon problem, the missing condition is a terminal condition. In an infinite-horizon problem, it is the **transversality condition**.

A family of Euler-consistent paths. To see the problem concretely, let $u(c) = \ln c$. Then (4.2) reads $1/c_t = \beta/c_{t+1}$, so $c_{t+1} = \beta c_t$ and $c_t = \beta^t c_0$. Iterating (4.1), we find that the cake left after period T is

$$k_{T+1} = k_0 - \sum_{t=0}^{T} c_t = k_0 - c_0 \frac{1 - \beta^{T+1}}{1 - \beta} \xrightarrow[T \to \infty]{} k_\infty \equiv k_0 - \frac{c_0}{1 - \beta}. \tag{4.3}$$

The limit depends on the initial choice c_0, and we can distinguish the following cases (Figure 4.1).

1. **Too much consumption,** $c_0 > (1-\beta)k_0$**.** Then $k_\infty < 0$, so the path asks for more cake than exists. At some finite date the cake runs out and the constraint $k_{t+1} \geq 0$ is violated. The path is *infeasible*.

2. **Too little consumption,** $c_0 < (1-\beta)k_0$**.** Then $k_\infty > 0$, so a positive amount of cake is never eaten. The path is feasible, but it cannot be optimal. Consider the alternative plan that eats an extra $\varepsilon \in (0, k_\infty]$ at date 0 and otherwise consumes exactly as before. It is still feasible, because the cake remaining at every date falls by only $\varepsilon \leq k_\infty$, and it raises lifetime utility by approximately $u'(c_0)\varepsilon > 0$. Leaving cake uneaten forever is pure waste.

3. **Exactly right,** $c_0 = (1-\beta)k_0$**.** Then $k_\infty = 0$, and the cake is exhausted, but only asymptotically. This is the optimal plan.

The Euler equation alone cannot distinguish between these paths, because each of them is consistent with optimal behavior *between any two consecutive dates*. What goes wrong with the paths in case 2 is a property of the path *as a whole*, since value is left unused "at infinity".

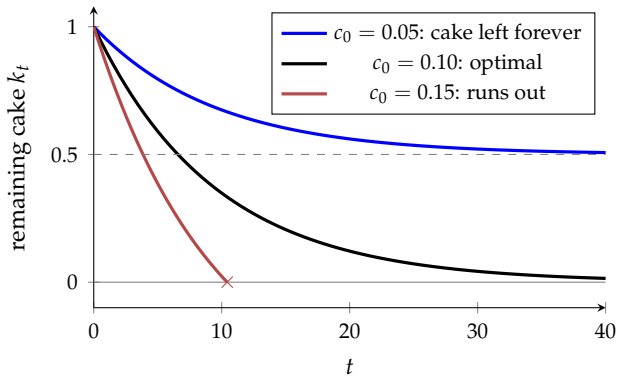


Figure 4.1: Three paths that satisfy the cake-eating Euler equation $c_{t+1} = \beta c_t$, with $u = \ln c$, $\beta = 0.9$ and $k_0 = 1$. Starting with too much consumption exhausts the cake in finite time (here at $t \approx 10.4$), which is infeasible. Starting with too little leaves half of the cake uneaten forever. Only $c_0 = (1-\beta)k_0 = 0.1$ is optimal, and it exhausts the cake asymptotically.

Where the transversality condition comes from. The terminal condition is easiest to understand in a problem that ends at a finite date T. We add the constraint $k_{T+1} \geq 0$, since the household cannot leave a negative cake,

and write the Lagrangian with multipliers $\beta^t \lambda_t$ on the resource constraints and $\beta^T \mu$ on the terminal constraint,

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \Big\{ u(c_t) + \lambda_t \big[k_t - c_t - k_{t+1} \big] \Big\} + \beta^T \mu \, k_{T+1}.$$

The first-order conditions for c_t and k_{t+1}, $t < T$, are exactly as before. The new conditions concern the last period,

$$\frac{\partial \mathcal{L}}{\partial c_T} = 0 : \;\; u'(c_T) = \lambda_T, \qquad \frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 : \;\; -\beta^T \lambda_T + \beta^T \mu = 0 \implies \mu = \lambda_T,$$

together with the Kuhn–Tucker conditions $\mu \geq 0$ and $\mu \, k_{T+1} = 0$. Since $\mu = \lambda_T = u'(c_T) > 0$, complementary slackness forces $k_{T+1} = 0$, so that with a finite horizon the household eats the whole cake by the last date. We can write the condition as

$$\beta^T u'(c_T) \, k_{T+1} = 0.$$

This condition states that the value, in units of date-0 utility, of what is left at the end is zero. Here k_{T+1} is the quantity of cake left, and $\beta^T u'(c_T)$ is the date-0 utility value of one unit of cake at date T. Either nothing is left, or what is left is worthless. With an infinite horizon there is no last date, and the natural analogue is the limit of this condition, the **transversality condition** (TVC),

$$\lim_{T \to \infty} \beta^T u'(c_T) \, k_{T+1} = 0.$$

The TVC selects the optimal path. With log utility, $\beta^T u'(c_T) = \beta^T / (\beta^T c_0) = 1/c_0$ is constant along every Euler-consistent path, so the TVC term is k_{T+1}/c_0, which converges to k_∞ / c_0 by (4.3). Among the feasible paths ($k_\infty \geq 0$), the TVC holds if and only if $k_\infty = 0$, that is,

$$c_0 = (1 - \beta) k_0.$$

The paths of case 2 violate the TVC precisely because they leave cake of positive value unused. Substituting back, we obtain

$$k_t = k_0 - c_0 \frac{1 - \beta^t}{1 - \beta} = \beta^t k_0, \qquad c_t = \beta^t c_0 = (1 - \beta) \beta^t k_0 = (1 - \beta) k_t.$$

The optimal **policy function** is therefore

$$c_t = (1 - \beta) k_t, \qquad k_{t+1} = \beta k_t,$$

so that each period the household eats the fraction $1 - \beta$ of what is left. A more patient household (higher β) eats a smaller fraction and depletes the cake more slowly.

Remark 4.1. The TVC is written in terms of the *value* $\beta^T u'(c_T)k_{T+1}$ rather than the *quantity* k_{T+1}. In the cake problem the two coincide, because $\beta^T u'(c_T)$ stays bounded away from zero. In the Ramsey model below, capital converges to a positive steady state k^*, so that $\lim k_{t+1} = k^* > 0$, and yet the path is optimal. What must vanish is the present value of the capital left at infinity,

$$\beta^t u'(c_t)k_{t+1} \approx \beta^t u'(c^*)k^* \to 0.$$

Leaving positive capital at infinity is therefore fine, whereas leaving capital whose *present value* does not vanish is not.

Example 4.1 (An Euler-consistent plan that violates the TVC)**.** Let $\beta = 0.9$ and $k_0 = 1$, and consider the "half-cake" plan $c_t = 0.05\,(0.9)^t$ (the blue line in Figure 4.1). It satisfies the Euler equation, since $c_{t+1}/c_t = 0.9 = \beta$, but $\sum_t c_t = 0.05/0.1 = 0.5$, so half of the cake is never eaten and $k_t \to 0.5$. The TVC term is $k_{T+1}/c_0 \to 0.5/0.05 = 10 \neq 0$. The optimal plan $c_t = 0.1\,(0.9)^t$ has $k_{T+1} = 0.9^{T+1}$ and TVC term $0.9^{T+1}/0.1 \to 0$.

The same answer by dynamic programming. Let $V(k)$ be the maximal lifetime utility from a cake of size k. The Bellman equation is

$$V(k) = \max_{0 \le k' \le k} \left\{ \ln(k - k') + \beta V(k') \right\}.$$

We guess $V(k) = E + B \ln k$. The FOC for k' is

$$\frac{1}{k - k'} = \frac{\beta B}{k'},$$

so $k' = \frac{\beta B}{1+\beta B} k$ and $k - k' = \frac{1}{1+\beta B} k$. Substituting into the Bellman equation, we find that the right-hand side equals a constant plus $(1 + \beta B) \ln k$. Matching the coefficients of $\ln k$ on both sides gives $B = 1 + \beta B$, so $B = 1/(1 - \beta)$ and $k' = \beta k$, which is the same policy as above.

From cake eating to Ramsey. The Ramsey model is a cake-eating problem in which the cake *grows*. Capital left for tomorrow produces output, and the gross return on one unit of "cake" is $f'(k_{t+1}) + 1 - \delta$ instead of 1. The Euler equation (4.2) then becomes

$$u'(c_t) = \beta u'(c_{t+1}) \left[f'(k_{t+1}) + 1 - \delta \right],$$

and the TVC keeps exactly the same form.

4.2 The Ramsey Model

We now study the Ramsey model (Ramsey, 1928; Cass, 1965; Koopmans, 1965).

4.2.1 Preferences

A representative household lives for $t = 0, 1, 2, \ldots$ and values a consumption stream $\{c_t\}_{t=0}^{\infty}$ by

$$\sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

The function u is strictly increasing and strictly concave, and it satisfies the Inada conditions $u'(0) = +\infty$ and $u'(\infty) = 0$. The *rate of time preference* ρ is defined by $\beta = 1/(1+\rho)$, or $\rho = (1-\beta)/\beta$, so that a utility unit tomorrow is worth $1/(1+\rho)$ utility units today. We mostly use CRRA utility (1.4), $u(c) = (c^{1-\sigma} - 1)/(1-\sigma)$, with EIS $1/\sigma$. For simplicity, the population is constant and normalized to one, so aggregate and per-capita variables coincide.

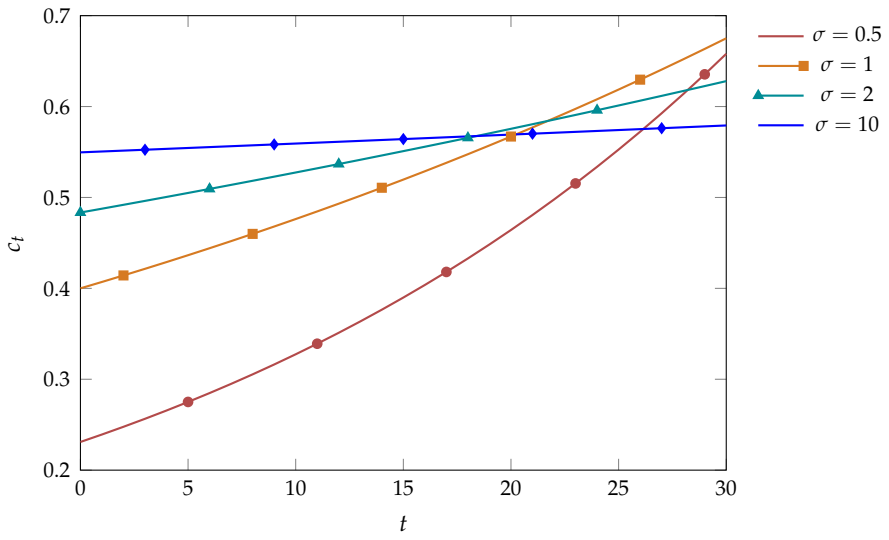


Figure 4.2: Consumption smoothing and the EIS. The figure shows households with the same lifetime wealth $\sum_t c_t/R^t = 10$, $\beta = 0.96$ and $R = 1.06$ (so that $\beta R = 1.0176 > 1$). The optimal path is $c_t = c_0(\beta R)^{t/\sigma}$ with $c_0 = 10\,[1 - (\beta R)^{1/\sigma}/R]$. A household with a low EIS (high σ) keeps consumption almost flat even though waiting pays, whereas a household with a high EIS tilts consumption steeply towards the future.

4.2.2 Technology and resources

Output is produced with a neoclassical technology, $y_t = f(k_t)$, and capital depreciates at rate $\delta \in (0, 1]$. Output plus undepreciated capital is split between consumption and next period's capital,

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t \qquad \Longleftrightarrow \qquad k_{t+1} - k_t = f(k_t) - \delta k_t - c_t.$$

In the Solow model, $c_t = (1-s)f(k_t)$ with s exogenous, whereas in the Ramsey model c_t is chosen.

4.3 The planner's problem

4.3.1 Finite horizon

To understand the role of the terminal condition, we first consider a planner who cares about periods $0, \dots, T$ and solves

$$\begin{aligned} &\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t) \\ &\text{s.t.} \quad c_t + k_{t+1} = f(k_t) + (1-\delta)k_t, \;\; k_{t+1} \geq 0, \;\; k_0 \text{ given.} \end{aligned}$$

We attach the multiplier $\beta^t \lambda_t$ to the period-t resource constraint (λ_t is a current-value multiplier, as explained below) and $\beta^T \mu$ to the terminal constraint $k_{T+1} \geq 0$,

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \Big\{ u(c_t) + \lambda_t \big[f(k_t) + (1-\delta)k_t - c_t - k_{t+1} \big] \Big\} + \beta^T \mu k_{T+1}.$$

The FOCs are

$$\begin{aligned} c_t: &\quad u'(c_t) = \lambda_t, && t = 0, \dots, T, \\ k_{t+1}: &\quad \lambda_t = \beta \lambda_{t+1} \big[f'(k_{t+1}) + 1 - \delta \big], && t = 0, \dots, T-1, \\ k_{T+1}: &\quad \lambda_T = \mu, \qquad \mu \geq 0, \; \mu k_{T+1} = 0. \end{aligned}$$

Since $\lambda_T = u'(c_T) > 0$, we have $\mu > 0$ and therefore $k_{T+1} = 0$. It is never optimal to leave capital unconsumed at the end of the world. The complementary-slackness condition can be written as $\beta^T u'(c_T) k_{T+1} = 0$, which says that the *value* of the terminal capital stock must be zero. The infinite-horizon *transversality condition* is the limit of this condition.

4.3.2 Infinite Horizon

The planner solves

$$\begin{aligned} &\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t) \\ &\text{s.t.} \quad c_t + k_{t+1} = f(k_t) + (1-\delta)k_t, \;\; k_{t+1} \geq 0, \;\; k_0 \text{ given.} \end{aligned}$$

There is one constraint for *each* period, hence infinitely many constraints, and each needs its own Lagrange multiplier. The literature uses two equivalent ways of attaching these multipliers, and we present them side by side.

The present-value Lagrangian. We attach a multiplier $\tilde{\lambda}_t$ to the period-t constraint, exactly as in a static problem,

$$\mathcal{L}^{PV} = \sum_{t=0}^{\infty} \Big\{ \beta^t u(c_t) + \tilde{\lambda}_t \big[f(k_t) + (1-\delta)k_t - c_t - k_{t+1} \big] \Big\}.$$

Because the objective is measured in units of date-0 utility, $\tilde{\lambda}_t$ is the increase in the *date-0* value of the plan if one more unit of the good were available at date t. It is a *present-value* shadow price.

The current-value Lagrangian. Alternatively, we can measure each multiplier in units of utility *of the period in which the constraint applies* by writing $\tilde{\lambda}_t = \beta^t \lambda_t$,

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ u(c_t) + \lambda_t \big[f(k_t) + (1-\delta)k_t - c_t - k_{t+1} \big] \Big\}.$$

Now λ_t is the value of one more unit of the good at date t, measured in date-t utility. It is a *current-value* shadow price. The two Lagrangians are the same function with differently labeled multipliers, so they deliver the same allocation. The current-value form is usually more convenient, because in a steady state λ_t converges to a constant, $\lambda^* = u'(c^*)$, whereas the present-value multiplier $\tilde{\lambda}_t = \beta^t \lambda_t$ shrinks to zero.

Locating each choice variable. The key step in deriving the first-order conditions is to locate every term that contains a given choice variable. Writing out the part of the current-value Lagrangian that involves periods t and $t+1$,

$$\begin{aligned} \mathcal{L} = \cdots + \beta^t \Big\{ u(c_t) + \lambda_t \big[f(k_t) + (1-\delta)k_t - c_t - k_{t+1} \big] \Big\} \\ + \beta^{t+1} \Big\{ u(c_{t+1}) + \lambda_{t+1} \big[f(k_{t+1}) + (1-\delta)k_{t+1} - c_{t+1} - k_{t+2} \big] \Big\} + \cdots \end{aligned}$$

we see that c_t appears only in the period-t term, while k_{t+1} appears twice. It enters the period-t constraint as a *use* of resources (capital installed for tomorrow), and the period-$(t+1)$ constraint as a *source* of resources (capital that produces output and is partly left over after depreciation). Differentiating, we obtain

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0: \quad \beta^t \big[u'(c_t) - \lambda_t \big] = 0 \implies u'(c_t) = \lambda_t, \tag{4.4}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0: \quad & -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} \big[f'(k_{t+1}) + 1 - \delta \big] = 0 \\ & \implies \lambda_t = \beta \lambda_{t+1} \big[f'(k_{t+1}) + 1 - \delta \big]. \end{aligned} \tag{4.5}$$

The same steps applied to $\mathcal{L}^{PV}$ give

$$\beta^t u'(c_t) = \tilde{\lambda}_t, \qquad \tilde{\lambda}_t = \tilde{\lambda}_{t+1}\big[f'(k_{t+1}) + 1 - \delta\big],$$

which become (4.4)–(4.5) after substituting $\tilde{\lambda}_t = \beta^t \lambda_t$ and dividing by β^t. In the present-value form, the discount factor sits inside the multiplier, because $\tilde{\lambda}_{t+1}$ already measures date-$(t+1)$ resources in date-0 utility, so no β appears in the second condition.

Remark 4.2 (A recipe). We handle every dynamic Lagrangian in these notes in the same way. We (i) write the constraint of each period with all its time subscripts, (ii) attach a multiplier to each constraint, $\beta^t \lambda_t$ in current value or $\tilde{\lambda}_t$ in present value, (iii) find, for each choice variable, *every* period in which it appears (a stock chosen at t for $t+1$ always appears in two consecutive constraints), (iv) differentiate and set the derivatives to zero, and (v) eliminate the multipliers to obtain an Euler equation and add the transversality condition.

The shadow price

The multiplier λ_t is the value, in utility units of period t, of one more unit of the good in period t. Condition (4.4) says that at the optimum a unit of the good is worth the same whether it is consumed (marginal utility $u'(c_t)$) or kept as resources. Condition (4.5) is an *asset-pricing equation* for capital, since a unit invested at t costs λ_t and delivers $f'(k_{t+1}) + 1 - \delta$ units at $t+1$, each worth λ_{t+1}, discounted by β.

4.3.3 The Euler equation

Substituting (4.4) at t and $t+1$ into (4.5), we obtain the Euler equation

$$u'(c_t) = \beta\, u'(c_{t+1}) \left[f'(k_{t+1}) + 1 - \delta\right] \tag{4.6}$$

or, equivalently,

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = R_{t+1} \equiv f'(k_{t+1}) + 1 - \delta.$$

- The left-hand side is the marginal utility cost of saving one more unit today.
- The right-hand side is the discounted marginal utility of the extra R_{t+1} units available tomorrow.
- Equivalently, the marginal rate of substitution between c_t and c_{t+1} equals the marginal rate of transformation R_{t+1}, so the household's willingness to trade consumption over time equals the technology's ability to do so.

With CRRA utility, the Euler equation reads $c_t^{-\sigma} = \beta c_{t+1}^{-\sigma} R_{t+1}$, so that

$$\frac{c_{t+1}}{c_t} = \big(\beta R_{t+1}\big)^{1/\sigma}$$

and, taking logs,

$$\ln \frac{c_{t+1}}{c_t} = \frac{1}{\sigma}\big[\ln R_{t+1} - \ln(1+\rho)\big] \approx \frac{1}{\sigma}\big[f'(k_{t+1}) - \delta - \rho\big],$$

using $\ln(1+x) \approx x$. Consumption grows when the net return on capital $f'(k_{t+1}) - \delta$ exceeds the rate of time preference ρ, and falls when it is below. The EIS $1/\sigma$ scales the response, and a household with a high σ wants a flat consumption path and reacts little to returns (Figure 4.2).

4.3.4 The transversality condition

The Euler equation and the resource constraint are two first-order difference equations in (k_t, c_t), but we have only one initial condition, k_0. The missing boundary condition is the **transversality condition** (TVC),

$$\lim_{t\to\infty} \beta^t u'(c_t)\, k_{t+1} = 0. \tag{4.7}$$

It says that the present value (in utility) of the capital stock left "at infinity" must be zero. Since $u'(c_t) = \lambda_t$, we can equivalently write it with the current-value multiplier, $\lim_{t\to\infty} \beta^t \lambda_t k_{t+1} = 0$, or with the present-value multiplier, $\lim_{t\to\infty} \tilde{\lambda}_t k_{t+1} = 0$. In both forms, the date-0 value of the terminal capital stock vanishes. As in the cake-eating problem, it is the limit of the terminal condition $\beta^T u'(c_T) k_{T+1} = 0$ of the finite-horizon problem. A path that keeps accumulating capital whose present value does not vanish is "dying rich", because the household could consume part of that capital at some date and be better off. For a concave problem like this one, the Euler equation, the resource constraint and the TVC are *sufficient* for optimality (Appendix A.3).

4.4 The steady state

In a steady state, $c_t = c_{t+1} = c^*$ and $k_t = k_{t+1} = k^*$. The Euler equation gives $1 = \beta[f'(k^*) + 1 - \delta]$, that is,

$$f'(k^*) = \rho + \delta \qquad \text{(modified golden rule)}, \qquad c^* = f(k^*) - \delta k^*.$$

With $f(k) = Ak^\alpha$, we obtain

$$k^* = \left(\frac{\alpha A}{\rho + \delta}\right)^{\frac{1}{1-\alpha}}, \qquad c^* = A(k^*)^\alpha - \delta k^*.$$

Why the modified golden rule?

In a steady state, consumption is constant, so the household is willing to postpone one unit of consumption only if the net return compensates its impatience exactly, $f'(k) - \delta = \rho$. The steady state is determined by preferences (ρ) and technology (f, δ) only. The EIS $1/\sigma$ does not appear, because it governs how *fast* the economy moves towards k^*, not where it ends.

Comparative statics. A more patient household ($\beta \uparrow$, $\rho \downarrow$) has a higher k^* and c^*. Faster depreciation lowers k^*, and higher TFP raises both k^* and c^*.

Dynamic efficiency. The golden rule maximizes steady-state consumption $f(k) - \delta k$, so $f'(k_{GR}) = \delta$. Because $\rho > 0$,

$$f'(k^*) = \rho + \delta > \delta = f'(k_{GR}) \quad \Longrightarrow \quad k^* < k_{GR}.$$

The Ramsey economy never over-accumulates. It does not go all the way to the golden rule, where steady-state consumption is highest, because getting there requires sacrificing consumption *now* for gains that arrive *later*, and an impatient household values the path, not only the destination. Over-accumulation, which we saw is possible in Solow and OLG economies, is ruled out by the TVC. In a steady state with $k > k_{GR}$, the net return would be below zero, and the household could eat some capital and be better off forever.

4.5 The decentralized economy

4.5.1 Households and firms

We now let a representative household own the capital stock, rent it to competitive firms at rental rate R_t^k, and supply one unit of labor at wage w_t. Let $r_t \equiv R_t^k - \delta$ be the net return on capital. The household's budget constraint is

$$c_t + a_{t+1} = (1 + r_t)a_t + w_t,$$

where a_t is wealth (here, capital). To prevent the household from borrowing indefinitely and never repaying, we impose a *no-Ponzi condition*,

$$\lim_{t \to \infty} \frac{a_{t+1}}{\prod_{s=1}^{t}(1 + r_s)} \geq 0.$$

The household's Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ u(c_t) + \lambda_t \big[(1 + r_t)a_t + w_t - c_t - a_{t+1} \big] \Big\},$$

with FOCs $u'(c_t) = \lambda_t$ and $\lambda_t = \beta\lambda_{t+1}(1 + r_{t+1})$, and hence

$$u'(c_t) = \beta(1 + r_{t+1})\, u'(c_{t+1}).$$

Firms solve $\max F(K, L) - R^k K - wL$, so that $R^k_t = f'(k_t)$ and $w_t = f(k_t) - k_t f'(k_t)$.

4.5.2 Equilibrium and efficiency

In equilibrium, $a_t = k_t$ and $1 + r_{t+1} = f'(k_{t+1}) + 1 - \delta$, so the household's Euler equation is exactly (4.6). Moreover, its budget constraint becomes the resource constraint,

$$c_t + k_{t+1} = (1 + f'(k_t) - \delta)k_t + f(k_t) - k_t f'(k_t) = f(k_t) + (1 - \delta)k_t.$$

The household's TVC is (4.7). Since the equilibrium satisfies the same equations as the planner's optimum, the two coincide.

Proposition 4.1 (Welfare theorems)**.** *The competitive equilibrium of the Ramsey economy is Pareto optimal and coincides with the planner's solution.*

This is the key difference from the OLG model. With a single infinitely-lived (dynastic) household, there is no "infinity of agents" and no room for Pareto-improving chains of transfers.

4.6 Dynamics: the phase diagram

4.6.1 The dynamic system

The optimal path solves

$$k_{t+1} = f(k_t) + (1 - \delta)k_t - c_t, \tag{4.8}$$

$$u'(c_t) = \beta u'(c_{t+1})\big[f'(k_{t+1}) + 1 - \delta\big], \tag{4.9}$$

with k_0 given and the TVC. Capital is a *state* (predetermined) variable, while consumption is a *control* ("jump") variable that can take any value at $t = 0$. To draw the phase diagram (Appendix A.5), we first find the locus along which each variable is constant and the direction in which that variable moves off the locus. We then combine the two loci in a single diagram.

4.6.2 The $\Delta k = 0$ locus

We start with capital. From (4.8), the change in the capital stock is

$$\Delta k_t \equiv k_{t+1} - k_t = f(k_t) - \delta k_t - c_t,$$

so capital is constant if and only if consumption equals output net of depreciation,

$$\Delta k_t = 0 \quad \Longleftrightarrow \quad c_t = f(k_t) - \delta k_t.$$

Figure 4.3 plots this locus. It is hump-shaped, starting at the origin, increasing while $f'(k) > \delta$, peaking at the golden rule k_{GR} where $f'(k_{GR}) = \delta$, and declining thereafter.

The locus splits the plane into two regions. At a point *above* the locus, consumption exceeds net output, so the economy eats into its capital stock and k falls. At a point *below* the locus, consumption falls short of net output, so the economy invests more than it needs to replace depreciated capital and k rises. The horizontal arrows in Figure 4.3 show these movements. Their length is proportional to $|\Delta k_t|$, so capital moves faster the further the economy is from the locus.

To see the trajectories implied by this locus on its own, we can imagine for a moment that consumption is held fixed at some level $\bar{c}$ below the peak of the hump (the dashed line in Figure 4.3). The line $c = \bar{c}$ cuts the locus twice, at $k_a < k_{GR} < k_b$. From any $k_0 \in (k_a, k_b)$, the economy lies below the locus and capital rises towards k_b. From any $k_0 > k_b$, it lies above the locus and capital falls towards k_b. From $k_0 < k_a$, capital falls and the economy eventually runs out of capital. This thought experiment describes the motion of k only. In the Ramsey model consumption moves as well, and we turn to its dynamics next.

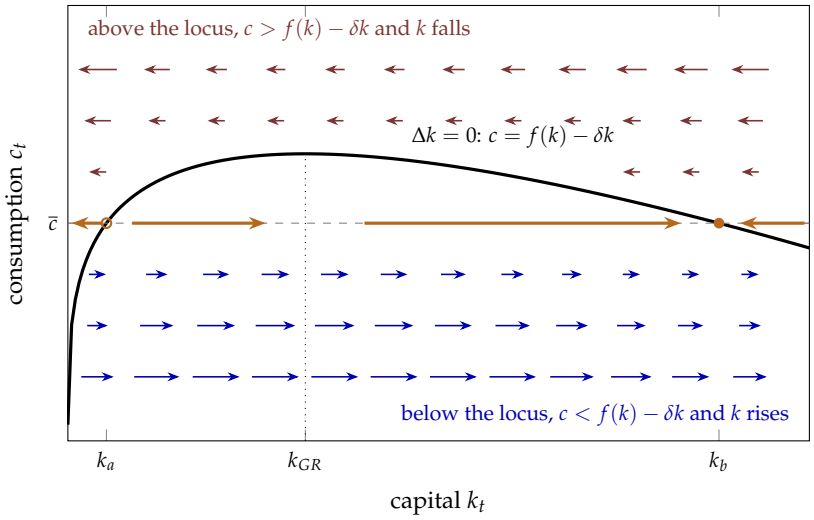


Figure 4.3: The $\Delta k = 0$ locus for $f(k) = k^{0.3}$ and $\delta = 0.1$. Above the locus capital falls (red arrows), and below it capital rises (blue arrows). Arrow lengths are proportional to $|\Delta k_t|$. The orange trajectories show how capital would move if consumption were held fixed at $\bar{c} = 0.85$. Capital converges to k_b from either side and moves away from k_a.

4.6.3 The $\Delta c = 0$ locus

We now turn to consumption. From (4.9), $c_{t+1} = c_t$ if and only if $\beta[f'(k_{t+1}) + 1 - \delta] = 1$, that is, if and only if the capital stock on which the return is earned equals the modified-golden-rule level,

$$\Delta c_t = 0 \quad \Longleftrightarrow \quad k_{t+1} = k^*.$$

Moreover, since both f' and u' are decreasing,

$$c_{t+1} > c_t \quad \Longleftrightarrow \quad f'(k_{t+1}) + 1 - \delta > \frac{1}{\beta} \quad \Longleftrightarrow \quad k_{t+1} < k^*.$$

Consumption rises when capital is below k^*, because its return then exceeds the rate of time preference, and it falls when capital is above k^*. As in the standard textbook diagram, the $\Delta c = 0$ locus is therefore the **vertical line** $k = k^*$ shown in Figure 4.4. To the *left* of the line, the vertical arrows point upwards and consumption rises. To the *right*, they point downwards and consumption falls. The locus does not depend on c, because the Euler equation pins down the *return* required for flat consumption, and this return depends only on capital.

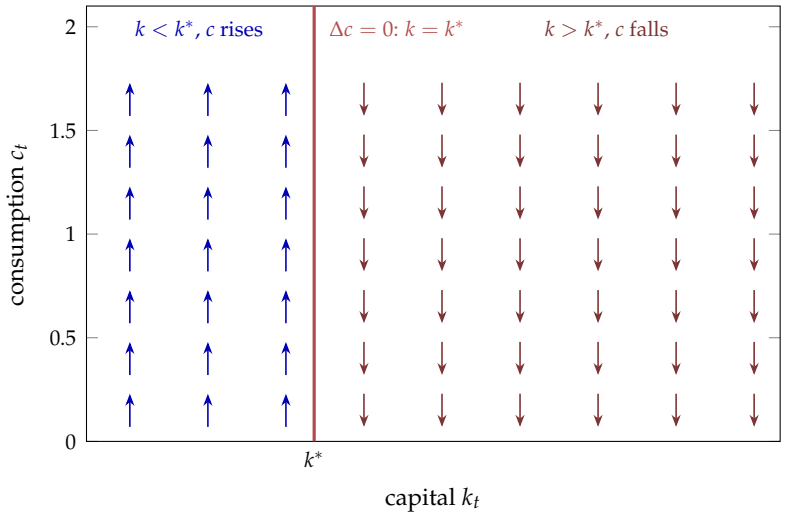


Figure 4.4: The $\Delta c = 0$ locus for $f(k) = k^{0.3}$, $\beta = 0.95$ and $\delta = 0.1$, with $k^* \approx 2.626$. To the left of k^*, the return on capital exceeds the rate of time preference and consumption rises (blue arrows). To the right, consumption falls (red arrows).

Remark 4.3 (Dating of capital on the horizontal axis). The vertical locus is exact when the capital on the horizontal axis is the capital that earns the return, k_{t+1} (capital carried into next period), and it is exact in continuous

time, where $\dot{c} = 0 \iff k = k^*$. If we instead put beginning-of-period capital k_t on the axis, the condition $k_{t+1} = k^*$ must be translated with (4.8) into

$$c_t = f(k_t) + (1 - \delta)k_t - k^*,$$

an upward-sloping curve that passes through the same steady state (k^*, c^*) and approaches the vertical line as the period becomes short. The steady state, the saddle-path property and the direction of motion near the steady state are the same in both representations, and we therefore use the standard vertical line.

4.6.4 The phase diagram and the saddle path

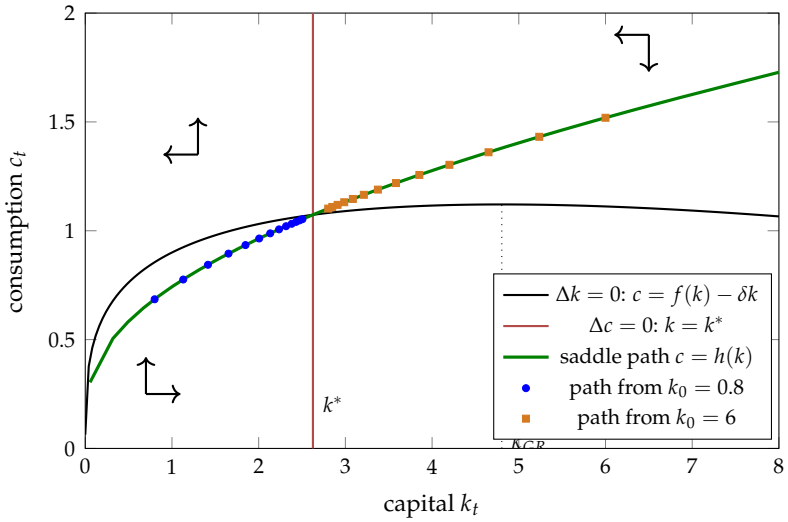


Figure 4.5: The discrete-time Ramsey phase diagram for $f(k) = k^{0.3}$, $\beta = 0.95$, $\delta = 0.1$ and $\sigma = 2$, with $k^* \approx 2.626$, $c^* \approx 1.073$ and $k_{GR} \approx 4.804$. The $\Delta c = 0$ locus is the vertical line $k = k^*$ (Remark 4.3), and the $\Delta k = 0$ locus is $c = f(k) - \delta k$. The saddle path was computed by iterating the system backwards from the steady state. Dots show optimal paths from $k_0 = 0.8$ and $k_0 = 6$ (every second period). Arrows indicate direction of motion in four representative regions.

Figure 4.5 superimposes the two loci of Figures 4.3 and 4.4. They intersect at the steady state (k^*, c^*) (and at the origin) and divide the plane into four regions. In each region, the direction of motion combines the horizontal arrow of Figure 4.3, given by the sign of Δk, with the vertical arrow of Figure 4.4, given by the sign of Δc. Below the $\Delta k = 0$ locus and to the left of k^*, for instance, both capital and consumption rise, so the economy moves north-east. Above the $\Delta k = 0$ locus and to the right of k^*, both fall, and

the economy moves south-west. In the two remaining regions, the arrows point north-west and south-east.

Proposition 4.2 (Saddle-path stability)**.** *For every $k_0 > 0$ there is a unique $c_0 = h(k_0)$ such that the path generated by (4.8)–(4.9) converges to (k^*, c^*). This path satisfies the TVC and is the optimal path. All other choices of c_0 generate paths that are either infeasible or violate the TVC.*

- If c_0 is *above* the saddle path, consumption is too high. Capital is run down, and in finite time the economy would need negative capital to continue on the Euler equation. Such a path is infeasible.
- If c_0 is *below* the saddle path, consumption is too low. Capital is accumulated beyond k^*, and eventually beyond k_{GR}, while consumption falls towards zero. The household is then "dying rich", the TVC fails, and the path is dominated by one that consumes more.

The function $c = h(k)$ that describes the saddle path is the optimal **policy function**. It is upward sloping, so richer economies consume more. Starting from $k_0 < k^*$, consumption and capital both increase monotonically along the saddle path, and starting from $k_0 > k^*$ both decrease.

4.7 Linearization and the speed of convergence

Let $\hat{k}_t \equiv k_t - k^*$ and $\hat{c}_t \equiv c_t - c^*$. Linearizing (4.8) around the steady state and using $f'(k^*) + 1 - \delta = 1/\beta$, we obtain

$$\hat{k}_{t+1} = \frac{1}{\beta}\hat{k}_t - \hat{c}_t.$$

To linearize (4.9), we differentiate both sides and find

$$\begin{aligned} u''(c^*)\hat{c}_t &= \beta u''(c^*)\big[f'(k^*) + 1 - \delta\big]\hat{c}_{t+1} + \beta u'(c^*)f''(k^*)\hat{k}_{t+1} \\ &= u''(c^*)\hat{c}_{t+1} + \beta u'(c^*)f''(k^*)\hat{k}_{t+1}. \end{aligned}$$

Dividing by $u''(c^*)$ and using $u'(c^*)/u''(c^*) = -c^*/\sigma$ (CRRA), we have

$$\hat{c}_{t+1} = \hat{c}_t - \kappa\,\hat{k}_{t+1}, \qquad \kappa \equiv -\frac{\beta c^* f''(k^*)}{\sigma} > 0.$$

Substituting $\hat{k}_{t+1}$ from the first equation gives

$$\hat{c}_{t+1} = (1+\kappa)\hat{c}_t - \frac{\kappa}{\beta}\hat{k}_t.$$

In matrix form,

$$\begin{pmatrix} \hat{k}_{t+1} \\ \hat{c}_{t+1} \end{pmatrix} = M \begin{pmatrix} \hat{k}_t \\ \hat{c}_t \end{pmatrix}, \qquad M = \begin{pmatrix} 1/\beta & -1 \\ -\kappa/\beta & 1+\kappa \end{pmatrix}.$$

The eigenvalues μ_1, μ_2 of M solve $\mu^2 - \mathrm{tr}(M)\mu + \det(M) = 0$ with

$$\det M = \frac{1+\kappa}{\beta} - \frac{\kappa}{\beta} = \frac{1}{\beta} > 1, \qquad \mathrm{tr}M = 1 + \frac{1}{\beta} + \kappa.$$

Let $P(\mu) = \mu^2 - \mathrm{tr}(M)\mu + \det M$. Then

$$P(1) = 1 - 1 - \frac{1}{\beta} - \kappa + \frac{1}{\beta} = -\kappa < 0,$$

while $P(0) = 1/\beta > 0$ and $P(\mu) \to +\infty$ as $\mu \to \infty$. Hence one root lies in $(0,1)$ and the other above 1,

$$0 < \mu_1 < 1 < \mu_2.$$

The steady state is a **saddle point** (Appendix A.4.4). Since there is one predetermined variable (k) and one unstable root, there is exactly one convergent path for each k_0 (the Blanchard–Kahn condition). Along it, $\hat{k}_{t+1} = \mu_1 \hat{k}_t$, and the first row of the system gives the slope of the linearized saddle path:

$$\hat{c}_t = \Big(\frac{1}{\beta} - \mu_1\Big)\hat{k}_t, \qquad \frac{1}{\beta} - \mu_1 > 0.$$

Intuition

The speed of convergence is $1 - \mu_1$. A higher σ (lower EIS) lowers κ, which moves μ_1 towards one, so households that dislike uneven consumption paths accumulate capital more slowly. In the example of Figure 4.5, $\kappa \approx 0.0207$, $\mu_1 \approx 0.888$ and $\mu_2 \approx 1.185$. About 11% of the gap to the steady state is closed each period (a half-life of about 5.8 periods), and the slope of the saddle path at the steady state is $1/\beta - \mu_1 \approx 0.165$.

4.8 Dynamic programming

The Lagrangian approach chooses the whole infinite sequence $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ at once. Dynamic programming takes a different route and breaks the infinite-horizon problem into a sequence of identical two-period problems, "today" versus "the rest of the future". We use this approach for the stochastic RBC model, and it is the basis of almost all numerical methods in macroeconomics.

4.8.1 From the sequence problem to the Bellman equation

Step 1: the value function. We define $V(k_0)$ as the *maximal* lifetime utility the planner can obtain when the economy starts with capital k_0:

$$V(k_0) \equiv \max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} = f(k_t) + (1-\delta)k_t \ \ \forall t.$$

The function V depends on the initial capital stock only, because k_0, the **state variable**, is all the planner needs to know about the past. The **control variable** is what the planner chooses each period, consumption c_t, or equivalently next period's capital k_{t+1}.

Step 2: split today from the future. We separate the first term of the sum from the others and factor β out of the remaining ones,

$$\sum_{t=0}^{\infty} \beta^t u(c_t) = u(c_0) + \beta \sum_{t=1}^{\infty} \beta^{t-1} u(c_t).$$

The choice at date 0 affects the future only through the capital stock it leaves, k_1. Whatever k_1 is chosen, it is optimal to continue from date 1 onwards in the best possible way given k_1. This is the **principle of optimality** of Bellman (1957), according to which an optimal plan has the property that, whatever the first decision, the remaining decisions are optimal given the state that the first decision produces. The maximal value of the continuation is therefore

$$\max_{\{c_t, k_{t+1}\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) = V(k_1).$$

The continuation value is the *same* function V because the problem that starts at date 1 with capital k_1 is identical to the one that starts at date 0 with capital k_0, with the same preferences, the same technology and the same infinite horizon ahead. In other words, the problem is *stationary*.

Step 3: the Bellman equation. Combining the two steps, we obtain

$$V(k_0) = \max_{c_0, k_1} \left\{ u(c_0) + \beta V(k_1) \right\} \quad \text{s.t.} \quad c_0 + k_1 = f(k_0) + (1-\delta)k_0.$$

Since nothing in this equation depends on the calendar date, we drop the time subscripts and denote by a prime the value of a variable next period. Substituting the constraint, the **Bellman equation** is

$$V(k) = \max_{0 \le k' \le f(k) + (1-\delta)k} \left\{ u\big(f(k) + (1-\delta)k - k'\big) + \beta V(k') \right\}. \tag{4.10}$$

Reading the Bellman equation

The value of having capital k today equals the best achievable sum of the utility of what is consumed today and the discounted value of the capital left for tomorrow. The function V summarizes, in one number for each k', everything the future has to offer. The trade-off is the familiar one, since a larger k' means less consumption today but a higher

continuation value tomorrow. We note that (4.10) is a *functional equation*, in which the unknown is the whole function V and appears on both sides.

The solution consists of the value function V and the **policy function** $k' = g(k)$, the maximizer in (4.10), or equivalently $c = h(k) \equiv f(k) + (1-\delta)k - g(k)$. Once the policy function is known, the entire optimal path follows from k_0 by iterating $k_{t+1} = g(k_t)$. In the phase diagram, $c = h(k)$ is precisely the saddle path.

4.8.2 FOC, envelope condition and the Euler equation

First-order condition. Differentiating the right-hand side of (4.10) with respect to k' gives

$$u'(c) = \beta V'(k'). \tag{4.11}$$

The left-hand side is the marginal cost of saving one more unit, namely the utility of the consumption given up. The right-hand side is the marginal benefit, the discounted increase in the value of the future from having one more unit of capital.

Envelope condition. Condition (4.11) involves the unknown derivative V'. To get rid of it, we differentiate both sides of (4.10) with respect to the state k, remembering that the optimal choice $k' = g(k)$ also depends on k,

$$V'(k) = u'(c)\big[f'(k) + 1 - \delta\big] + \underbrace{\big[-u'(c) + \beta V'(k')\big]}_{=\,0 \text{ by } (4.11)} g'(k).$$

The indirect effect through the choice vanishes, because at the optimum a marginal change in k' has no first-order effect on the value. This is the **envelope theorem**, and what remains is the **envelope condition**

$$V'(k) = u'(c)\big[f'(k) + 1 - \delta\big]. \tag{4.12}$$

One more unit of capital today yields $f'(k) + 1 - \delta$ more units of the good today, each worth $u'(c)$ when consumed. Comparing with the Lagrangian, we see that

$$V'(k_t) = \lambda_t \left[f'(k_t) + 1 - \delta\right],$$

so the derivative of the value function is the shadow value of capital.

The Euler equation. We shift (4.12) one period forward, $V'(k') = u'(c')[f'(k') + 1 - \delta]$, and substitute it into (4.11) to obtain

$$u'(c) = \beta u'(c')\big[f'(k') + 1 - \delta\big],$$

which is the same Euler equation (4.6) as before. Dynamic programming and the Lagrangian are two routes to the same answer. The Lagrangian delivers the optimality conditions most directly, while dynamic programming also delivers a recipe for computing the policy function and extends naturally to problems with uncertainty (Chapter 5), where the Bellman equation becomes

$$V(k, A) = \max_{k'} \Big\{ u(c) + \beta \, \mathbb{E}\big[V(k', A') \mid A \big] \Big\}.$$

Remark 4.4 (A recipe). To solve a problem by dynamic programming, we (i) identify the state variables (what the decision maker inherits from the past) and the controls (what is chosen today), (ii) write the law of motion of the state, (iii) write the Bellman equation "value today = max of payoff today $+ \beta \times$ value of tomorrow's state", (iv) take the first-order condition for the control, (v) use the envelope condition to eliminate V', and (vi) combine (iv) and (v) to obtain the Euler equation or, if a closed form exists, guess a functional form for V or for the policy function and verify it.

4.8.3 A closed-form solution: Brock and Mirman (1972)

The following example, due to Brock and Mirman (1972), is one of the very few cases in which the Bellman equation can be solved by hand.

Let

$$u(c) = \ln c, \qquad f(k) = A k^{\alpha}, \qquad \delta = 1.$$

Full depreciation is a reasonable approximation when a period is long. The resource constraint becomes $c_t + k_{t+1} = A k_t^{\alpha}$, and the Bellman equation is

$$V(k) = \max_{k'} \Big\{ \ln(A k^{\alpha} - k') + \beta V(k') \Big\}.$$

Guess and verify the value function. We guess $V(k) = E + B \ln k$ with unknown constants E and B. The FOC is

$$\frac{1}{A k^{\alpha} - k'} = \frac{\beta B}{k'} \quad \Longrightarrow \quad k' = \frac{\beta B}{1 + \beta B} A k^{\alpha}, \qquad c = \frac{1}{1 + \beta B} A k^{\alpha}.$$

Substituting back, we obtain

$$E + B \ln k = \ln A + \alpha \ln k - \ln(1 + \beta B) + \beta E + \beta B \Big[\ln \frac{\beta B}{1 + \beta B} + \ln A + \alpha \ln k \Big].$$

Both sides are of the form "constant + coefficient $\times \ln k$", so the guess is consistent. Matching the coefficients on $\ln k$ gives

$$B = \alpha + \alpha \beta B \quad \Longrightarrow \quad B = \frac{\alpha}{1 - \alpha \beta}.$$

Then $\beta B/(1+\beta B) = \alpha\beta$, and the policy functions are

$$k_{t+1} = \alpha\beta\, Ak_t^\alpha, \qquad c_t = (1-\alpha\beta)\, Ak_t^\alpha.$$

Matching the constants gives

$$E = \frac{1}{1-\beta}\Big[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta}\ln(\alpha\beta) + \frac{\ln A}{1-\alpha\beta}\Big].$$

Guess and verify the policy function. Alternatively, we can guess $k_{t+1} = s\, Ak_t^\alpha$ and $c_t = (1-s)Ak_t^\alpha$ and use the Euler equation

$$\frac{1}{c_t} = \beta\frac{1}{c_{t+1}}\alpha Ak_{t+1}^{\alpha-1}.$$

The two sides are

$$\text{RHS} = \frac{\alpha\beta Ak_{t+1}^{\alpha-1}}{(1-s)Ak_{t+1}^\alpha} = \frac{\alpha\beta}{(1-s)k_{t+1}} = \frac{\alpha\beta}{(1-s)s\, Ak_t^\alpha},$$
$$\text{LHS} = \frac{1}{(1-s)Ak_t^\alpha}.$$

They are equal for all k_t if and only if $s = \alpha\beta$.

The TVC. Along the optimal path, the transversality condition holds because

$$\beta^t u'(c_t)k_{t+1} = \beta^t\frac{\alpha\beta y_t}{(1-\alpha\beta)y_t} = \frac{\alpha\beta}{1-\alpha\beta}\beta^t \to 0.$$

Ramsey meets Solow

The optimal policy is to save a constant fraction $s^* = \alpha\beta$ of output. The Brock–Mirman economy therefore *is* a Solow economy with $\delta = 1$, except that the saving rate is now derived. It is higher when the household is more patient (β) and when capital is more productive (α). It does not depend on A, because with log utility the income and substitution effects of a higher return cancel. The steady state is $k^* = (\alpha\beta A)^{1/(1-\alpha)}$, and indeed $f'(k^*) = \alpha A(k^*)^{\alpha-1} = 1/\beta = \rho + \delta$ with $\delta = 1$. Since $\ln k_{t+1} = \ln(\alpha\beta A) + \alpha\ln k_t$, we have $\ln k_t - \ln k^* = \alpha^t(\ln k_0 - \ln k^*)$, and convergence is monotone.

Example 4.2. Let $\alpha = 0.3$, $\beta = 0.95$, $A = 1$ and $k_0 = 0.1k^*$. Then $s^* = 0.285$, $k^* \approx 0.1664$, $y^* \approx 0.5839$ and $c^* \approx 0.4175$, and the transition is as follows.

t	k_t	y_t	c_t	k_{t+1}
0	0.0166	0.2927	0.2093	0.0834
1	0.0834	0.4746	0.3394	0.1353
2	0.1353	0.5487	0.3923	0.1564
3	0.1564	0.5731	0.4098	0.1633
4	0.1633	0.5807	0.4152	0.1655
5	0.1655	0.5830	0.4168	0.1661
∞	0.1664	0.5839	0.4175	0.1664

4.8.4 Beyond the closed form

With $\delta < 1$ or $\sigma \neq 1$, there is no closed form, and we solve the Bellman equation numerically by **value function iteration**. Starting from any V_0, we compute

$$V_{j+1}(k) = \max_{k'} \left\{ u\big(f(k) + (1-\delta)k - k'\big) + \beta V_j(k') \right\}$$

on a grid of capital values and iterate until convergence. Convergence is guaranteed because the Bellman operator is a contraction (Appendix A.3). Alternatively, as in Figure 4.5, we can trace the saddle path by iterating the Euler equation and resource constraint backwards from points close to the steady state on the linearized stable arm.

4.9 Comparative dynamics

Suppose that at date 0, with the economy in its steady state, β falls unexpectedly and permanently (so ρ rises). The new steady state has $f'(k^*_{\text{new}}) = \rho_{\text{new}} + \delta$, so $k^*_{\text{new}} < k^*_{\text{old}}$. The $\Delta k = 0$ locus does not depend on β, while the vertical $\Delta c = 0$ locus $k = k^*$ shifts *left* because k^* falls. Capital cannot jump, so consumption jumps onto the new saddle path. Since the economy must now run down its capital ($k_0 = k^*_{\text{old}} > k^*_{\text{new}}$), consumption must exceed net output, $c_0 > f(k_0) - \delta k_0 = c^*_{\text{old}}$. Consumption *jumps up* on impact and then declines, together with capital, to the new, lower steady state. Impatience raises consumption today at the cost of lower consumption in the long run.

Suppose now that A rises permanently. Both loci shift. The $\Delta k = 0$ locus moves up, because there is more net output at every k, and the vertical $\Delta c = 0$ locus moves right, because k^* rises when f' is higher at every k. In the long run, k^* and c^* are higher. On impact, k is given, and consumption can rise or fall because of two opposing forces.

- The *income (wealth) effect*. The household is permanently richer and wants to consume more now.
- The *substitution effect*. The return to saving is higher, which makes it attractive to postpone consumption and build capital.

If σ is high (low EIS), the household wants a flat path, the wealth effect dominates and c_0 rises. If σ is low (high EIS), c_0 may fall as the household invests heavily to reach the new steady state quickly. In the Brock–Mirman case, $c_0 = (1-\alpha\beta)Ak_0^{\alpha}$ rises in proportion to A.

4.10 Subsistence consumption in the Ramsey model

Let $u(c) = \frac{(c-\bar{c})^{1-\sigma}-1}{1-\sigma}$ with $\bar{c} \geq 0$ (Stone–Geary), and define *effective* consumption $\tilde{c}_t \equiv c_t - \bar{c}$. The Euler equation is $\tilde{c}_t^{-\sigma} = \beta \tilde{c}_{t+1}^{-\sigma} R_{t+1}$, so effective consumption obeys the CRRA law,

$$\tilde{c}_{t+1} = (\beta R_{t+1})^{1/\sigma} \tilde{c}_t,$$

while actual consumption grows at

$$\frac{c_{t+1}}{c_t} = \frac{\tilde{c}_t}{c_t}(\beta R_{t+1})^{1/\sigma} + \frac{\bar{c}}{c_t} \cdot 1,$$

a weighted average of the CRRA growth factor and 1, with weight $\bar{c}/c_t$ on 1. The EIS,

$$\mathrm{EIS}(c) = -\frac{u'(c)}{cu''(c)} = \frac{c-\bar{c}}{\sigma c} = \frac{1}{\sigma}\Big(1 - \frac{\bar{c}}{c}\Big),$$

is increasing in c. It tends to 0 near subsistence and to $1/\sigma$ for the rich. Poor households cannot afford to postpone consumption even when returns are high, so poor economies accumulate capital slowly. The steady state is unchanged, $f'(k^*) = \rho + \delta$, provided that it is feasible, $c^* = f(k^*) - \delta k^* \geq \bar{c}$.

	Solow	**OLG (Diamond)**	**Ramsey**
Saving	exogenous s	chosen, finite life	chosen, infinite horizon
Steady state	$sf(k) = (n+\delta)k$	$(1+n)k = s(w,R)$	$f'(k) = \rho + \delta$
Uniqueness	yes (concave f)	not guaranteed	yes
Dynamic efficiency	not guaranteed	not guaranteed	always
Welfare theorems	—	may fail	hold
Dynamics	monotone convergence	1-D difference equation	saddle path
Long-run growth	exogenous	exogenous	exogenous

Chapter 5

The Real Business Cycle Model

Roadmap

The real business cycle (RBC) model of Kydland and Prescott (1982) and King et al. (1988) is the Ramsey model with two additions. Households choose how much to *work*, and total factor productivity is hit by *random shocks*. The model asks a bold question, namely whether a frictionless, efficient economy driven only by technology shocks can reproduce the main features of business cycles. We set up the model, derive and interpret the intratemporal (consumption–leisure) and intertemporal (Euler) conditions, compute the steady state and calibrate it, log-linearize every equation step by step, solve the linear system by the method of undetermined coefficients, analyze permanent and temporary productivity shocks with simple diagrams, study impulse responses, and evaluate the model's fit to US business-cycle moments. Along the way we meet the Frisch elasticity of labor supply and see how the persistence of shocks shapes the responses.

Business cycles are fluctuations of aggregate variables around their trend. After removing the trend (for instance with the filter of Hodrick and Prescott, 1997), quarterly data for the United States and many other economies show a remarkably stable set of regularities.

1. **Co-movement.** Consumption, investment, hours worked and labor productivity are all *procyclical*, in the sense that they move with output.

2. **Relative volatility.** Consumption is *less* volatile than output, investment is two to three times *more* volatile than output, and hours are about as volatile as output.

3. **Persistence.** All series are highly autocorrelated, so a quarter of high output tends to be followed by another.

4. **Prices.** The real wage and labor productivity are less volatile than output and only weakly correlated with it, while the real interest rate is smooth and roughly acyclical.

Table 5.1 in Section 5.7 documents these facts for the US economy. The RBC model tries to explain these facts within the neoclassical growth model, using the *same* model and parameters that explain long-run growth.

5.1 The model

5.1.1 Technology

A representative competitive firm produces with

$$Y_t = A_t F(K_t, N_t) = A_t K_t^{\alpha} N_t^{1-\alpha}, \qquad 0 < \alpha < 1,$$

where N_t is hours worked and A_t is total factor productivity (TFP). TFP follows a stationary first-order autoregressive process in logs,

$$\ln A_t = \rho_A \ln A_{t-1} + s_A \varepsilon_t, \qquad \varepsilon_t \sim \text{i.i.d. } N(0,1), \quad |\rho_A| < 1.$$

The parameter ρ_A measures the *persistence* of shocks. With $\rho_A = 0.95$, 95% of a shock survives into the next quarter, and half of it is gone after $\ln 0.5 / \ln 0.95 \approx 13.5$ quarters. The firm rents capital at R_t^k and hires labor at w_t, and profit maximization gives

$$w_t = A_t F_N(K_t, N_t) = (1-\alpha)\frac{Y_t}{N_t},$$
$$R_t^k = A_t F_K(K_t, N_t) = \alpha \frac{Y_t}{K_t}.$$

5.1.2 Households

The representative household maximizes expected lifetime utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t, N_t), \qquad u(C, N) = \frac{C^{1-\sigma} - 1}{1-\sigma} - \chi \frac{N^{1+\varphi}}{1+\varphi},$$

with $\sigma > 0$, $\varphi \geq 0$ and $\chi > 0$. The time endowment is normalized to one, so $1 - N_t$ is leisure, and $\mathbb{E}_t$ denotes the expectation conditional on information available at t. The household owns the capital stock and faces

$$C_t + I_t = w_t N_t + R_t^k K_t, \qquad K_{t+1} = (1-\delta)K_t + I_t.$$

Parameters of preferences.

- The parameter $1/\sigma$ is the EIS. With a high σ, households strongly prefer smooth consumption, and with a low σ, consumption reacts more to the interest rate. The case $\sigma = 1$ is log utility.
- The parameter $1/\varphi$ is the **Frisch elasticity** of labor supply, derived below. With a high φ, hours react little to wages, and with a low φ, hours are very elastic.
- The parameter χ scales the disutility of work, and we choose it to match average hours.

5.1.3 The household's optimality conditions

Combining the two constraints gives

$$C_t + K_{t+1} = w_t N_t + (R_t^k + 1 - \delta)K_t.$$

In a stochastic problem, the household chooses a *plan*, that is, a choice for every possible history of shocks. It is convenient to attach a multiplier $\beta^t \lambda_t$ to the period-t budget constraint in every state,

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big\{ u(C_t, N_t) + \lambda_t \big[w_t N_t + (R_t^k + 1 - \delta)K_t - C_t - K_{t+1} \big] \Big\}.$$

The choice of K_{t+1} is made at t, but its payoff arrives at $t+1$ and is uncertain. The FOCs are

$$C_t: \quad C_t^{-\sigma} = \lambda_t, \tag{5.1}$$

$$N_t: \quad \chi N_t^{\varphi} = \lambda_t w_t, \tag{5.2}$$

$$K_{t+1}: \quad \lambda_t = \beta \mathbb{E}_t \Big[\lambda_{t+1} \big(R_{t+1}^k + 1 - \delta \big) \Big]. \tag{5.3}$$

The intratemporal condition. Dividing (5.2) by (5.1), we obtain

$$\frac{\chi N_t^{\varphi}}{C_t^{-\sigma}} = w_t \quad \Longleftrightarrow \quad \chi C_t^{\sigma} N_t^{\varphi} = w_t \tag{5.4}$$

The left-hand side is the marginal rate of substitution between leisure and consumption, that is, the number of units of consumption the household requires to give up one more unit of leisure. The right-hand side is the market rate at which an hour of work is converted into consumption. Working one more hour costs χN^{φ} utils and yields w units of goods worth $wC^{-\sigma}$ utils.

The Euler equation. Substituting (5.1) into (5.3), we obtain

$$C_t^{-\sigma} = \beta \mathbb{E}_t \left[C_{t+1}^{-\sigma} \left(R_{t+1}^k + 1 - \delta \right) \right] \tag{5.5}$$

This is the stochastic version of the Ramsey Euler equation. The marginal utility of consuming today equals the *expected* discounted marginal utility of saving one more unit, earning the uncertain gross return $R_{t+1}^k + 1 - \delta$, and consuming the proceeds tomorrow.

The Frisch elasticity. We write (5.2) as

$$\ln N_t = \frac{1}{\varphi} \left[\ln \lambda_t + \ln w_t - \ln \chi \right].$$

Holding the marginal utility of wealth λ_t constant, we have

$$\left. \frac{\partial \ln N_t}{\partial \ln w_t} \right|_{\lambda \text{ constant}} = \frac{1}{\varphi}.$$

This is the **Frisch elasticity**, the response of hours to a wage change that does not change the household's lifetime wealth (as with a short-lived wage increase). It captures the pure substitution effect over time. A *permanent* wage increase also makes the household richer and lowers λ_t as consumption rises, and this wealth effect pushes hours down. The distinction is the key to understanding why temporary and persistent shocks have different effects on hours.

Transversality. As in Chapter 4, the transversality condition is

$$\lim_{t \to \infty} \mathbb{E}_0 \beta^t \lambda_t K_{t+1} = 0.$$

5.1.4 Competitive equilibrium

Definition 5.1. Given K_0 and the stochastic process for A_t, a competitive equilibrium is a set of state-contingent prices $\{w_t, R_t^k\}$ and allocations $\{C_t, N_t, K_{t+1}, Y_t, I_t\}$ such that households and firms optimize and markets clear, $Y_t = C_t + I_t$.

The equilibrium is characterized by eight equations in eight variables

$(Y, C, I, N, K, w, R^k, A)$,

Euler:	$C_t^{-\sigma} = \beta \mathbb{E}_t\big[C_{t+1}^{-\sigma}(R_{t+1}^k + 1 - \delta)\big]$
Labor supply:	$\chi C_t^{\sigma} N_t^{\varphi} = w_t$
Capital accumulation:	$K_{t+1} = (1-\delta)K_t + I_t$
Production:	$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}$
Labor demand:	$w_t = (1-\alpha) Y_t / N_t$
Capital demand:	$R_t^k = \alpha Y_t / K_t$
Resource constraint:	$Y_t = C_t + I_t$
TFP:	$\ln A_t = \rho_A \ln A_{t-1} + s_A \varepsilon_t$

There are no externalities, distortions or market power, so by the First Welfare Theorem the equilibrium coincides with the solution of a planner who maximizes the household's utility subject to the resource constraint. The planner's FOCs are (5.4) and (5.5) with $w = AF_N$ and $R^k = AF_K$ substituted in.

5.2 A special case with closed-form solution

Before turning to approximations, it is instructive to look at a case that can be solved exactly. Let $\delta = 1$, $u = \ln C + \chi \ln(1 - N)$ (log leisure instead of the power specification), and $Y = AK^{\alpha} N^{1-\alpha}$. We guess

$$C_t = (1 - \alpha\beta) Y_t, \qquad K_{t+1} = \alpha\beta Y_t, \qquad N_t = \overline{N}.$$

Euler equation. With $\delta = 1$, we have $R_{t+1}^k + 1 - \delta = \alpha Y_{t+1} / K_{t+1}$, so that

$$\beta \mathbb{E}_t\Big[\frac{1}{C_{t+1}} \frac{\alpha Y_{t+1}}{K_{t+1}}\Big] = \beta \mathbb{E}_t\Big[\frac{\alpha Y_{t+1}}{(1-\alpha\beta) Y_{t+1}\, \alpha\beta Y_t}\Big] = \frac{1}{(1-\alpha\beta) Y_t} = \frac{1}{C_t}.$$

The randomness of Y_{t+1} cancels, and the guess satisfies the Euler equation state by state.

Intratemporal condition. The condition

$$\frac{\chi}{1 - N_t} C_t = w_t = (1-\alpha)\frac{Y_t}{N_t}$$

gives $\chi(1 - \alpha\beta) N_t = (1-\alpha)(1 - N_t)$, so hours are constant at

$$\overline{N} = \frac{1-\alpha}{(1-\alpha) + \chi(1-\alpha\beta)}.$$

What the exact solution teaches

- **Propagation.** Output satisfies

$$\ln Y_{t+1} = \ln A_{t+1} + \alpha \ln(\alpha\beta) + \alpha \ln Y_t + (1-\alpha)\ln \overline{N}.$$

  Output therefore inherits the persistence of TFP *and* adds its own through capital. Applying $(1-\rho_A L)$, with L the lag operator, to both sides shows that $\ln Y$ follows an AR(2) process with roots α and ρ_A. Capital accumulation is a propagation mechanism, but a weak one when α is about one third.

- **Hours do not move.** With log utility and full depreciation, the income and substitution effects of a productivity shock on labor supply cancel exactly. To make hours fluctuate, the model must break this knife-edge. As we now show, $\delta < 1$ together with persistent but not permanent shocks makes the substitution effect dominate.

5.3 The steady state and calibration

5.3.1 The non-stochastic steady state

We set $\varepsilon_t = 0$ for all t, so that $A = 1$, and look for constant values. We proceed step by step.

Step 1 (rental rate). The Euler equation with $C_{t+1} = C_t$ gives

$$\overline{R}^k = \frac{1}{\beta} - 1 + \delta.$$

The return to capital is pinned down by impatience and depreciation alone (modified golden rule).

Step 2 (capital–labor ratio). From $\overline{R}^k = \alpha(K/N)^{\alpha-1}$, we obtain

$$\frac{K}{N} = \left(\frac{\alpha}{\overline{R}^k}\right)^{\frac{1}{1-\alpha}}.$$

Step 3 (wage and output per hour). The wage is $w = (1-\alpha)(K/N)^{\alpha}$ and output per hour is $Y/N = (K/N)^{\alpha}$.

Step 4 (investment). From $K = (1-\delta)K + I$, we have $I = \delta K$, so $I/N = \delta K/N$.

Step 5 (consumption). The resource constraint gives $C/N = Y/N - \delta K/N$.

Step 6 (hours and χ). The ratios above do not depend on N. We choose the level of hours to match the data, in which people work about one third

of their time endowment, so $N = 1/3$. We then back out the preference parameter from the labor-supply condition,

$$\chi = \frac{w}{C^{\sigma} N^{\varphi}}.$$

Great ratios. The investment–output ratio is

$$\frac{I}{Y} = \frac{\delta K}{Y} = \frac{\delta \alpha}{\overline{R}^{k}} = \frac{\alpha \delta}{1/\beta - 1 + \delta},$$

using $\overline{R}^{k} = \alpha Y/K$. It depends only on α, β and δ.

5.3.2 Calibration

Calibration chooses parameters so that the model's steady state matches long-run averages in the data. For a quarterly model, we use the following values.

Parameter	Target / source	Value
α	capital share of income	$1/3$
β	average annual real return $\approx 4\%$	0.99
δ	annual depreciation $\approx 10\%$	0.025
σ	log utility (balanced growth)	1
φ	Frisch elasticity of 1	1
N	one third of time worked	$1/3$ ($\Rightarrow \chi \approx 7.87$)
ρ_A	persistence of measured TFP	0.95

The implied steady state is $\overline{R}^{k} \approx 0.0351$, $K/N \approx 29.26$, $w \approx 2.054$, $Y \approx 1.027$, $K \approx 9.755$, $I \approx 0.244$ and $C \approx 0.783$. These values give $I/Y \approx 0.237$, $C/Y \approx 0.763$ and $K/Y \approx 9.50$ in quarterly terms (about 2.4 in annual terms), all of which are close to the data.

5.4 Log-linearization

5.4.1 Uhlig's method

The equilibrium conditions are nonlinear stochastic difference equations without a closed-form solution. We approximate them by linear equations in *log-deviations from the steady state*,

$$\widetilde{x}_t \equiv \ln x_t - \ln \overline{x} \approx \frac{x_t - \overline{x}}{\overline{x}},$$

the percentage deviation of x_t from its steady state. A convenient method (Uhlig, 1999) substitutes $x_t = \overline{X} e^{\widetilde{x}_t}$ and uses the first-order approximations

$$e^{\widetilde{x}_t} \approx 1 + \widetilde{x}_t, \qquad e^{\widetilde{x}_t + a\widetilde{Y}_t} \approx 1 + \widetilde{x}_t + a\widetilde{Y}_t, \qquad \widetilde{x}_t \widetilde{Y}_t \approx 0.$$

A useful shortcut is the following. For a *multiplicative* equation, we simply take logs and subtract the steady-state version. For an *additive* equation, we weight each term by its steady-state level.

5.4.2 Linearization

Production. Taking logs gives $\ln Y_t = \ln A_t + \alpha \ln K_t + (1-\alpha)\ln N_t$, and subtracting the steady state ($\ln \overline{A} = 0$) yields

$$\widetilde{Y}_t = \widetilde{A}_t + \alpha \widetilde{K}_t + (1-\alpha)\widetilde{N}_t.$$

Factor demands. From $w_t = (1-\alpha)Y_t/N_t$ and $R^k_t = \alpha Y_t/K_t$, we obtain

$$\widetilde{w}_t = \widetilde{Y}_t - \widetilde{N}_t, \qquad \widetilde{R}^k_t = \widetilde{Y}_t - \widetilde{K}_t.$$

Labor supply. The log of the labor-supply condition, $\ln \chi + \sigma \ln C_t + \varphi \ln N_t = \ln w_t$, gives

$$\sigma \widetilde{C}_t + \varphi \widetilde{N}_t = \widetilde{w}_t.$$

TFP. Since $\widetilde{A}_t = \ln A_t$, we have $\widetilde{A}_t = \rho_A \widetilde{A}_{t-1} + s_A \varepsilon_t$.

Resource constraint. We write $\overline{Y} e^{\widetilde{Y}_t} = \overline{C} e^{\widetilde{C}_t} + \overline{I} e^{\widetilde{I}_t}$. Using $e^x \approx 1 + x$ and $\overline{Y} = \overline{C} + \overline{I}$, we obtain

$$\overline{Y}\widetilde{Y}_t = \overline{C}\widetilde{C}_t + \overline{I}\widetilde{I}_t.$$

Capital accumulation. We write $\overline{K} e^{\widetilde{K}_{t+1}} = (1-\delta)\overline{K} e^{\widetilde{K}_t} + \overline{I} e^{\widetilde{I}_t}$. Linearizing and using $\overline{I} = \delta \overline{K}$, we obtain

$$\overline{K}\widetilde{K}_{t+1} = (1-\delta)\overline{K}\widetilde{K}_t + \delta \overline{K}\widetilde{I}_t \quad \Longrightarrow \quad \widetilde{K}_{t+1} = (1-\delta)\widetilde{K}_t + \delta \widetilde{I}_t.$$

Euler equation. We write $C_t^{-\sigma} = \beta \mathbb{E}_t[C_{t+1}^{-\sigma}(R^k_{t+1} + 1 - \delta)]$ as

$$\overline{C}^{-\sigma} e^{-\sigma \widetilde{C}_t} = \beta \mathbb{E}_t \Big[\overline{C}^{-\sigma} e^{-\sigma \widetilde{C}_{t+1}} \big(\overline{R}^k e^{\widetilde{R}^k_{t+1}} + 1 - \delta \big) \Big].$$

Dividing by $\overline{C}^{-\sigma}$ and linearizing the right-hand side, we have

$$\begin{aligned} 1 - \sigma \widetilde{C}_t &\approx \beta \mathbb{E}_t \Big[(1 - \sigma \widetilde{C}_{t+1}) \big(\overline{R}^k + 1 - \delta + \overline{R}^k \widetilde{R}^k_{t+1} \big) \Big] \\ &\approx \beta (\overline{R}^k + 1 - \delta) \big(1 - \sigma \mathbb{E}_t \widetilde{C}_{t+1} \big) + \beta \overline{R}^k \, \mathbb{E}_t \widetilde{R}^k_{t+1}, \end{aligned}$$

where the second line drops the second-order term $\widetilde{C}_{t+1}\widetilde{R}^k_{t+1}$. Since $\beta(\overline{R}^k + 1 - \delta) = 1$ in the steady state, we obtain

$$\sigma \, \mathbb{E}_t \big(\widetilde{C}_{t+1} - \widetilde{C}_t \big) = \beta \overline{R}^k \, \mathbb{E}_t \widetilde{R}^k_{t+1}.$$

Expected consumption growth is proportional to the expected deviation of the return on capital from its steady state, with coefficient $\beta\overline{R}^k/\sigma$, so the EIS $1/\sigma$ again governs the strength of the response.

5.4.3 The linear system

Collecting the linearized equations, we obtain the system

$$\begin{aligned}
&\sigma\mathbb{E}_t(\widetilde{C}_{t+1}-\widetilde{C}_t)=\beta\overline{R}^k\,\mathbb{E}_t\widetilde{R}^k_{t+1}, && \sigma\widetilde{C}_t+\varphi\widetilde{N}_t=\widetilde{w}_t,\\
&\widetilde{K}_{t+1}=(1-\delta)\widetilde{K}_t+\delta\widetilde{I}_t, && \widetilde{Y}_t=\widetilde{A}_t+\alpha\widetilde{K}_t+(1-\alpha)\widetilde{N}_t,\\
&\widetilde{R}^k_t=\widetilde{Y}_t-\widetilde{K}_t, && \widetilde{w}_t=\widetilde{Y}_t-\widetilde{N}_t,\\
&\overline{Y}\widetilde{Y}_t=\overline{C}\widetilde{C}_t+\overline{I}\widetilde{I}_t, && \widetilde{A}_t=\rho_A\widetilde{A}_{t-1}+s_A\varepsilon_t.
\end{aligned}$$

This is a system of eight linear equations in eight log-deviations. The variables are of three types, namely the *endogenous state* $\widetilde{K}_t$ (predetermined), the *exogenous state* $\widetilde{A}_t$, and *jump* variables $\widetilde{Y},\widetilde{C},\widetilde{N},\widetilde{w},\widetilde{R}^k,\widetilde{I}$.

5.5 Solving the linear model

5.5.1 Reducing the system

Hours. Combining labor supply $\widetilde{w}=\sigma\widetilde{C}+\varphi\widetilde{N}$ with labor demand $\widetilde{w}=\widetilde{Y}-\widetilde{N}$, we find

$$\widetilde{N}_t=\frac{\widetilde{Y}_t-\sigma\widetilde{C}_t}{1+\varphi}.$$

Hours rise with output (the wage is higher) and fall with consumption (the wealth effect).

Output. Substituting into the production function and solving for $\widetilde{Y}_t$, we obtain

$$\widetilde{Y}_t\Big[1-\frac{1-\alpha}{1+\varphi}\Big]=\widetilde{A}_t+\alpha\widetilde{K}_t-\frac{(1-\alpha)\sigma}{1+\varphi}\widetilde{C}_t,$$

and since $1-\frac{1-\alpha}{1+\varphi}=\frac{\alpha+\varphi}{1+\varphi}$,

$$\widetilde{Y}_t=\psi_A\widetilde{A}_t+\psi_K\widetilde{K}_t-\psi_C\widetilde{C}_t,$$

$$\psi_A=\frac{1+\varphi}{\alpha+\varphi},\qquad \psi_K=\frac{\alpha(1+\varphi)}{\alpha+\varphi},\qquad \psi_C=\frac{(1-\alpha)\sigma}{\alpha+\varphi}.$$

We note that $\psi_A>1$. Because hours respond, output reacts more than one-for-one to TFP, so the labor market *amplifies* shocks.

Capital. The resource constraint gives $\widetilde{I}_t = (\overline{Y}\widetilde{Y}_t - \overline{C}\widetilde{C}_t)/\overline{I}$. Using $\delta/\overline{I} = 1/\overline{K}$, we have

$$\begin{aligned}\widetilde{K}_{t+1} &= \Big(1 - \delta + \frac{\overline{Y}}{\overline{K}}\psi_K\Big)\widetilde{K}_t + \frac{\overline{Y}}{\overline{K}}\psi_A\widetilde{A}_t - \Big(\frac{\overline{Y}}{\overline{K}}\psi_C + \frac{\overline{C}}{\overline{K}}\Big)\widetilde{C}_t \\ &\equiv \gamma_K\widetilde{K}_t + \gamma_A\widetilde{A}_t - \gamma_C\widetilde{C}_t.\end{aligned}$$

Euler. Using $\widetilde{R}^k_{t+1} = \widetilde{Y}_{t+1} - \widetilde{K}_{t+1}$ and the expression for output, we obtain

$$\sigma\mathbb{E}_t(\widetilde{C}_{t+1} - \widetilde{C}_t) = \beta\overline{R}^k\,\mathbb{E}_t\Big[\psi_A\widetilde{A}_{t+1} + (\psi_K - 1)\widetilde{K}_{t+1} - \psi_C\widetilde{C}_{t+1}\Big],$$

with $\mathbb{E}_t\widetilde{A}_{t+1} = \rho_A\widetilde{A}_t$.

We now have a system of two difference equations in $(\widetilde{K}_t, \widetilde{C}_t)$ driven by $\widetilde{A}_t$, which is the stochastic analogue of the Ramsey phase diagram. As in Section 4.7, the system has one stable root, one unstable root and one predetermined variable. The *Blanchard–Kahn condition* is therefore satisfied, and there is a unique stable solution.

5.5.2 The method of undetermined coefficients

We guess that the solution is linear in the state variables,

$$\widetilde{C}_t = \nu_{CK}\widetilde{K}_t + \nu_{CA}\widetilde{A}_t, \qquad \widetilde{K}_{t+1} = \nu_{KK}\widetilde{K}_t + \nu_{KA}\widetilde{A}_t.$$

Substituting the first guess into the capital equation gives $\nu_{KK} = \gamma_K - \gamma_C\nu_{CK}$ and $\nu_{KA} = \gamma_A - \gamma_C\nu_{CA}$. Substituting both guesses into the Euler equation and requiring it to hold for *all* values of $\widetilde{K}_t$ and $\widetilde{A}_t$ gives two equations.

- Matching the coefficients on $\widetilde{K}_t$ gives a *quadratic* equation in ν_{CK}:

  $$\sigma\nu_{CK}(\nu_{KK} - 1) = \beta\overline{R}^k\big[(\psi_K - 1) - \psi_C\nu_{CK}\big]\nu_{KK}, \qquad \nu_{KK} = \gamma_K - \gamma_C\nu_{CK}.$$

  One root implies $|\nu_{KK}| < 1$ (stable), the other $|\nu_{KK}| > 1$ (explosive, violating the TVC). We keep the stable one.

- Matching the coefficients on $\widetilde{A}_t$ gives a *linear* equation in ν_{CA} (given ν_{CK}):

  $$\begin{aligned}&\sigma\big[\nu_{CK}\nu_{KA} + (\rho_A - 1)\nu_{CA}\big] \\ &\qquad = \beta\overline{R}^k\Big[\psi_A\rho_A + (\psi_K - 1)\nu_{KA} - \psi_C\big(\nu_{CK}\nu_{KA} + \rho_A\nu_{CA}\big)\Big].\end{aligned}$$

With the calibration above ($\alpha = 1/3$, $\beta = 0.99$, $\delta = 0.025$, $\sigma = \varphi = 1$, $\rho_A = 0.95$), we find $\psi_A = 1.5$, $\psi_K = \psi_C = 0.5$ and

$$\nu_{CK} \approx 0.557, \qquad \nu_{CA} \approx 0.384, \qquad \nu_{KK} \approx 0.954, \qquad \nu_{KA} \approx 0.107.$$

The other root of the quadratic, $\nu_{CK} \approx -0.237$, gives $\nu_{KK} \approx 1.059 > 1$ and is discarded. All other variables follow from the static equations. In matrix form, the general method writes the system as

$$x_t = P x_{t-1} + Q z_t, \qquad y_t = R x_{t-1} + S z_t$$

for states x, shocks z and jump variables y, and the Blanchard–Kahn method obtains the same solution from the eigen-decomposition of the system matrix.

5.6 Impulse responses

With the system solved, we can simulate how the economy responds to a shock. We hit the economy with a one-time shock ε_0 that raises TFP by 1% and trace how the other variables react over time (Figure 5.1).

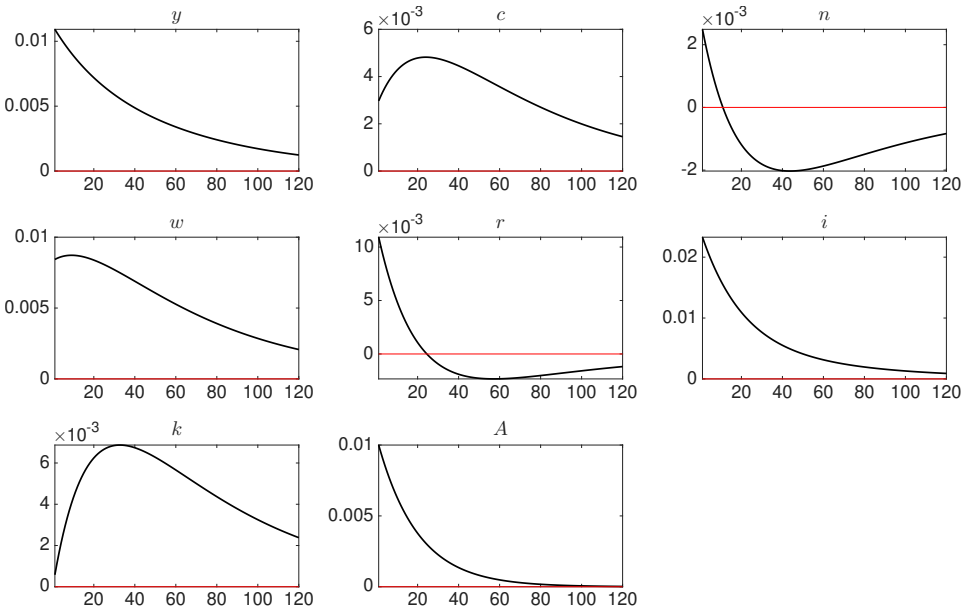


Figure 5.1: Impulse responses to a one-time 1% TFP shock in the baseline RBC model.

- **Output** rises on impact by about 1.31%, more than the 1% rise in TFP, because hours rise too (amplification).
- **Hours** rise because the wage is temporarily high, which makes working now and enjoying leisure later attractive. This is intertemporal substitution, governed by the Frisch elasticity. Later, when TFP has largely returned to normal but the household is richer (higher capital), the wealth effect pushes hours slightly *below* the steady state.

- **Consumption** rises by much less than output and keeps rising for a while, because the household smooths consumption, as the permanent-income logic of the Euler equation dictates.
- **Investment** absorbs most of the extra output and rises by about 4.3%. It is the variable through which the household saves the temporary windfall. This is why investment is so much more volatile than consumption.
- **Capital** builds up slowly and peaks about 20 quarters after the shock. Through capital, the shock *propagates* and keeps output above trend after TFP has largely reverted.
- The **wage** rises with labor productivity, and the **rental rate** jumps on impact (TFP is high and capital is fixed) and then falls below its steady state as capital accumulates.

5.7 Goodness of fit: model versus data

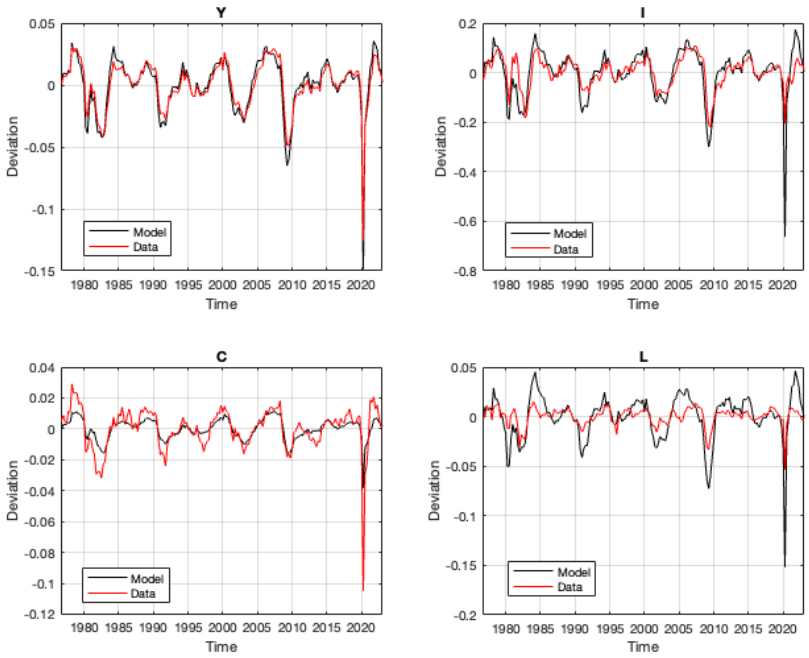


Figure 5.2: The RBC model and US macroeconomic data, 1975–2022.

5.7.1 How the RBC model is evaluated

The calibration strategy of the RBC literature has a particular logic. The parameters are chosen to match *long-run* features of the data, such as average factor shares, the average real interest rate, the average investment–output ratio, average hours. The same model, with the same parameters, is then used to explain *short-run* fluctuations, which played no role in choosing the parameters. Loosely speaking, long-run averages "identify" the parameters, while business-cycle moments are "over-identifying restrictions" that can be used to test the model. The test is informal. We simulate the model, filter the simulated series exactly as the data, and compare a small set of second moments, namely standard deviations, correlations with output and autocorrelations.

The data. Table 5.1 reports these moments for the US economy, computed by Sims (2024) from quarterly data over 1947Q1–2022Q3. All series are in logs (except the real interest rate) and detrended with the Hodrick and Prescott (1997) filter with smoothing parameter 1600.

The model. The model moments in the same table are taken from the same source and come from the basic RBC model of this chapter with log utility in consumption and leisure, $u = \ln C + \theta \ln(1 - N)$, which differs from our baseline only in the form of the disutility of work. The parameters are $\alpha = 1/3$, $\beta = 0.99$, $\delta = 0.02$ (to match an investment–output ratio of about 22%), θ set to match $\overline{N} = 1/3$, and a TFP process estimated on detrended measured TFP, $\rho_A = 0.979$ and $s_A = 0.009$. We note that with $\overline{N} = 1/3$ these preferences imply a Frisch elasticity of $(1 - \overline{N})/\overline{N} = 2$ (see the exercises).

Table 5.1: HP-filtered business-cycle moments in US data (1947Q1–2022Q3) and in the basic RBC model. Source, Sims (2024).

	US data				RBC model			
Series	s.d.	rel. s.d.	corr$(\cdot, Y)$	autocorr.	s.d.	rel. s.d.	corr$(\cdot, Y)$	autocorr.
Output Y	0.0166	1.00	1.00	0.79	0.0164	1.00	1.00	0.72
Consumption C	0.0110	0.66	0.76	0.67	0.0063	0.38	0.95	0.77
Investment I	0.0444	2.67	0.76	0.86	0.0533	3.25	0.99	0.71
Hours N	0.0211	1.27	0.87	0.82	0.0070	0.43	0.98	0.71
Labor productivity Y/N	0.0114	0.69	0.27	0.70	0.0096	0.58	0.99	0.74
Real wage w	0.0115	0.69	−0.01	0.71	0.0096	0.58	0.99	0.74
Real interest rate r	0.0043	0.26	0.00	0.47	0.0005	0.03	0.97	0.71
TFP A	0.0127	0.76	0.78	0.76	0.0117	0.71	1.00	0.72

5.7.2 What the model gets right

Given that the model has a single shock and was calibrated to long-run averages only, its performance is remarkable.

- **Output volatility.** The model generates a standard deviation of output of 1.64%, almost exactly the 1.66% in the data. Measured TFP is the only driving force, so this says that technology shocks of the measured size *can* account for the amplitude of the business cycle.
- **Relative volatilities of consumption and investment.** Consumption is less volatile than output and investment two to three times more volatile, as in the data. This is the consumption-smoothing logic of the Euler equation, since a persistent but not permanent rise in productivity is a temporary windfall, most of which is saved and invested.
- **TFP, productivity and wages.** The relative volatilities of TFP, labor productivity and the real wage are close to those in the data, and the real interest rate is correctly much less volatile than output.
- **Persistence.** The autocorrelations of all series are close to those in the data (around 0.7).
- **Co-movement.** Consumption, investment, hours and TFP are strongly procyclical, as in the data.

5.7.3 What the model gets wrong

- **Hours are not volatile enough.** In the data, hours are more volatile than output (relative s.d. 1.27), whereas in the model they are less than half as volatile as output (0.43), despite a Frisch elasticity of 2, already well above most microeconomic estimates. Lowering the Frisch elasticity to more plausible values makes the problem worse. Proposed remedies include indivisible labor (Hansen, 1985), in which individuals work either full time or not at all, which makes the *aggregate* labor supply much more elastic than individual labor supply.
- **Correlations with output are too high.** With a single shock, every variable is almost perfectly correlated with output (correlations between 0.95 and 1). In the data, consumption and investment are less tightly linked to output (0.76), labor productivity is only weakly procyclical (0.27) and the real wage and the real interest rate are acyclical. This failure is less damaging than it looks, because adding a second source of fluctuations that moves output and productivity in opposite directions (for instance, shocks to labor supply or to government spending) lowers these correlations.

- **Prices.** The model's real interest rate is far too smooth (0.03 against 0.26) and strongly procyclical, whereas it is acyclical in the data. Wages and productivity are perfectly proportional in the model ($\widetilde{w} = \widetilde{Y} - \widetilde{N}$), so their moments are identical, which is not the case in the data.
- **Weak internal propagation.** The persistence of output in the model comes almost entirely from the assumed persistence of TFP. As the exact solution of this chapter showed, capital accumulation adds only an autoregressive root of about α, and with an i.i.d. TFP process the model generates almost no persistence of its own.
- **Measurement of TFP.** The Solow residual used to calibrate the shocks mixes true technology with procyclical variations in factor utilization and labor hoarding. Part of measured TFP volatility is therefore an endogenous response to other shocks, not an exogenous driving force.

The fit has deteriorated over time. The moments have changed considerably since the RBC model was proposed in the early 1980s. In data ending in 1983, hours and output are about equally volatile, labor productivity is clearly procyclical (correlation 0.57) and the real interest rate is mildly countercyclical. After the mid-1980s, during the "Great Moderation", output became less volatile, hours became much more volatile relative to output, and labor productivity and the real wage became *countercyclical* (Sims, 2024). The basic RBC model fits the earlier sample considerably better than the later one.

From RBC to DSGE. Despite these limitations, the RBC model is the backbone of modern dynamic stochastic general-equilibrium (DSGE) models (King and Rebelo, 1999). New Keynesian models add nominal rigidities and monetary policy, and other extensions add more shocks (government spending, investment-specific technology, preferences), labor-market search frictions, financial frictions, habit formation and variable capital utilization. The methodology, namely optimizing agents, general equilibrium, log-linearization, calibration or estimation, and impulse responses, is the same.

Chapter 6

Endogenous Growth

In the Solow and Ramsey models, diminishing returns to capital eventually put a stop to accumulation. As we have seen, the marginal product of capital falls until it equals $n + \delta$ (Solow) or $\rho + \delta$ (Ramsey), and long-run growth of income per worker then requires exogenous technical progress. To make growth endogenous, the model needs a mechanism that allows the economy to escape diminishing returns.

6.1 The AK Model

To remove diminishing returns, we let the production function be

$$Y_t = AK_t,$$

where K is physical capital. A representative household with CRRA utility solves

$$\max \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma} - 1}{1-\sigma}$$

subject to

$$C_t + K_{t+1} = AK_t + (1-\delta)K_t.$$

Let $R \equiv 1 + A - \delta$ be the gross return on capital. The Euler equation is $C_t^{-\sigma} = \beta R C_{t+1}^{-\sigma}$, so consumption grows at a *constant* gross rate

$$\gamma \equiv \frac{C_{t+1}}{C_t} = \left[\beta(1 + A - \delta)\right]^{1/\sigma}.$$

Because the marginal product of capital is the constant A, the return never falls and neither does consumption growth. To pin down the level of consumption, we iterate the resource constraint forward and obtain

$$K_0 = \sum_{t=0}^{T} \frac{C_t}{(1 + A - \delta)^{t+1}} + \frac{K_{T+1}}{(1 + A - \delta)^{T+1}}.$$

The TVC implies that the last term vanishes as $T \to \infty$, so that

$$(1 + A - \delta)K_0 = \sum_{t=0}^{\infty} \frac{C_t}{(1 + A - \delta)^t}$$
$$= C_0 \sum_{t=0}^{\infty} \Big(\frac{\gamma}{(1 + A - \delta)} \Big)^t = \frac{C_0}{1 - \gamma/(1 + A - \delta)},$$

and therefore

$$C_0 = \big[(1 + A - \delta) - \gamma\big] K_0.$$

This requires $\gamma < 1 + A - \delta$, or equivalently $\beta(1 + A - \delta)^{1-\sigma} < 1$, which ensures that utility is bounded. Then

$$K_{t+1} = (1 + A - \delta)K_t - \big[(1 + A - \delta) - \gamma\big] K_t = \gamma K_t,$$

so capital, output and consumption all grow at the same rate $\gamma - 1$ from date 0 on. The saving rate is constant and equal to

$$s = \frac{Y - C}{Y} = \frac{\gamma - 1 + \delta}{A}.$$

Intuition

The growth rate depends on preferences (β, σ) and technology (A, δ). The AK model delivers perpetual endogenous growth, but it is silent about *where* non-diminishing returns come from. The models that follow address this question. They rest on the idea that knowledge is *non-rival*, since one person's use does not prevent another's, so a larger stock of ideas can raise productivity without bound. Ideas are, however, produced by people who must be paid, which requires some degree of *market power*.

6.2 The Romer Model

In the model of Romer (1990), growth is driven by purposeful research that creates new varieties of intermediate goods. This process is called *horizontal innovation*, because research expands the range of available inputs rather than improving the quality of an existing input. Each invention takes the form of a blueprint (or patent) that describes how to produce a particular intermediate good. A perpetual patent grants its owner the exclusive right to produce that variety. The resulting monopoly profits provide the private reward for invention.

It is useful to distinguish the stock of inventions from their market value. The variable N_t measures the number, or more precisely the measure, of varieties available for production in period t. By contrast, v_t is the

market value of the patent on one variety. In other words, N_t measures the range of available technologies, whereas v_t measures the financial value of owning one of them.

The economy consists of households, a competitive final-good sector, monopolistic producers of differentiated intermediate goods, and a research sector with free entry. Population is constant and equal to L. Labor can move freely between final-good production and research, so both activities pay the same wage. The final good is the numeraire. It can be consumed or invested in physical capital.

The timing of events is the following. Production in period t uses the N_t varieties already available. Research undertaken during that period creates $N_{t+1} - N_t$ additional varieties, which become available for production in period $t + 1$. Their patents are sold at date t for v_t each and begin paying profits in period $t + 1$.

6.2.1 Households

The representative household supplies labor inelastically and chooses consumption and saving to solve

$$\max_{\{C_t, F_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln C_t, \qquad \beta = \frac{1}{1+\rho}, \qquad \rho > 0.$$

A larger β, or equivalently a smaller ρ, means that the household places greater weight on future consumption.

The budget constraint is

$$F_{t+1} + C_t = (1 + r_t)F_t + W_t L,$$

where F_t denotes the amount invested at date $t-1$ and carried into period t, while F_{t+1} is the amount invested at date t. The net return earned between dates $t-1$ and t is r_t. Financial wealth includes ownership of physical capital and ownership claims on patented intermediate-good firms.

The Euler equation is

$$\frac{C_{t+1}}{C_t} = \beta(1 + r_{t+1}) = \frac{1 + r_{t+1}}{1 + \rho}.$$

Saving sacrifices one unit of current consumption in exchange for $1 + r_{t+1}$ units of future consumption. The household chooses saving so that the utility cost of this sacrifice equals its discounted utility benefit.

On a balanced growth path (BGP), consumption grows at a constant rate g and the interest rate is constant, so that

$$\frac{C_{t+1}}{C_t} = 1 + g.$$

Consequently,

$$1 + r = (1 + \rho)(1 + g) \qquad \Longleftrightarrow \qquad r = \rho + (1 + \rho)g,$$

which for small growth and discount rates is approximately $r = \rho + g$.

6.2.2 The final-good sector

A competitive firm produces the final good using labor and a continuum of differentiated intermediate inputs,

$$Y_t = L_{Y,t}^{1-\alpha} \int_0^{N_t} x_t(i)^\alpha \, di, \qquad 0 < \alpha < 1.$$

The index i identifies a particular variety, and $x_t(i)$ is the quantity of that variety used in production. Because varieties form a continuum, N_t need not be an integer.

The firm solves

$$\max_{L_{Y,t}, \{x_t(i)\}} \left\{ Y_t - W_t L_{Y,t} - \int_0^{N_t} P_t(i) x_t(i) \, di \right\}.$$

Its first-order conditions are

$$\begin{aligned} W_t &= (1 - \alpha) \frac{Y_t}{L_{Y,t}}, \\ P_t(i) &= \alpha L_{Y,t}^{1-\alpha} x_t(i)^{\alpha - 1}. \end{aligned} \tag{6.1}$$

Equation (6.1) is the inverse demand curve facing the producer of variety i, which we can equivalently write as

$$x_t(i) = L_{Y,t} \left(\frac{\alpha}{P_t(i)} \right)^{1/(1-\alpha)}.$$

In a symmetric allocation, every available variety is used in the same quantity, $x_t(i) = x_t$, and output is

$$Y_t = L_{Y,t}^{1-\alpha} N_t x_t^\alpha.$$

As we show below, one unit of capital supplies one unit of intermediate input services, so aggregate capital satisfies

$$K_t = \int_0^{N_t} K_t(i) \, di = N_t x_t.$$

Substituting $x_t = K_t / N_t$ gives

$$Y_t = L_{Y,t}^{1-\alpha} N_t^{1-\alpha} K_t^\alpha.$$

6.2.3 The intermediate-goods sector

Each variety is supplied by a single firm that owns its patent. The firm rents physical capital and converts its services one-for-one into intermediate input services,

$$x_t(i) = K_t(i).$$

This formulation treats $x_t(i)$ as the flow of services supplied by capital assigned to variety i. Physical capital depreciates at rate δ.

Let R_t^k be the rental payment per unit of capital. Since one unit of capital also leaves $1-\delta$ units of undepreciated capital, its gross return is $R_t^k + 1 - \delta$. No-arbitrage with other assets requires

$$R_t^k + 1 - \delta = 1 + r_t,$$

so

$$R_t^k = r_t + \delta.$$

Taking R_t^k and $L_{Y,t}$ as given, the producer of variety i solves

$$\begin{aligned}\max_{x_t(i)} \pi_t(i) &= P_t(i)x_t(i) - R_t^k x_t(i) \\ &= \alpha L_{Y,t}^{1-\alpha} x_t(i)^\alpha - R_t^k x_t(i).\end{aligned}$$

The first-order condition is

$$\alpha^2 L_{Y,t}^{1-\alpha} x_t(i)^{\alpha-1} = R_t^k.$$

Combining this condition with (6.1) yields

$$P_t(i) = \frac{R_t^k}{\alpha}.$$

The price is therefore a constant multiple of marginal cost. The *gross markup factor* is $1/\alpha$, while the markup as a fraction of marginal cost is $(1-\alpha)/\alpha$.

All varieties face the same demand and production costs, so that in a symmetric equilibrium

$$x_t = L_{Y,t}\left(\frac{\alpha^2}{R_t^k}\right)^{1/(1-\alpha)}.$$

The operating profit of each firm is

$$\pi_t = (P_t - R_t^k)x_t = (1-\alpha)P_t x_t = \alpha(1-\alpha)L_{Y,t}^{1-\alpha} x_t^\alpha = \alpha(1-\alpha)\frac{Y_t}{N_t}. \quad (6.2)$$

Aggregate monopoly profits are therefore

$$N_t \pi_t = \alpha(1-\alpha)Y_t.$$

These profits are measured after paying for capital services. They are the income attributable to the exclusive production right, rather than the return on physical capital itself. We assume that the firm distributes all operating profits to its shareholders, so its dividend in period t equals π_t.

An entrant must acquire the patent before producing a new variety. It can finance this purchase by issuing shares to households. Because the firm rents its physical capital, the value of its equity is the value of its patent. Shares in the firm and ownership of the patent are alternative ways of describing the same claim to future profits and should not be counted as separate assets.

6.2.4 The R&D sector

Research employs $L_{R,t}$ workers to develop new blueprints according to

$$N_{t+1} - N_t = \theta N_t L_{R,t}, \qquad \theta > 0.$$

The left-hand side measures the flow of new varieties created during period t. Dividing by the existing stock gives the growth rate of varieties,

$$g_{N,t} \equiv \frac{N_{t+1} - N_t}{N_t} = \theta L_{R,t}. \tag{6.3}$$

Each researcher produces θN_t new blueprints. Research becomes more productive as the stock of existing knowledge grows, because past discoveries provide ideas and methods on which subsequent research can build. An individual research firm takes N_t as given and does not pay previous inventors for this contribution to its research productivity. This is an *intertemporal knowledge spillover*.

Patent protection and knowledge spillovers concern different rights. A patent grants the exclusive right to produce a particular variety, while the knowledge embodied in previous inventions can still help researchers develop other varieties. The model assumes that this knowledge does not become obsolete and that its contribution to research productivity is proportional to N_t.

Let v_t denote the date-t market price of the exclusive right to produce one variety and receive its future monopoly profits. This is an asset price, measured in units of the final good per patent. For an existing patent, v_t is its value after the period-t dividend has been paid. A patent created during period t has the same value because it first becomes productive in period $t + 1$. Under symmetry, patents have identical future profit streams regardless of when they were invented.

The patent is valuable because its owner can either retain it and collect future dividends or sell it to another investor. It does not itself supply

physical capital. Its owner must still rent the capital needed for intermediate production, and that cost has already been deducted in the calculation of π_t.

An investor who purchases a patent for v_t at date t receives the dividend π_{t+1} in period $t+1$ and then owns a patent worth v_{t+1}. The gross return on this investment is therefore

$$\frac{\pi_{t+1} + v_{t+1}}{v_t}.$$

In this deterministic economy, this return must equal the gross return on other assets, which gives the no-arbitrage condition

$$1 + r_{t+1} = \frac{\pi_{t+1} + v_{t+1}}{v_t}.$$

Equivalently,

$$r_{t+1} = \underbrace{\frac{\pi_{t+1}}{v_t}}_{\text{dividend yield}} + \underbrace{\frac{v_{t+1} - v_t}{v_t}}_{\text{capital gain or loss}}.$$

The dividend is the cash income distributed by the firm, and the capital gain is the increase in the market value of the patent, so a fall in v_{t+1} is a capital loss.

A patent can therefore provide a positive return even when its price does not rise. If $v_{t+1} = v_t$, the entire return consists of dividends. Conversely, a patent whose price is expected to rise can offer the required return with a lower dividend yield.

6.2.5 The present value of monopoly profits

Rearranging the no-arbitrage condition gives

$$v_t = \frac{\pi_{t+1} + v_{t+1}}{1 + r_{t+1}}.$$

Iterating forward for T periods yields

$$v_t = \sum_{j=1}^{T} \frac{\pi_{t+j}}{\prod_{s=1}^{j}(1 + r_{t+s})} + \frac{v_{t+T}}{\prod_{s=1}^{T}(1 + r_{t+s})}.$$

We impose the no-bubble condition

$$\lim_{T \to \infty} \frac{v_{t+T}}{\prod_{s=1}^{T}(1 + r_{t+s})} = 0,$$

so that the patent price equals the discounted value of all future dividends,

$$v_t = \sum_{j=1}^{\infty} \frac{\pi_{t+j}}{\prod_{s=1}^{j}(1 + r_{t+s})}.$$

If the interest rate is constant, this becomes

$$v_t = \sum_{j=1}^{\infty} \frac{\pi_{t+j}}{(1+r)^j}.$$

The resale value in the one-period return equation is therefore consistent with valuing the patent entirely from its dividends, because the next buyer pays for the dividends that remain after the sale. The no-bubble condition rules out a price component sustained solely by the prospect of reselling the patent at a higher price forever.

Patent values on a balanced growth path. Consider a symmetric BGP with constant labor allocation and a constant interest rate. Since $R^k = r + \delta$ is constant, the quantity produced by each intermediate firm is constant,

$$x = L_{Y,t} \left(\frac{\alpha^2}{r+\delta} \right)^{1/(1-\alpha)}.$$

It follows that

$$\frac{K_t}{N_t} = x, \qquad \frac{Y_t}{N_t} = L_{Y,t}^{1-\alpha} x^{\alpha}$$

are constant. Aggregate capital and output therefore grow at the same rate as the number of varieties.

By (6.2), profit per variety is also constant,

$$\pi = \alpha(1-\alpha)\frac{Y_t}{N_t}.$$

For $r > 0$, the patent is consequently a perpetuity with value[1]

$$v = \sum_{j=1}^{\infty} \frac{\pi}{(1+r)^j} = \frac{\pi}{r}.$$

[1]Suppose that the patent pays a constant profit π every period forever, beginning next period, and that the interest rate is constant at $r > 0$. Its present value is

$$v = \sum_{j=1}^{\infty} \frac{\pi}{(1+r)^j}.$$

Let $q = 1/(1+r)$. Since $0 < q < 1$, the geometric-series formula gives

$$\sum_{j=1}^{\infty} q^j = \frac{q}{1-q}.$$

Therefore,

$$v = \pi \frac{1/(1+r)}{1-1/(1+r)} = \pi \frac{1/(1+r)}{r/(1+r)} = \frac{\pi}{r}.$$

The patent has a constant price and pays a constant dividend, so its return is entirely the dividend yield,

$$r = \frac{\pi}{v}.$$

A constant value per patent is compatible with aggregate growth. The economy grows by adding more varieties and more firms, rather than by continually increasing the size and profits of each existing firm. Aggregate profits $N_t\pi$ grow because the number of profit-generating varieties grows.

The result $v = \pi/r$ requires constant profits per patent. It should not be applied to aggregate profits, which increase with N_t, or to a transition during which profits and interest rates vary.

6.2.6 R&D Firms

A research firm hiring $L_{R,t}$ workers pays $W_t L_{R,t}$ and creates $\theta N_t L_{R,t}$ patents. Selling each patent for v_t gives research profits

$$\Pi_{R,t} = v_t \theta N_t L_{R,t} - W_t L_{R,t} = (\theta N_t v_t - W_t)\, L_{R,t}.$$

Free entry implies the complementary-slackness conditions

$$\begin{aligned} \theta N_t v_t &\leq W_t, \\ L_{R,t} &\geq 0, \\ (W_t - \theta N_t v_t)\, L_{R,t} &= 0. \end{aligned}$$

Whenever research is active, we therefore have

$$\theta N_t v_t = W_t \qquad \Longleftrightarrow \qquad v_t = \frac{W_t}{\theta N_t}.$$

If the patent price exceeded its invention cost, research would be profitable and entry would increase demand for researchers. Equilibrium wages and patent values must adjust until that profit opportunity disappears. If the patent price were below its invention cost, no firm would undertake research.

Zero profits in research do not imply zero profits in intermediate production. The intermediate firm earns positive operating profits, but it must first purchase the patent. Its purchase price capitalizes those future profits and compensates the resources used to create the invention.

Financing variety expansion. Households ultimately finance innovation by acquiring claims on patented firms. An entrant can issue shares, use the proceeds to purchase a newly developed blueprint, and distribute its subsequent operating profits as dividends. An inventor who retains the patent instead holds an asset with the same market value.

With active research, total expenditure on new patents satisfies

$$v_t(N_{t+1} - N_t) = W_t L_{R,t}.$$

The left-hand side is the value of newly created patent assets, and the right-hand side is the wage bill required to create them. Purchasing an existing patent transfers ownership and finances no new invention. Purchasing a newly issued claim can finance research and expand the set of varieties.

On the BGP, free entry is consistent with a constant patent value even though wages rise,

$$W_t = \theta N_t v. \tag{6.4}$$

A higher wage raises the cost of hiring a researcher, but a larger knowledge stock raises the number of patents that researcher can produce. Along the BGP, these effects offset each other, leaving the cost and value of one patent constant.

6.2.7 Equilibrium

All varieties face the same demand and the same cost, so $x_t(i) = x_t$, $P_t(i) = P_t$ and $\pi_t(i) = \pi_t$. Capital-market clearing gives $N_t x_t = K_t$, and output is

$$Y_t = N_t L_{Y,t}^{1-\alpha} x_t^{\alpha},$$

so that

$$P_t x_t = \alpha L_{Y,t}^{1-\alpha} x_t^{\alpha} = \alpha \frac{Y_t}{N_t}$$

and

$$\pi_t = (P_t - R_t^k) x_t = (1-\alpha) P_t x_t = \alpha(1-\alpha)\frac{Y_t}{N_t}.$$

Substituting $x_t = K_t / N_t$, we obtain

$$Y_t = L_{Y,t}^{1-\alpha} N_t \Big(\frac{K_t}{N_t}\Big)^{\alpha} = N_t^{1-\alpha} K_t^{\alpha} L_{Y,t}^{1-\alpha}.$$

For given N, there are constant returns to K and $L_{Y,t}$, but $N_t^{1-\alpha}$ acts as TFP. Increasing the number of varieties raises productivity, so there are increasing returns to K, $L_{Y,t}$ and N together. This is what makes sustained growth possible.

With intermediate inputs supplied by capital services, aggregate physical capital evolves according to

$$K_{t+1} = (1-\delta)K_t + I_t,$$

and the final-good resource constraint is

$$C_t + I_t = Y_t.$$

6.2.8 Balanced growth with variety expansion

To show that capital, output and consumption grow at the same rate, we consider a balanced growth path with a constant interest rate r and a constant allocation of labor,

$$L_{Y,t} = L_Y, \qquad L_{R,t} = L_R, \qquad L_Y + L_R = L.$$

These conditions imply that production per variety is constant, while aggregate production expands through the creation of new varieties.

The research technology then implies

$$N_{t+1} = N_t + \theta N_t L_R = (1 + \theta L_R) N_t.$$

Hence, by (6.3), the number of varieties grows at the constant rate

$$g_N \equiv \frac{N_{t+1} - N_t}{N_t} = \theta L_R \equiv g.$$

We now show that capital also grows with the number of varieties. The optimal quantity supplied by each intermediate producer satisfies

$$\alpha^2 L_Y^{1-\alpha} x_t^{\alpha-1} = r + \delta.$$

Because r and L_Y are constant, production per variety is constant and equal to

$$x_t = L_Y \left(\frac{\alpha^2}{r+\delta} \right)^{1/(1-\alpha)} \equiv x.$$

Each variety requires x units of capital, so aggregate capital is

$$K_t = N_t x.$$

Consequently,

$$\frac{K_{t+1}}{K_t} = \frac{N_{t+1} x}{N_t x} = 1 + g,$$

which establishes that $g_K = g_N = g$. Under symmetry, aggregate output is

$$Y_t = L_Y^{1-\alpha} N_t x^{\alpha}.$$

Since L_Y and x are constant,

$$\frac{Y_{t+1}}{Y_t} = \frac{L_Y^{1-\alpha} N_{t+1} x^{\alpha}}{L_Y^{1-\alpha} N_t x^{\alpha}} = 1 + g.$$

We therefore have $g_Y = g_K = g_N = g$, and output per variety and the capital–output ratio are constant,

$$\frac{Y_t}{N_t} = L_Y^{1-\alpha} x^{\alpha}, \qquad \frac{K_t}{Y_t} = \frac{x^{1-\alpha}}{L_Y^{1-\alpha}}.$$

Consumption also grows at the same rate. To see this, we recall that capital accumulation and final-good market clearing require

$$\begin{aligned} K_{t+1} &= (1-\delta)K_t + I_t, \\ C_t + I_t &= Y_t. \end{aligned}$$

Because $K_{t+1} = (1+g)K_t$, investment must satisfy

$$I_t = K_{t+1} - (1-\delta)K_t = (g+\delta)K_t.$$

Investment both equips new varieties and replaces depreciated capital. Substituting into the resource constraint gives

$$C_t = Y_t - (g+\delta)K_t = \left[L_Y^{1-\alpha}x^{\alpha} - (g+\delta)x\right] N_t.$$

Provided that the expression in brackets is positive, consumption is therefore a constant multiple of the number of varieties. It follows that

$$\frac{C_{t+1}}{C_t} = \frac{N_{t+1}}{N_t} = 1+g.$$

Combining these results, we obtain

$$g_C = g_Y = g_K = g_N = \theta L_R.$$

For this allocation to be an equilibrium, consumption growth must also satisfy the household's Euler equation,

$$1+g = \frac{1+r}{1+\rho}.$$

The common growth rate and the interest rate must therefore satisfy

$$r = \rho + (1+\rho)g. \tag{6.5}$$

Intuitively, a steeper consumption path requires more saving today, and households save more only if the return is higher. From (6.4) and $r = \pi/v$, we also have

$$r = \frac{\theta N \pi}{W} = \frac{\theta N \alpha(1-\alpha)Y/N}{(1-\alpha)Y/L_Y} = \alpha\theta L_Y,$$

and since $L_Y = L - L_R = L - g/\theta$, we obtain

$$r = \alpha(\theta L - g). \tag{6.6}$$

Higher growth requires more research labor, which leaves fewer production workers to generate patent profits. With lower patent profits, the no-arbitrage condition implies that the return on physical capital must also be

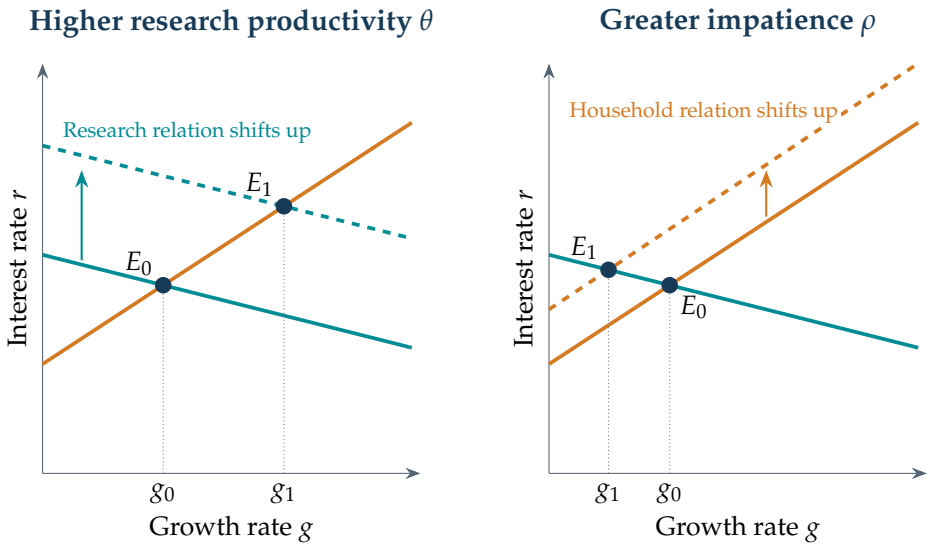


Figure 6.1: The BGP and its comparative statics. The BGP lies at the intersection of the household relation (6.5) and the research relation (6.6).

lower. The BGP is thus pinned down by the two equations (6.5) and (6.6), as Figure 6.1 shows.

Higher growth is therefore associated with more productive research θ, a larger profit share of variety-specific output α, a larger market L (the scale effect), and more patient consumers (a lower ρ, since impatience lowers the value of future profits).

During a transition, changes in the interest rate or labor allocation can change capital and output per variety, so capital, output, consumption and varieties need not grow at the same rate. Figure 6.2 gives an example.

6.2.9 Solving for research labor

The last step is to solve for research effort on the BGP. We combine the following conditions.

Step 1. Free entry: $\theta N_t v = W_t = (1-\alpha)Y_t/L_{Y,t}$.

Step 2. Patent value: $v = \pi/r = \alpha(1-\alpha)(Y_t/N_t)/r$.

Step 3. Interest rate: $r = \rho + (1+\rho)\theta L_{R,t}$.

Substituting Step 2 into Step 1 and eliminating Y_t/N_t, we obtain

$$\alpha\theta L_{Y,t} = r.$$

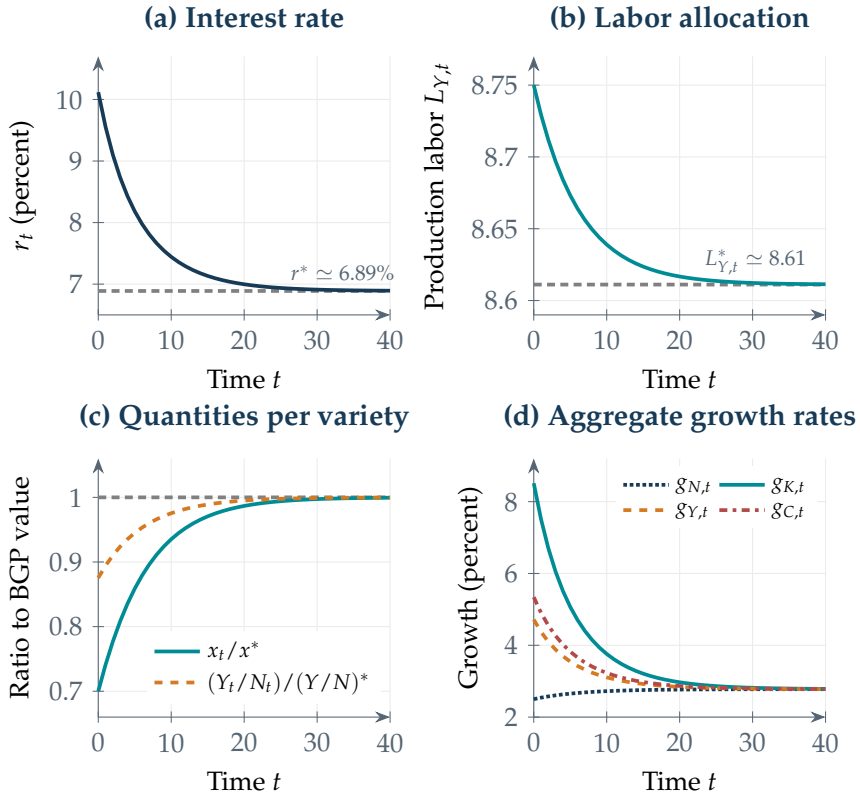


Figure 6.2: Capital initially grows faster than varieties because capital per variety is rising. Output growth also depends on the changing labor allocation, while consumption growth reflects the changing interest rate. As x_t, $L_{Y,t}$, and r_t settle at their BGP values, all four growth rates converge to the same value.

Since $L_{R,t} = L - L_{Y,t}$, solving for $L_{R,t}$ yields

$$L_{R,t} = \frac{\alpha\theta L - \rho}{\theta(1+\alpha+\rho)},$$

$$g = \theta L_{R,t} = \frac{\alpha\theta L - \rho}{1+\alpha+\rho}.$$

Research is active when $\alpha\theta L > \rho$. Otherwise, $L_{R,t} = 0$ and there is no growth.

Intuition

Free entry and $v = \pi/r$ give $r = \theta N\pi/W$. Hiring one researcher costs W and creates θN patents, each paying the perpetual profit flow π, so $\theta N\pi/W$ is the *dividend yield* of research, and in equilibrium it must equal the interest rate. Substituting $\pi = \alpha(1-\alpha)Y/N$ and $W = (1-$

$\alpha)Y/L_{Y,t}$, we find that the yield is $\alpha\theta L_{Y,t}$. It is high when many workers are in production, because then the market for each new variety (and so its profit) is large relative to the cost of a researcher. Faster growth raises r through the Euler equation, which lowers the present value of profits and hence research incentives. This is the force that pins down a finite growth rate.

The scale effect

In the Romer model, the growth rate increases with the *level* of the population,

$$g = \frac{\alpha\theta L - \rho}{1 + \alpha + \rho}.$$

Larger economies should grow faster, and an economy whose population grows should see its growth rate rise without bound.

How Figure 6.2 is computed. Steps 1–3 above use $v = \pi/r$, which is valid only when profits per patent and the interest rate are constant. The formula for $L_{R,t}$ therefore gives research labor *on the BGP*. Away from it, profits and the interest rate change over time, the patent value is no longer π/r, and research labor has to be computed jointly with the rest of the transition. This is how the paths in Figure 6.2 were obtained.

Parameters and BGP. The figure uses $\alpha = 0.4$, $\theta = 0.02$, $L = 10$, $\rho = 0.04$ and $\delta = 0.06$, so $\alpha\theta L = 0.08 > \rho$ and research is active. The formulas above give $L_R^* = 1.389$, $L_Y^* = 8.611$, $g^* = 2.78\%$ and $r^* = 6.89\%$, and the BGP capital per variety is

$$x^* = L_Y^* \left(\frac{\alpha^2}{r^* + \delta} \right)^{1/(1-\alpha)} = 12.35.$$

Variables per variety. The aggregates K_t, Y_t and C_t grow forever, so we work with ratios to N_t, which settle down on the BGP,

$$x_t = \frac{K_t}{N_t}, \qquad y_t = \frac{Y_t}{N_t} = L_{Y,t}^{1-\alpha} x_t^{\alpha}, \qquad c_t = \frac{C_t}{N_t}.$$

Timing. At the start of period t the stocks K_t and N_t are inherited from the past, so x_t is a *state* (predetermined) variable. During period t, households choose consumption C_t, and free entry into research determines how labor is split between production and research, $L_{Y,t} + L_{R,t} = L$. These are *jump* variables, since nothing ties them to the past. Production then uses the N_t existing varieties, the researchers create $\theta N_t L_{R,t}$ patents that become productive in $t + 1$, and the output not consumed is added to the capital stock.

The equations. Using $W_t = (1-\alpha)Y_t/L_{Y,t}$ and $\pi_t = \alpha(1-\alpha)Y_t/N_t$, we can write the equilibrium conditions of the previous subsections in terms of $(x_t, L_{Y,t}, c_t)$ only. The *static* conditions determine the period-t prices given x_t and $L_{Y,t}$,

$$r_t = \alpha^2 L_{Y,t}^{1-\alpha} x_t^{\alpha-1} - \delta \qquad \text{(rental rate } R_t^k = r_t + \delta\text{)}, \tag{6.7}$$

$$v_t = \frac{W_t}{\theta N_t} = \frac{(1-\alpha)L_{Y,t}^{-\alpha} x_t^{\alpha}}{\theta} \qquad \text{(free entry)}, \tag{6.8}$$

$$\pi_t = \alpha(1-\alpha) L_{Y,t}^{1-\alpha} x_t^{\alpha} \qquad \text{(profit per patent)}.$$

The *dynamic* conditions link period t to period $t+1$,

$$N_{t+1} = (1+\theta L_{R,t}) N_t \qquad \text{(research technology)}, \tag{6.9}$$

$$(1+\theta L_{R,t})\, x_{t+1} = (1-\delta)x_t + y_t - c_t \qquad \text{(capital accumulation)}, \tag{6.10}$$

$$(1+\theta L_{R,t}) \frac{c_{t+1}}{c_t} = \frac{1+r_{t+1}}{1+\rho} \qquad \text{(Euler equation)}, \tag{6.11}$$

$$1 + r_{t+1} = \left(1 + \alpha\theta L_{Y,t+1}\right) \frac{v_{t+1}}{v_t} \qquad \text{(no-arbitrage)}. \tag{6.12}$$

Equation (6.10) is $K_{t+1} = (1-\delta)K_t + Y_t - C_t$ divided by N_t, using $K_{t+1}/N_t = x_{t+1}N_{t+1}/N_t$. Equation (6.11) is the Euler equation $C_{t+1}/C_t = (1+r_{t+1})/(1+\rho)$ with $C_t = c_t N_t$. Equation (6.12) is the no-arbitrage condition $(1+r_{t+1})v_t = \pi_{t+1} + v_{t+1}$, after using $\pi_{t+1}/v_{t+1} = \alpha\theta L_{Y,t+1}$, which follows from (6.8). The patent earns the interest rate through a dividend yield, $\alpha\theta L_{Y,t+1}$ (per unit of v_{t+1}), and a capital gain, v_{t+1}/v_t. On the BGP, v is constant and (6.12) reduces to $r = \alpha\theta L_Y$, the condition obtained in Steps 1–2.

At $t = 0$ we are given K_0 and N_0. We normalize $N_0 = 1$ and set $K_0 = 0.7\,x^*$, so the economy starts with too little capital per variety, $x_0 = 0.7x^* = 8.64$. We are *not* given $L_{Y,0}$ and c_0. If we knew them, everything else would follow by recursion. Conditions (6.7)–(6.8) give the date-0 prices, (6.9)–(6.10) give N_1 and x_1, and (6.11)–(6.12) give two equations that determine c_1 and $L_{Y,1}$, and so on. The difficulty is that, as in the Ramsey model, almost every choice of $(L_{Y,0}, c_0)$ produces a path that eventually explodes. Either capital per variety collapses to zero, or consumption per variety does, violating the transversality and no-bubble conditions. Only one choice puts the economy on the *saddle path* that converges to the BGP. The two initial jumps are therefore pinned down by a *terminal* condition, not an initial one.

Solution method. We choose a horizon T long enough for the economy to be practically on the BGP by then (we use $T = 300$). The unknowns are $L_{Y,0}, \ldots, L_{Y,T-1}$ and $x_1, \ldots, x_T$, with x_0 given and the terminal values set to the BGP, $x_{T+1} = x^*$ and $L_{Y,T} = L_Y^*$. Given these unknowns, (6.10) delivers c_t for every t, and (6.7)–(6.8) deliver r_t and v_t. The Euler equation (6.11)

and the no-arbitrage condition (6.12) must hold at $t = 0, \dots, T-1$. This gives $2T$ nonlinear equations in $2T$ unknowns, which a standard Newton solver handles easily from the initial guess $L_{Y,t} = L_Y^*$ and x_t rising smoothly from x_0 to x^*. Lengthening T leaves the first 40 periods unchanged, so the terminal condition does not drive the results. Table 6.1 shows the first periods of the solution.

Table 6.1: The first periods of the transition in Figure 6.2, with $N_0 = 1$ and growth rates $g_{Z,t} = Z_{t+1}/Z_t - 1$.

t	x_t	$L_{Y,t}$	$L_{R,t}$	r_t (%)	v_t	c_t/y_t	$g_{N,t}$ (%)	$g_{K,t}$ (%)	$g_{Y,t}$ (%)	$g_{C,t}$ (%)
0	8.64	8.750	1.250	10.12	29.85	0.856	2.50	8.51	4.72	5.34
1	9.15	8.729	1.271	9.55	30.57	0.861	2.54	7.51	4.37	4.90
2	9.59	8.712	1.288	9.10	31.18	0.865	2.58	6.70	4.10	4.55
10	11.55	8.639	1.361	7.44	33.69	0.884	2.72	3.76	3.11	3.23
BGP	12.35	8.611	1.389	6.89	34.65	0.891	2.78	2.78	2.78	2.78

Period 0, step by step. The solver returns $L_{Y,0} = 8.750$ and $c_0 = 7.45$, which is 16% below its BGP value. Given these values, the period unfolds as follows.

- By (6.7), capital is scarce, so its rental rate is high, with $r_0 = 10.12\%$ against $r^* = 6.89\%$.
- Output per variety is $y_0 = L_{Y,0}^{0.6} x_0^{0.4} = 8.71$ and investment is $y_0 - c_0 = 1.25$.
- Research employs $L_{R,0} = 1.250$ workers, fewer than on the BGP, so by (6.9) varieties grow by only $g_{N,0} = \theta L_{R,0} = 2.50\%$ and $N_1 = 1.025$.
- By (6.10), $x_1 = (0.94 \times 8.64 + 1.25)/1.025 = 9.15$. Capital grows by $g_{K,0} = (1 + g_{N,0}) x_1/x_0 - 1 = 8.51\%$, much faster than varieties, because capital per variety is rising.
- The patent value is $v_0 = 29.85$, which is *not* $\pi_0/r_0 = 2.09/0.1012 = 20.65$. Future profits per patent are rising and future interest rates are falling, so the patent is worth more than a perpetuity at today's profit and interest rate.
- We can check that (6.11) and (6.12) hold between $t = 0$ and $t = 1$. For example, $(1 + 0.008 \times 8.729) \times 30.57/29.85 = 1.0955 = 1 + r_1$.

Period 1 then starts from the new state $x_1 = 9.15$, and the same logic repeats. Equivalently, the saddle path defines two *policy functions*, $L_{Y,t} = \ell(x_t)$ and $c_t = c(x_t)$, and the whole transition collapses to a one-dimensional map $x_{t+1} = h(x_t)$ that converges monotonically to x^*. Along it, the gap $x^* - x_t$ shrinks by about 15% per period.

Why the paths look as they do.

- **Interest rate.** When capital per variety is low, its marginal product is high, so r_t starts high and falls as x_t rises (panel (a)).
- **Labor allocation.** A high interest rate makes the future profits of a patent less valuable. To earn the high return required by (6.12), the dividend yield $\alpha\theta L_{Y,t+1}$ must be high. Hence more workers stay in production and fewer do research than on the BGP (panel (b)). The capital gain $v_{t+1}/v_t > 1$, which comes from the rising wage per unit of knowledge as x_t grows, provides the rest of the return. As r_t falls, workers move gradually into research.
- **Growth rates.** Since $g_{N,t} = \theta L_{R,t}$, variety growth starts *below* g^* and rises toward it. Capital grows fastest because capital per variety is catching up. Output grows more slowly than capital because $\alpha < 1$. Consumption growth is governed by the interest rate through (6.11), so it starts high and falls with r_t (panel (d)).
- **Convergence.** As $x_t \to x^*$, the ratios x_t, y_t, c_t become constant. Then $L_{Y,t} \to L_Y^*$ and $r_t \to r^*$, and all four growth rates converge to the common BGP value $g^* = \theta L_R^* = 2.78\%$.

6.3 The Jones Model

6.3.1 The scale-effect problem

Since the 1950s, the number of scientists and engineers in the United States and other advanced economies has increased many-fold, yet growth rates of TFP and income per capita have remained roughly constant. Jones (1995) concluded that the strong scale effect of the Romer model is rejected by the data. His fix is to weaken the knowledge spillover.

6.3.2 Set-up

We retain the preceding model's production technology and patent structure, but allow population to grow and weaken the knowledge spillover,

$$\begin{aligned} L_{t+1} &= (1+n)L_t, \qquad n > 0, \\ N_{t+1} - N_t &= \theta N_t^{\phi} L_{R,t}, \qquad \theta > 0, \quad 0 \le \phi < 1. \end{aligned} \tag{6.13}$$

This is a Jones-style extension with research labor entering linearly. On an interior BGP, research uses a constant population share $s_R \in (0,1)$,

$$L_{R,t} = s_R L_t, \qquad L_{Y,t} = (1 - s_R)L_t.$$

Both research and production employment therefore grow at rate n.

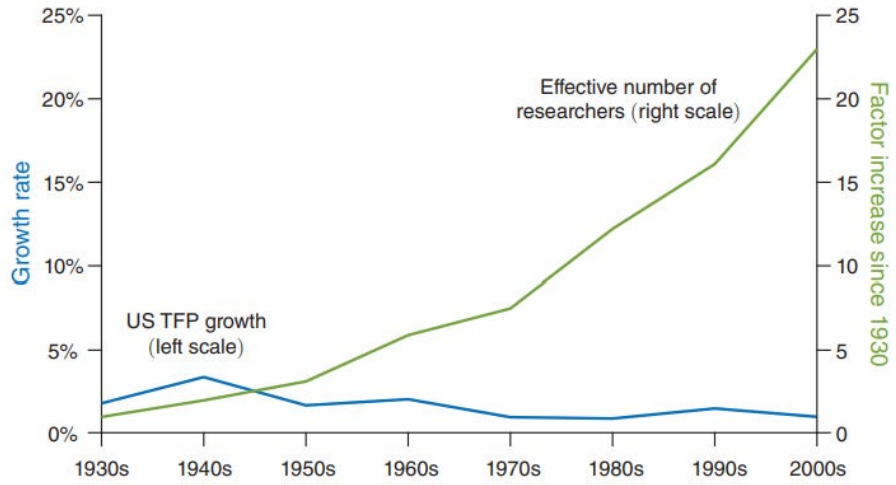


Figure 6.3: Research employment and productivity growth, illustrating the absence of scale effects in the data.

Dividing (6.13) by N_t gives

$$g_{N,t} = \theta s_R L_t N_t^{\phi-1}.$$

Existing knowledge still raises the number of ideas a researcher produces when $\phi > 0$, but less than proportionally. There are thus diminishing returns to the knowledge spillover, or, in a popular phrase, "ideas are getting harder to find".

For a constant positive variety growth rate g_N, we compare adjacent dates,

$$\begin{aligned} 1 = \frac{g_{N,t+1}}{g_{N,t}} &= \left(\frac{N_{t+1}}{N_t}\right)^{\phi-1} \frac{L_{R,t+1}}{L_{R,t}} \\ &= (1+g_N)^{\phi-1}(1+n). \end{aligned}$$

Rearranging yields

$$1 + g_N = (1+n)^{1/(1-\phi)} \qquad \Longleftrightarrow \qquad g_N = (1+n)^{1/(1-\phi)} - 1,$$

or, equivalently, $\log(1+g_N) = \log(1+n)/(1-\phi)$. For fixed ϕ and sufficiently small rates, $g_N \simeq n/(1-\phi)$. This approximation can become inaccurate when ϕ is close to one.

Semi-endogenous growth

The long-run growth rate of income per worker depends only on the population growth rate n and the degree of knowledge spillovers ϕ. There is no scale effect in growth rates (larger economies are richer, not faster growing), and if $n = 0$, growth eventually stops. Policies such as R&D subsidies affect the *level* of income but not its long-run growth rate.

6.3.3 Equilibrium R&D

Let $S_R \equiv L_{R,t}/L_{Y,t}$, the ratio of research to production labor, constant on the BGP.

- **Profits grow.** Profit per variety, $\pi_t = \alpha(1-\alpha)Y_t/N_t$, grows at rate n, because Y grows at $(1+g_N)(1+n)$ and N at $1+g_N$.
- **Patent value.** The patent value is

$$v_t = \sum_{j\geq 1} \frac{\pi_{t+j}}{(1+r)^j} = \frac{\pi_{t+1}}{1+r} \sum_{j\geq 0} \left(\frac{1+n}{1+r}\right)^j = \frac{\pi_{t+1}}{r-n}.$$

- **Interest rate.** Consumption per capita grows at g_N and aggregate consumption at $(1+g_N)(1+n)$. With log utility over aggregate consumption, $1+r = (1+\rho)(1+g_N)(1+n)$, so that $r - n = (1+n)[\rho + (1+\rho)g_N]$.
- **Free entry.** Free entry requires $\theta N_t^{\phi} v_t = W_t = (1-\alpha)Y_t/L_{Y,t}$, and the research technology gives $\theta N_t^{\phi-1} = g_N/L_{R,t}$.

Substituting $v_t = \alpha(1-\alpha)(1+n)(Y_t/N_t)/(r-n)$ into the free-entry condition, we obtain

$$\theta N_t^{\phi-1}\alpha(1+n)L_{Y,t} = r - n \quad \Longrightarrow \quad \frac{\alpha(1+n)g_N}{S_R} = (1+n)\big[\rho + (1+\rho)g_N\big],$$

$$S_R = \frac{\alpha g_N}{\rho + (1+\rho)g_N} \approx \frac{\alpha n}{(1-\phi)\rho + n}.$$

The ratio S_R increases with n, α and ϕ, decreases with ρ, and does not depend on the size of the economy.

6.4 Is there too little research? The planner's problem

6.4.1 Set-up

A planner maximizes $\sum_t \beta^t \ln C_t$ subject to

$$Y_t = N_t^{1-\alpha} K_t^{\alpha} L_{Y,t}^{1-\alpha}, \qquad K_{t+1} = Y_t - C_t + (1-\delta)K_t,$$
$$N_{t+1} = N_t + \theta N_t^{\phi} L_{R,t}, \qquad L_{Y,t} + L_{R,t} = L_t.$$

The planner uses the *aggregate* production function. The planner does not need to price intermediate goods and is therefore not affected by the monopoly markup. With current-value multipliers λ_t (capital) and μ_t (varieties), the Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ \ln C_t + \lambda_t \big[N_t^{1-\alpha} K_t^{\alpha} L_{Y,t}^{1-\alpha} + (1-\delta)K_t - C_t - K_{t+1} \big] + \mu_t \big[N_t + \theta N_t^{\phi} (L_t - L_{Y,t}) - N_{t+1} \big] \Big\}.$$

6.4.2 First-order conditions

$$C_t: \quad \frac{1}{C_t} = \lambda_t, \tag{P1}$$

$$L_{Y,t}: \quad \lambda_t (1-\alpha) \frac{Y_t}{L_{Y,t}} = \mu_t \theta N_t^{\phi}, \tag{P2}$$

$$K_{t+1}: \quad \lambda_t = \beta \lambda_{t+1} \Big(\alpha \frac{Y_{t+1}}{K_{t+1}} + 1 - \delta \Big), \tag{P3}$$

$$N_{t+1}: \quad \mu_t = \beta \Big[\lambda_{t+1} (1-\alpha) \frac{Y_{t+1}}{N_{t+1}} + \mu_{t+1} \big(1 + \phi \theta N_{t+1}^{\phi-1} L_{R,t+1} \big) \Big]. \tag{P4}$$

- Condition (P2) says that moving a worker from production to research loses that worker's marginal product in output (valued at λ_t) and gains θN^{ϕ} new varieties (valued at μ_t).
- Condition (P3) is the Euler equation, with the *social* marginal product of capital $\alpha Y/K$. In the market, capital earns only $\alpha^2 Y/K$ because of the markup.
- Condition (P4) treats a variety as an asset. Its value today equals the discounted value tomorrow of (i) its *full* marginal product in production, $(1-\alpha)Y/N$, (ii) its own continuation value, and (iii) its contribution to future research, through the spillover ϕg_N.

6.4.3 The optimal research share

On the BGP, $\lambda_t Y_t \equiv y$ is constant (by (P1), since C and Y grow at the same rate), and $\phi\theta N^{\phi-1} L_{R,t} = \phi g_N$. From (P2), using $\theta N_t^{\phi-1} = g_N / L_{R,t}$,

$$m \equiv \mu_t N_t = (1-\alpha) y \frac{N_t^{1-\phi}}{\theta L_{Y,t}} = (1-\alpha) y \frac{S_R}{g_N},$$

which is constant. Multiplying (P4) by $N_{t+1} = (1+g_N)N_t$ gives

$$\begin{aligned} m(1+g_N) &= \beta\big[(1-\alpha)y + m(1+\phi g_N)\big] \\ \Longrightarrow \quad m\big[(1+\rho)(1+g_N) - 1 - \phi g_N\big] &= (1-\alpha)y, \end{aligned}$$

that is, $m[\rho + (1+\rho-\phi)g_N] = (1-\alpha)y$. Substituting m, we obtain

$$S_R^* = \frac{g_N}{\rho + (1+\rho-\phi)g_N} \;\approx\; \frac{n}{(1-\phi)(\rho+n)}.$$

6.4.4 Comparison and the sources of underinvestment

Comparing the market and the optimal research shares, we find

$$\frac{S_R}{S_R^*} = \alpha \, \frac{\rho + (1+\rho-\phi)g_N}{\rho + (1+\rho)g_N} \;\le\; \alpha < 1 \qquad (\phi \ge 0).$$

The market under-invests in research

The gap reflects the following externalities.

1. **Surplus appropriability.** The social value of a new variety is its full marginal product $(1-\alpha)Y/N$, but its owner captures only the profit $\pi = \alpha(1-\alpha)Y/N$, a fraction α of it. Consumers of the new variety obtain a surplus the inventor cannot charge for.
2. **Intertemporal knowledge spillover** ($\phi > 0$). A new idea makes future research more productive, a benefit that no private inventor is paid for (the term $-\phi g_N$ in the planner's formula).

With $\phi = 0$, $S_R = \alpha S_R^*$.

Special case: Romer. With $\phi = 1$ and $n = 0$, we have $S_R^* = g/(\rho(1+g))$ with $g = \theta L_{R,t}^*$, that is, $\theta L_{Y,t}^* = \rho(1 + \theta L_{R,t}^*)$, which gives

$$L_{R,t}^* = \frac{\theta L - \rho}{\theta(1+\rho)} \qquad \text{versus} \qquad L_{R,t} = \frac{\alpha\theta L - \rho}{\theta(1+\alpha+\rho)}.$$

The market value has a smaller numerator and a larger denominator, so $L_{R,t} < L_{R,t}^*$ and the market growth rate is below the optimal one.

6.4.5 R&D subsidies

Suppose that the government pays a fraction ς of the wage bill of researchers. Free entry becomes $\theta N_t^{\phi} v_t = (1-\varsigma) W_t$, and repeating the derivation in the Jones model, where g_N is independent of policy, gives

$$S_R(\varsigma) = \frac{1}{1-\varsigma} \cdot \frac{\alpha g_N}{\rho + (1+\rho) g_N}.$$

Setting $S_R(\varsigma^*) = S_R^*$, we obtain

$$\varsigma^* = 1 - \alpha \frac{\rho + (1+\rho-\phi) g_N}{\rho + (1+\rho) g_N} \qquad \overset{\phi=0}{\Longrightarrow} \qquad \varsigma^* = 1 - \alpha.$$

Example 6.1. Let $n = 0.01$, $\phi = 0.5$, $\rho = 0.02$ and $\alpha = 1/3$. Then $g_N = 1.01^2 - 1 = 0.0201$, $S_R \approx 0.165$ (14% of labor in research), $S_R^* \approx 0.660$ (40%), and $\varsigma^* \approx 0.75$. Calibrated models of this type typically imply that the market spends far too little on research and that optimal subsidies are large.

Remark 6.1. A full implementation of the optimum in the Romer–Jones economy also requires correcting the *static* distortion due to monopoly pricing (capital earns $\alpha^2 Y/K$ instead of $\alpha Y/K$), for instance by subsidizing purchases of intermediate goods at rate $1-\alpha$. The R&D subsidy above corrects only the research margin.

6.5 The Schumpeterian Model

6.5.1 Overview

Schumpeter (1942) described growth as a process of *creative destruction*: new, better products replace old ones, and the firms that made the old products lose their rents. Aghion and Howitt (1992) and Grossman and Helpman (1991) turned this idea into a growth model of *vertical* innovation, in which research does not add new varieties, as in the Romer model, but improves the *quality* of an existing input.

- In the Romer model, a new variety is added to the existing ones, and every patent yields profits forever.
- In the Schumpeterian model, a new quality replaces the previous one, and a patent yields profits only until the next innovation displaces its owner.

The economy consists of households, a competitive final-good sector, a single monopolist who holds the patent on the best available quality of the intermediate good (the *incumbent*), and a research sector with free entry.

Population is constant and equal to L. Labor can move freely between intermediate-good production and research, so both activities pay the same wage w_t, and

$$L_{x,t} + L_{R,t} = L.$$

The timing of events is the following. In period t, the incumbent produces with the leading quality A_t, and research firms hire $L_{R,t}$ workers. If research succeeds, the innovator obtains a patent on the improved quality γA_t, which is used from period $t+1$ on, and the period-t incumbent is displaced. If research fails, the period-t incumbent keeps its position in $t+1$.

6.5.2 Households

The representative household is risk neutral and solves

$$\max \ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t C_t, \qquad \beta = \frac{1}{1+\rho}, \quad \rho > 0.$$

With linear utility the marginal utility of consumption is constant and equal to one, so the Euler equation for any safe asset with net return r is $1 = \beta(1+r)$, that is,

$$r = \rho.$$

The interest rate is pinned down by impatience alone and does not respond to growth. Because households are risk neutral, a risky asset, such as a patent that may be destroyed, must offer an *expected* return equal to ρ. There is no capital in this model, so all output is consumed, $C_t = Y_t$.

6.5.3 The final-good sector

A competitive firm produces the final good from the intermediate good x_t and a fixed factor M (for instance land or entrepreneurship), normalized to $M = 1$, according to

$$Y_t = A_t x_t^{\alpha} M^{1-\alpha} = A_t x_t^{\alpha}, \qquad 0 < \alpha < 1.$$

Here A_t is the quality of the leading version of the intermediate good, and a better input makes every unit of x_t more productive. Taking the intermediate price p_t as given, the firm maximizes $A_t x_t^{\alpha} - p_t x_t$, which gives the inverse demand curve

$$p_t = \alpha A_t x_t^{\alpha-1}. \tag{6.14}$$

The price elasticity of this demand is $-\mathrm{d}\ln x / \mathrm{d}\ln p = 1/(1-\alpha) > 1$. Spending on the intermediate good is $p_t x_t = \alpha Y_t$, and the fixed factor earns the remaining $(1-\alpha)Y_t$.

6.5.4 The intermediate-good monopolist

The incumbent produces the intermediate good one-for-one from labor, so its marginal cost is the wage w_t. Taking the demand curve (6.14) as given, it solves

$$\max_{x_t} \ \pi_t = \big(\alpha A_t x_t^{\alpha-1} - w_t\big) x_t = \alpha A_t x_t^{\alpha} - w_t x_t.$$

The first-order condition is

$$\alpha^2 A_t x_t^{\alpha-1} = w_t \quad \Longleftrightarrow \quad p_t = \frac{w_t}{\alpha}.$$

As in the Romer model, the price is a constant markup $1/\alpha$ over marginal cost, because demand has the constant elasticity $1/(1-\alpha)$. All production labor works for the incumbent, $x_t = L_{x,t}$, so

$$\begin{aligned} w_t &= \alpha^2 A_t L_{x,t}^{\alpha-1}, \\ \pi_t &= (p_t - w_t) x_t = (1-\alpha) p_t x_t = \alpha(1-\alpha) A_t L_{x,t}^{\alpha} = \alpha(1-\alpha) Y_t. \end{aligned} \tag{6.15}$$

Output is therefore divided between the fixed factor, which receives $(1-\alpha)Y_t$, production workers, who receive $w_t L_{x,t} = \alpha^2 Y_t$, and the incumbent, who receives the monopoly profit $\alpha(1-\alpha)Y_t$. Both the wage and the profit are proportional to the current quality A_t, and their ratio

$$\frac{\pi_t}{w_t} = \frac{1-\alpha}{\alpha} L_{x,t} \tag{6.16}$$

does not depend on A_t.

6.5.5 The innovation technology

Each worker employed in research at date t discovers an improved quality with probability $\lambda > 0$. With $L_{R,t}$ researchers, the probability that an innovation occurs is[2]

$$z_t = \lambda L_{R,t}.$$

An innovation raises the quality of the intermediate good by the factor $\gamma > 1$, the *size* of the innovation. The leading quality therefore evolves as a random "quality ladder",

$$A_{t+1} = \begin{cases} \gamma A_t & \text{with probability } z_t, \\ A_t & \text{with probability } 1 - z_t. \end{cases}$$

Each step up the ladder is an act of creative destruction, in which the new incumbent earns profits on the higher quality, and the patent of the previous incumbent becomes worthless.

[2] We assume $\lambda L_{R,t} \leq 1$. We can think of a large number of research firms, each with a small probability of success, and z_t is the probability that one of them succeeds.

6.5.6 The value of a patent

Let V_t be the value, at date t, of the patent on the current leading quality A_t, measured *including* the profit earned in period t. The incumbent earns π_t today. Next period, nobody innovates with probability $1-z$, in which case the incumbent keeps the patent, which is then worth V_{t+1}. With probability z, a research firm innovates, the incumbent is displaced and its patent is worth 0. Discounting the expected value at the interest rate ρ, we obtain

$$V_t = \pi_t + \beta\big[(1-z)V_{t+1} + z\cdot 0\big] = \pi_t + \beta(1-z)V_{t+1}.$$

As long as the incumbent survives, its quality stays at A_t and, with a constant labor allocation, its profit stays at $\pi(A_t) = \alpha(1-\alpha)A_t L_x^{\alpha}$. The value of the patent on a given quality is therefore constant over its life, $V_{t+1} = V_t \equiv V(A_t)$, and

$$\begin{aligned} V(A) &= \pi(A) + \beta(1-z)V(A) \\ \Longrightarrow \quad V(A) &= \frac{\pi(A)}{1-\beta(1-z)} = \pi(A)\,\frac{1+\rho}{\rho+z}, \end{aligned} \tag{6.17}$$

where the last step uses $1-\beta(1-z) = \frac{(1+\rho)-(1-z)}{1+\rho} = \frac{\rho+z}{1+\rho}$. Because $\pi(A)$ is proportional to A, so is the value, $V(\gamma A) = \gamma V(A)$.

Creative destruction as an extra discount rate

We define the effective discount rate ρ_{eff} by $1+\rho_{\text{eff}} = 1/[\beta(1-z)]$, so that

$$\rho_{\text{eff}} = \frac{\rho+z}{1-z} \approx \rho + z.$$

The patent is discounted not only for impatience (ρ) but also for the risk of being destroyed (z). A patent survives each period with probability $1-z$, so its expected life is $1/z$ periods. The faster the economy innovates, the shorter the life of each patent and the lower its value. This is the equilibrium force that limits research. In the Romer model, patents never become obsolete and are discounted at the interest rate only.

6.5.7 Research arbitrage

A research firm that hires $L_{R,t}$ workers at date t pays $w_t L_{R,t}$ and succeeds with probability $\lambda L_{R,t}$. If it succeeds, it owns the patent on quality γA_t from period $t+1$ on. That patent is worth $V(\gamma A_t) = \gamma V(A_t)$ at $t+1$, hence $\beta\gamma V(A_t)$ today. Expected research profits are

$$\Pi_{R,t} = \lambda L_{R,t}\,\beta\gamma V(A_t) - w_t L_{R,t} = \big[\beta\lambda\gamma V(A_t) - w_t\big]L_{R,t}.$$

Free entry implies

$$\beta\lambda\gamma V(A_t) \le w_t, \qquad L_{R,t} \ge 0, \qquad \big[w_t - \beta\lambda\gamma V(A_t)\big]L_{R,t} = 0.$$

Whenever research is active, the expected value of a researcher's work equals the wage. Using (6.17) and $\beta(1+\rho)=1$, we obtain

$$w_t = \beta\lambda\gamma V(A_t) = \lambda\gamma\,\frac{\pi(A_t)}{\rho+z}. \tag{6.18}$$

The same condition can be read as *no arbitrage in the labor market*. A worker earns w_t for sure in production, and the expected value of what the worker produces in research must be the same.

6.5.8 Equilibrium

A stationary equilibrium is a constant labor allocation $(L_x, L_{R,t})$, an innovation probability z, and a wage and profit, both proportional to A_t, such that the incumbent maximizes profits (6.15), research firms make zero expected profits (6.18), and the labor market clears.

Step 1: research arbitrage. Dividing both sides of (6.18) by $\pi(A_t)$ and using the ratio (6.16), $w_t/\pi_t = \alpha/[(1-\alpha)L_x]$, we find that the quality A_t cancels,

$$\frac{\alpha}{(1-\alpha)L_x} = \frac{\lambda\gamma}{\rho+z} \quad\Longleftrightarrow\quad z = \frac{\lambda\gamma(1-\alpha)}{\alpha}L_x - \rho. \tag{6.19}$$

This *research-arbitrage* relation is increasing in L_x, because more workers in production mean a larger market for the intermediate good, hence a larger profit $\pi \propto L_x^{\alpha}$ relative to the wage, which makes research more attractive.

Step 2: labor-market clearing. With $L_{R,t} = L - L_x$, we have

$$z = \lambda(L - L_x). \tag{6.20}$$

This relation is decreasing in L_x, since every worker in production is a worker not doing research.

Step 3: solution. Equating (6.19) and (6.20) gives $\alpha(\rho+\lambda L-\lambda L_x) = \lambda\gamma(1-\alpha)L_x$, so that

$$L_x = \frac{\alpha(\rho+\lambda L)}{\lambda\big[\alpha+\gamma(1-\alpha)\big]}, \qquad z = \lambda(L-L_x) = \lambda L - \frac{\alpha(\rho+\lambda L)}{\alpha+\gamma(1-\alpha)}. \tag{6.21}$$

Research is active ($L_x < L$, that is, $z>0$) if and only if $\alpha\rho < \lambda\gamma(1-\alpha)L$. We also assume parameters such that $z \le 1$.

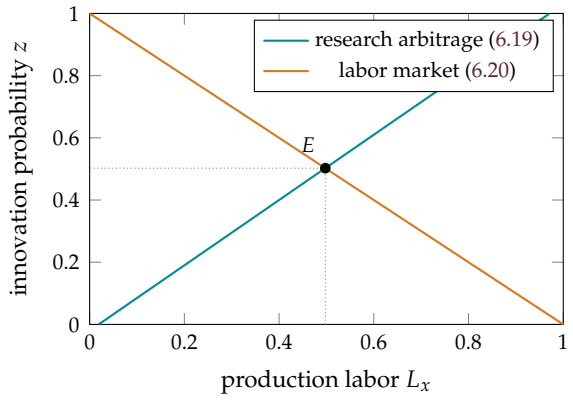


Figure 6.4: Equilibrium of the Schumpeterian model for $\alpha = 0.5$, $\gamma = 1.05$, $\lambda = 1$, $L = 1$ and $\rho = 0.02$. The research-arbitrage line $z = \lambda\gamma(1-\alpha)L_x/\alpha - \rho$ and the labor-market line $z = \lambda(L - L_x)$ cross at $L_x \approx 0.498$, $z \approx 0.502$.

6.5.9 Growth

One sector. Output is $Y_t = A_t L_x^{\alpha}$ with L_x constant, so it grows exactly with quality,

$$\frac{Y_{t+1}}{Y_t} = \frac{A_{t+1}}{A_t} = \begin{cases} \gamma & \text{with probability } z, \\ 1 & \text{with probability } 1 - z. \end{cases}$$

Growth is random, with periods without innovation alternating with jumps of size γ. The expected growth factor and the expected log growth rate are

$$\mathbb{E}\left[\frac{Y_{t+1}}{Y_t}\right] = z\gamma + (1-z) = 1 + z(\gamma - 1),$$
$$\mathbb{E}\left[\ln \frac{Y_{t+1}}{Y_t}\right] = z \ln \gamma + (1-z)\ln 1 = z \ln \gamma.$$

Many sectors. We now let the final good be produced from a continuum of intermediate goods $j \in [0,1]$, each with its own quality A_{jt}, its own incumbent and its own research, according to the Cobb–Douglas aggregator $\ln Y_t = \int_0^1 \ln Y_{jt}\,\mathrm{d}j$ with $Y_{jt} = A_{jt} L_x^{\alpha}$. We assume that the sectors are symmetric, so that the same research-arbitrage condition holds in every sector and each one innovates with the same probability z, independently of the others. Then

$$\begin{aligned} \ln Y_{t+1} - \ln Y_t &= \int_0^1 \big(\ln A_{j,t+1} - \ln A_{jt}\big)\,\mathrm{d}j \\ &= \ln \gamma \times \big(\text{fraction of sectors that innovate at } t\big). \end{aligned}$$

With a continuum of independent sectors, the law of large numbers implies that the fraction that innovates is exactly z. Aggregate growth is therefore *deterministic*,

$$g \equiv \ln Y_{t+1} - \ln Y_t = z \ln \gamma.$$

Growth in the Schumpeterian model

Long-run growth equals the *frequency* of innovations times their *size*,

$$g = z \ln \gamma, \qquad z = \lambda L - \frac{\alpha(\rho + \lambda L)}{\alpha + \gamma(1-\alpha)}.$$

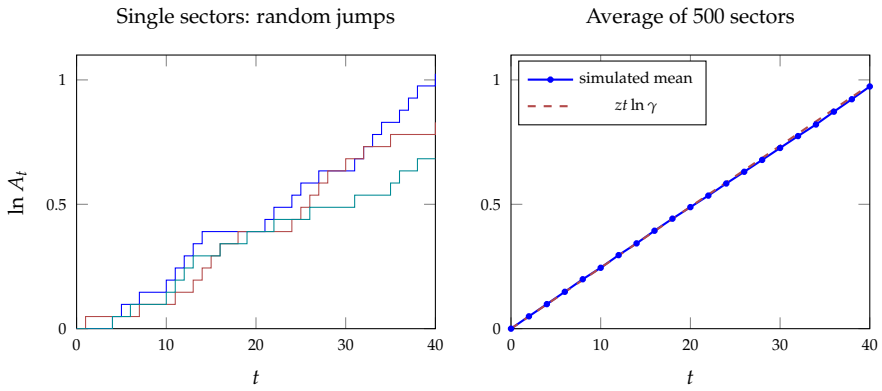


Figure 6.5: Quality ladders simulated with $z = 0.5024$ and $\gamma = 1.05$. In the left panel, individual sectors climb the ladder by random steps of size $\ln \gamma$. In the right panel, the average over 500 independent sectors grows almost exactly at the deterministic rate $z \ln \gamma$.

6.5.10 Comparative statics

We write (6.21) as $z = \lambda L - \alpha(\rho + \lambda L)/D$ with $D \equiv \alpha + \gamma(1-\alpha) > \alpha$. Then

$$\frac{\partial z}{\partial \lambda} = L\left(1 - \frac{\alpha}{D}\right) > 0, \qquad \frac{\partial z}{\partial L} = \lambda\left(1 - \frac{\alpha}{D}\right) > 0,$$
$$\frac{\partial z}{\partial \gamma} = \frac{\alpha(1-\alpha)(\rho + \lambda L)}{D^2} > 0, \qquad \frac{\partial z}{\partial \rho} = -\frac{\alpha}{D} < 0,$$
$$\frac{\partial z}{\partial \alpha} = -\frac{\gamma(\rho + \lambda L)}{D^2} < 0,$$

where the last derivative uses $\frac{\mathrm{d}}{\mathrm{d}\alpha}\frac{\alpha}{D} = \frac{D - \alpha(1-\gamma)}{D^2} = \frac{\gamma}{D^2}$.

Parameter	Effect on z and g	Intuition
$\lambda \uparrow$	$\uparrow$	research more productive
$\gamma \uparrow$	$\uparrow$	bigger prize per innovation (and larger steps, so g rises further through $\ln \gamma$)
$L \uparrow$	$\uparrow$	larger market (scale effect)
$\rho \uparrow$	$\downarrow$	future rents are worth less
$\alpha \uparrow$	$\downarrow$	smaller markup $1/\alpha$, smaller rents

As in the Romer model, there is a *scale effect*, since a larger labor force raises the innovation rate and growth. We also see that more product-market competition, that is, a smaller markup or a higher α, *reduces* innovation, because it lowers the rents that reward innovators.

6.5.11 A worked example

Let $\alpha = 0.5$, $\gamma = 1.05$, $\lambda = 1$, $L = 1$ and $\rho = 0.02$, and normalize the current quality to $A_t = 1$.

- *Labor allocation.* Production labor is

$$L_x = \frac{0.5\,(0.02 + 1)}{0.5 + 1.05 \times 0.5} = \frac{0.51}{1.025} \approx 0.498,$$

  so $L_{R,t} \approx 0.502$ and $z = \lambda L_{R,t} \approx 0.502$, and an innovation occurs about every other period.

- *Growth.* The growth rate is $g = z \ln \gamma \approx 0.502 \times 0.0488 \approx 2.45\%$ per period.

- *Wage and profit.* We have $w = \alpha^2 L_x^{\alpha-1} \approx 0.354$ and $\pi = \alpha(1-\alpha)L_x^{\alpha} \approx 0.176$.

- *Patent value.* The patent is worth $V = \pi(1+\rho)/(\rho+z) \approx 0.176 \times 1.02/0.522 \approx 0.344$, and free entry holds since $\beta\lambda\gamma V = 1.05 \times 0.344/1.02 \approx 0.354 = w$.

- *Effective discount rate.* We have $\rho_{\text{eff}} = (\rho+z)/(1-z) \approx 105\%$, and the expected life of a patent is $1/z \approx 2$ periods.

6.5.12 Romer versus Schumpeter

	Romer (1990)	**Aghion–Howitt (1992)**
Innovation	new varieties (horizontal)	better qualities (vertical)
Life of a patent	forever	until the next innovation (expected $1/z$)
Patent value	$v = \pi/r$	$V = \pi(1+\rho)/(\rho+z)$
Free entry	$\theta N v = W$	$w = \beta\lambda\gamma V$
Growth rate	$\theta L_R = \frac{\alpha\theta L - \rho}{1+\alpha+\rho}$	$z \ln\gamma$, deterministic only with many sectors
Scale effect	yes	yes

6.5.13 Welfare: the planner's problem

The planner chooses a constant allocation L_x to maximize the expected present value of output, since households are risk neutral,

$$\max_{L_x} \ \mathbb{E}\sum_{t=0}^{\infty} \beta^t A_t L_x^{\alpha}.$$

Since innovations arrive independently with probability z each period, $\mathbb{E}A_{t+1} = A_t[1 + z(\gamma-1)]$, so $\mathbb{E}A_t = A_0 G^t$ with $G \equiv 1 + z(\gamma-1)$. If $\beta G < 1$, we obtain

$$W(L_x) = \frac{A_0 L_x^{\alpha}}{1-\beta G}, \qquad z = \lambda(L - L_x).$$

Taking logs and differentiating with respect to L_x, recalling that $\partial z/\partial L_x = -\lambda$, we find

$$\frac{\alpha}{L_x} = \frac{\beta\lambda(\gamma-1)}{1-\beta G} \quad\Longrightarrow\quad \frac{\alpha}{L_x^*} = \frac{\lambda(\gamma-1)}{\rho - z^*(\gamma-1)}$$

where we used $(1-\beta G)/\beta = \rho - z(\gamma-1)$. Solving with $z^* = \lambda(L - L_x^*)$ gives

$$L_x^* = \frac{\alpha\big[\rho - \lambda(\gamma-1)L\big]}{\lambda(\gamma-1)(1-\alpha)},$$

which is interior when $\alpha\rho < \lambda(\gamma-1)L < \rho$.

6.5.14 Under- or over-investment in research

We write the market condition in the same form as the planner's,

$$\text{Market:}\quad \frac{\alpha}{L_x} = \frac{\lambda\,\gamma\,(1-\alpha)}{\rho+z}, \qquad\qquad \text{Planner:}\quad \frac{\alpha}{L_x^*} = \frac{\lambda\,(\gamma-1)}{\rho - z^*(\gamma-1)}.$$

In both conditions, the left side is the marginal cost of research (output lost by moving a worker out of production, in proportional terms) and the right side is the marginal benefit. The differences between them reveal the following effects.

1. **Appropriability** (market too low), captured by the factor $(1-\alpha)$. The innovator captures only the monopoly rent, not the whole increase in output.

2. **Intertemporal spillover** (market too low), captured by the denominators. The private innovator's rents end with the next innovation (discount rate $\rho + z$), whereas society keeps the higher quality forever and future innovators build on it, so expected growth $z(\gamma - 1)$ *lowers* the social discount rate.

3. **Business stealing** (market too high), captured by the numerators γ versus $\gamma - 1$. The private prize is the whole value of the leading position, γA, but society gains only the increment $(\gamma - 1)A$, and the rest is a transfer from the displaced incumbent.

The net effect is ambiguous.

Example 6.2. Let $L = 1$ and $\rho = 0.05$.

	α	γ	λ	z (market)	z^* (planner)	$z \ln \gamma$	$z^* \ln \gamma$
Large steps	0.5	1.20	0.2	0.086	0.150	1.6%	2.7%
Small steps	0.2	1.02	0.8	0.633	0.375	1.3%	0.7%

With large innovations, appropriability and spillovers dominate and the market innovates too little. The subsidy that implements the optimum, $1 - \varsigma^* = \lambda\gamma(1-\alpha)L_x^*/[\alpha(\rho + z^*)]$, is $\varsigma^* = 0.70$. With small innovations and large markups, business stealing dominates and the market innovates too much, so implementing the optimum would require *taxing* research.

Appendix A

Mathematical Appendix

Roadmap

This appendix collects the mathematical tools used in the notes, all in discrete time. It covers elasticities, growth rates and approximations, static optimization with the Kuhn–Tucker conditions and the envelope theorem, dynamic optimization with Lagrangians and with dynamic programming, single and systems of difference equations, and phase diagrams. The emphasis is on how to use the tools and how to read the results economically, not on proofs.

A.1 Important Concepts

A.1.1 Elasticities

For a differentiable function $y = h(x)$ with $x, y > 0$, the elasticity of y with respect to x is

$$\varepsilon_{y,x} = \frac{\mathrm{d}y/y}{\mathrm{d}x/x} = \frac{\mathrm{d}\ln y}{\mathrm{d}\ln x} = \frac{x\,h'(x)}{h(x)}.$$

It is unit-free and measures the percentage change in y for a one-percent change in x. The table below lists the elasticities used in these notes.

Elasticity	Definition	Interpretation
Output elasticity of capital	$\dfrac{kf'(k)}{f(k)}$	capital share under competition, $= \alpha$ for Cobb–Douglas
Elasticity of substitution	$\dfrac{\mathrm{d}\ln(K/L)}{\mathrm{d}\ln(F_L/F_K)}$	ease of substituting K for L, $= 1$ for Cobb–Douglas
Elasticity of marginal utility	$\sigma(c) = -\dfrac{cu''(c)}{u'(c)}$	curvature of u, relative risk aversion
EIS	$\dfrac{\mathrm{d}\ln(c_{t+1}/c_t)}{\mathrm{d}\ln R} = \dfrac{1}{\sigma(c)}$	willingness to shift consumption over time
Frisch elasticity	$\dfrac{\partial \ln N}{\partial \ln w}\Big\vert_{\lambda}$	labor response holding marginal utility of wealth fixed, $= 1/\varphi$
Price elasticity of demand	$-\dfrac{\mathrm{d}\ln x}{\mathrm{d}\ln P}$	determines the monopoly markup $\varepsilon/(\varepsilon-1)$

Useful rules. The elasticity of a product is the sum of elasticities, so that if $y = x^a z^b$, then $\mathrm{d}\ln y = a\,\mathrm{d}\ln x + b\,\mathrm{d}\ln z$. For a power function $y = Ax^a$, the elasticity is the constant a.

A.1.2 Growth rates in discrete time

Let $g_x \equiv x_{t+1}/x_t - 1$. The following rules are exact,

$$1 + g_{xy} = (1+g_x)(1+g_y), \qquad 1 + g_{x/y} = \frac{1+g_x}{1+g_y}, \qquad 1 + g_{x^a} = (1+g_x)^a.$$

For small growth rates, $\ln(1+g) \approx g$, so $g_{xy} \approx g_x + g_y$, $g_{x/y} \approx g_x - g_y$ and $\ln x_{t+1} - \ln x_t \approx g_x$. These approximations are exact in continuous time. In discrete time, they hold up to second-order terms such as $g_x g_y$ (for example, $n + g + ng \approx n + g$ in the Solow model).

A.1.3 Taylor approximations and log-linearization

The first-order Taylor expansion of h around $\overline{x}$ is

$$h(x) \approx h(\overline{x}) + h'(\overline{x})(x - \overline{x}),$$

and for several variables

$$h(x,y) \approx h(\overline{x},\overline{y}) + h_x(x - \overline{x}) + h_y(y - \overline{y}).$$

Log-linearization applies the same idea to $\tilde{x} \equiv \ln x - \ln \overline{x} \approx (x - \overline{x})/\overline{x}$. We write $x = \overline{x}e^{\tilde{x}}$ and use $e^{\tilde{x}} \approx 1 + \tilde{x}$ and $\tilde{x}\tilde{y} \approx 0$. For a product $y = x^a z^b$, we have $\tilde{y} = a\tilde{x} + b\tilde{z}$ exactly, and for a sum $y = x + z$, we have $\overline{y}\tilde{y} \approx \overline{x}\tilde{x} + \overline{z}\tilde{z}$.

A.1.4 The implicit function theorem

If $\Psi(y, x) = 0$ defines y implicitly as a function of x near a point where $\Psi_y \neq 0$, then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\Psi_x}{\Psi_y}.$$

With several parameters, $\partial y/\partial x_i = -\Psi_{x_i}/\Psi_y$. This is how we derived the properties of the OLG saving function and the effect of debt on the steady state.

A.2 Static optimization

A.2.1 Equality constraints: the Lagrangian

To maximize $f(x)$, $x \in \mathbb{R}^n$, subject to $g_j(x) = b_j$, $j = 1, \dots, m$, we form the Lagrangian

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^{m} \lambda_j \big[b_j - g_j(x) \big].$$

At an interior optimum (with linearly independent constraint gradients), $\partial \mathcal{L}/\partial x_i = 0$ for all i and the constraints hold. If f is concave and the g_j are linear (or the problem is otherwise concave), these conditions are also sufficient.

Meaning of the multiplier. Let $f^*(b)$ be the maximized value. Then

$$\frac{\partial f^*}{\partial b_j} = \lambda_j,$$

so the multiplier is the *shadow price* of the constraint, the increase in the maximal objective when the constraint is relaxed by one unit. In the two-period consumption problem, $\lambda = u'(c_0)$ is the marginal value of wealth.

A.2.2 Inequality constraints: Kuhn–Tucker

To maximize $f(x)$ subject to $g_j(x) \le b_j$, we form the same Lagrangian with $\lambda_j \ge 0$. The Kuhn–Tucker conditions are

$$\frac{\partial \mathcal{L}}{\partial x_i} = 0, \qquad \lambda_j \ge 0, \qquad g_j(x) \le b_j, \qquad \lambda_j \big[b_j - g_j(x) \big] = 0.$$

The last condition is **complementary slackness**, which says that either the constraint binds or its shadow price is zero. Non-negativity constraints $x_i \ge 0$ can be handled by adding $\mu_i x_i$ to the Lagrangian with $\mu_i \ge 0$ and $\mu_i x_i = 0$. Then $\partial f/\partial x_i + \mu_i = 0$, so at a corner $x_i = 0$ the marginal benefit of x_i is non-positive. This is how we found the education threshold in the OLG human-capital model (Section 3.8.1).

A.2.3 The envelope theorem

Let $V(a) = \max_x f(x, a)$, with maximizer $x^*(a)$. Then

$$V'(a) = \frac{\partial f}{\partial a}\big(x^*(a), a\big).$$

Because x^* is optimal, a small change in x has no first-order effect on the objective ($f_x = 0$), so only the *direct* effect of the parameter matters. With constraints, $V'(a) = \partial \mathcal{L}/\partial a$ evaluated at the optimum. The envelope theorem is the key step in deriving the Euler equation from the Bellman equation.

A.3 Dynamic optimization

A.3.1 The general problem

A typical dynamic problem has a **state** variable x_t (for example capital, inherited from the past) and a **control** variable c_t (for example consumption, chosen freely each period),

$$\max_{\{c_t, x_{t+1}\}} \sum_{t=0}^{T} \beta^t U(x_t, c_t) \quad \text{s.t.} \quad x_{t+1} = h(x_t, c_t), \quad x_0 \text{ given.}$$

The constraint $x_{t+1} = h(x_t, c_t)$ is the **law of motion** (transition equation) of the state.

A.3.2 Method 1: the Lagrangian

With current-value multipliers $\beta^t \lambda_t$, the Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \Big\{ U(x_t, c_t) + \lambda_t \big[h(x_t, c_t) - x_{t+1} \big] \Big\}.$$

The first-order conditions are

$$\begin{aligned} c_t: \quad & U_c(x_t, c_t) + \lambda_t h_c(x_t, c_t) = 0, \\ x_{t+1}: \quad & -\lambda_t + \beta \big[U_x(x_{t+1}, c_{t+1}) + \lambda_{t+1} h_x(x_{t+1}, c_{t+1}) \big] = 0. \end{aligned}$$

Eliminating the multipliers yields an **Euler equation**, a second-order difference equation in the state (or a first-order system in state and control). The solution is pinned down by the initial condition x_0 and a terminal condition. With a finite horizon and $x_{T+1} \geq 0$, the terminal condition is $\beta^T \lambda_T x_{T+1} = 0$.

Recipe. We (1) write the Lagrangian, (2) differentiate with respect to c_t and x_{t+1}, remembering that x_{t+1} appears in two periods, (3) eliminate the multipliers, (4) interpret the result as "marginal cost today = discounted marginal benefit tomorrow", and (5) add the boundary conditions.

A.3.3 The infinite horizon and the transversality condition

With $T = \infty$, the terminal condition becomes the **transversality condition**

$$\lim_{t\to\infty} \beta^t \lambda_t x_{t+1} = 0.$$

Theorem A.1 (Sufficiency)**.** *If U is concave, h is concave in (x, c) and the multipliers are non-negative, then a path satisfying the first-order conditions, the law of motion, x_0 and the transversality condition is optimal.*

Example A.1 (Cake eating)**.** A household owns a cake of size x_0 and solves $\max \sum_t \beta^t \ln c_t$ subject to $x_{t+1} = x_t - c_t$. The FOCs are $1/c_t = \lambda_t$ and $\lambda_t = \beta\lambda_{t+1}$, so the Euler equation is $c_{t+1} = \beta c_t$, that is, $c_t = \beta^t c_0$. *Every* c_0 satisfies the Euler equation, and the question is which one is optimal. Iterating the law of motion gives

$$x_{t+1} = x_0 - c_0 \sum_{s=0}^{t} \beta^s \to x_0 - \frac{c_0}{1-\beta}.$$

The TVC reads

$$\lim_{t\to\infty} \beta^t \frac{1}{\beta^t c_0} x_{t+1} = \frac{1}{c_0}\Big(x_0 - \frac{c_0}{1-\beta}\Big) = 0 \quad\Longrightarrow\quad c_0 = (1-\beta)x_0.$$

A plan with $c_0 < (1-\beta)x_0$ (for instance, "eat half the cake and follow the Euler equation forever") leaves cake uneaten forever and violates the TVC, while a plan with $c_0 > (1-\beta)x_0$ runs out of cake in finite time.

A.3.4 Method 2: dynamic programming

The Bellman equation. For a stationary infinite-horizon problem, we define the value function $V(x)$ as the maximal value attainable from state x. By Bellman's *principle of optimality* (an optimal plan remains optimal from any intermediate date onward),

$$V(x) = \max_c \Big\{ U(x,c) + \beta V\big(h(x,c)\big) \Big\}.$$

This converts an infinite-dimensional problem (choosing a whole sequence) into a one-period problem in which the future is summarized by V. The solution is a **policy function** $c = \pi(x)$.

From the Bellman equation to the Euler equation. The FOC is $U_c + \beta V'(x')h_c = 0$, and the envelope theorem gives $V'(x) = U_x + \beta V'(x')h_x$. Combining the two, shifted appropriately, delivers the same Euler equation as the Lagrangian. In the common case $x' = h(x) - c$ (as in the Ramsey model, with $U = u(c)$), the FOC is $u'(c) = \beta V'(x')$ and the envelope condition is $V'(x) = u'(c)h'(x)$, so $u'(c) = \beta u'(c')h'(x')$.

Guess and verify. When the structure is log-linear, we guess a value function of the same form. For cake eating, we guess $V(x) = a + b \ln x$. The FOC

$$\frac{1}{c} = \frac{\beta b}{x - c}$$

gives $c = \frac{x}{1+\beta b}$. After substituting, the coefficient on $\ln x$ on the right side is $1 + \beta b$, so $b = 1 + \beta b$, $b = 1/(1-\beta)$ and $c = (1-\beta)x$, which is the same answer as with the Lagrangian and the TVC.

Contraction and value function iteration. We define the Bellman operator

$$(TV)(x) = \max_c \{U(x,c) + \beta V(h(x,c))\}.$$

Under standard conditions (bounded continuous U, compact feasible sets), T is a *contraction* with modulus β, by Blackwell's sufficient conditions of monotonicity and discounting. By the contraction mapping theorem, T has a unique fixed point V, and iterating $V_{j+1} = TV_j$ from any starting V_0 converges to it. This justifies value function iteration as a numerical method.

Uncertainty. With a random shock z_t following a Markov process, the Bellman equation becomes

$$V(x,z) = \max_c \Big\{ U(x,c,z) + \beta \mathbb{E}\big[V(x',z') \mid z\big] \Big\},$$

and the Euler equation holds in expectation, as in the RBC model.

A.4 Difference equations

A difference equation links the value of a variable at one date to its values at earlier dates. Its *order* is the largest lag involved.

A.4.1 First-order linear equations

Consider $x_{t+1} = a x_t + b$. If $a \neq 1$, the **fixed point** (steady state) is $x^* = b/(1-a)$. Subtracting $x^* = a x^* + b$ gives

$$x_{t+1} - x^* = a(x_t - x^*) \quad \Longrightarrow \quad x_t = x^* + a^t (x_0 - x^*).$$

- If $|a| < 1$, the steady state is *globally stable*, with $x_t \to x^*$ for any x_0. Convergence is monotone if $0 < a < 1$ and oscillating (damped) if $-1 < a < 0$.
- If $|a| > 1$, the steady state is unstable, and x_t diverges unless $x_0 = x^*$ (explosively if $a > 1$, with explosive oscillations if $a < -1$).

- If $a = 1$, then $x_t = x_0 + bt$, with no steady state unless $b = 0$. If $a = -1$, the solution is a two-period cycle.

The fraction of the gap closed per period is $1 - a$ (for $0 < a < 1$), and the half-life is $\ln 2/(-\ln a)$. An example is $\ln k_{t+1} = \ln(sA/(1+n)) + \alpha \ln k_t$ in the Solow model with $\delta = 1$.

A.4.2 First-order nonlinear equations

Consider $x_{t+1} = g(x_t)$. A steady state satisfies $x^* = g(x^*)$, which graphically is an intersection of the graph of g with the 45° line. Linearizing around x^* gives

$$x_{t+1} - x^* \approx g'(x^*)(x_t - x^*),$$

which leads to the following result.

Proposition A.2 (Local stability)**.** *A steady state x^* of $x_{t+1} = g(x_t)$ is locally asymptotically stable if $|g'(x^*)| < 1$ and unstable if $|g'(x^*)| > 1$. Convergence is locally monotone if $0 < g'(x^*) < 1$ and oscillating if $-1 < g'(x^*) < 0$.*

The 45°-line diagram. We plot g and the 45° line. From x_0 on the horizontal axis, we go up to the graph of g to read x_1, then horizontally to the 45° line to transfer x_1 to the horizontal axis, then up to g again, and so on (the "staircase" of Figure 2.2). Where g lies above the 45° line, x increases, and where it lies below, x decreases.

Global results for monotone maps. If g is increasing, paths are monotone and cannot jump over a steady state. If, in addition, g is concave with $g(0) = 0$, $g'(0) > 1$ and $g'(\infty) < 1$, there is a unique positive steady state and it attracts every $x_0 > 0$ (as in the Solow and Diamond models). If g is S-shaped or discontinuous, there can be several steady states, which then alternate between stable and unstable, and the unstable ones are thresholds separating basins of attraction (poverty traps).

A.4.3 Second-order linear equations

Consider $x_{t+2} + a_1 x_{t+1} + a_2 x_t = 0$. The trial solution $x_t = \mu^t$ works if μ solves the **characteristic equation**

$$\mu^2 + a_1 \mu + a_2 = 0.$$

- With two distinct real roots μ_1, μ_2, the solution is $x_t = C_1 \mu_1^t + C_2 \mu_2^t$.
- With a repeated root μ, the solution is $x_t = (C_1 + C_2 t)\mu^t$.
- With complex roots $re^{\pm i\omega}$, the solution $x_t = r^t\big(C_1 \cos \omega t + C_2 \sin \omega t\big)$ oscillates, and the oscillations are damped if $r < 1$.

The constants are fixed by two boundary conditions. The solution converges to zero for all initial conditions if and only if both roots have modulus less than one, which is equivalent to

$$|a_2| < 1, \qquad 1 + a_1 + a_2 > 0, \qquad 1 - a_1 + a_2 > 0.$$

A non-homogeneous equation $x_{t+2} + a_1 x_{t+1} + a_2 x_t = b$ has the particular solution $x^* = b/(1 + a_1 + a_2)$, and the general solution is x^* plus the homogeneous solution. Euler equations, once the control is eliminated, are second-order equations in the state.

A.4.4 Systems of linear difference equations

Consider $z_{t+1} = Mz_t$ with $z_t \in \mathbb{R}^n$. If M has n linearly independent eigenvectors v_i with eigenvalues μ_i ($Mv_i = \mu_i v_i$), we can write $M = P\Lambda P^{-1}$ with $P = [v_1 \cdots v_n]$ and $\Lambda = \operatorname{diag}(\mu_i)$. Then $z_t = P\Lambda^t P^{-1} z_0$, that is,

$$z_t = \sum_{i=1}^{n} c_i \mu_i^t v_i, \qquad (c_1, \ldots, c_n)' = P^{-1} z_0.$$

Each eigenvector defines a direction along which the system moves like a scalar first-order equation with coefficient μ_i.

The 2×2 case. The eigenvalues solve $\mu^2 - \operatorname{tr}(M)\mu + \det(M) = 0$, so $\mu_1 + \mu_2 = \operatorname{tr} M$ and $\mu_1 \mu_2 = \det M$. Let $P(\mu) \equiv \mu^2 - \operatorname{tr}(M)\mu + \det(M)$.

- A **sink** (stable node or focus) has both $|\mu_i| < 1$, or equivalently $|\det M| < 1$ and $|\operatorname{tr} M| < 1 + \det M$. Every path converges.
- A **source** has both $|\mu_i| > 1$. Every path diverges, except the steady state itself.
- A **saddle** has $|\mu_1| < 1 < |\mu_2|$. With real roots, a sufficient condition is $P(1) < 0$, which places one root on each side of 1, as in the Ramsey model. Only paths starting on the **stable arm**, the line through the steady state in the direction of v_1, converge, and all others diverge.

State and jump variables: the Blanchard–Kahn condition. In economic models, some variables are *predetermined*, such as capital, whose value at t is inherited. Others are *jump* (forward-looking) variables, such as consumption and prices, which are free to take any value at t. Convergence pins down the jump variables, which must be chosen so that the system starts on the stable arm, that is, so that the coefficients on all unstable eigenvalues are zero. A unique convergent solution exists when

$$\#\{\text{eigenvalues with } |\mu| > 1\} = \#\{\text{jump variables}\}.$$

With fewer unstable roots, there are infinitely many convergent paths (indeterminacy), and with more, there is none.

Linearization of nonlinear systems. For $z_{t+1} = G(z_t)$ with steady state z^*, the linearized system is $z_{t+1} - z^* \approx J(z_t - z^*)$, where J is the Jacobian matrix of G at z^*. If no eigenvalue of J has modulus exactly one, the nonlinear system behaves locally like its linearization (Hartman–Grobman theorem). A saddle remains a saddle, and the stable arm of the linear system is tangent to the stable manifold of the nonlinear one (the saddle path).

A.5 Phase diagrams in discrete time

A phase diagram describes the dynamics of a two-dimensional system graphically. Consider

$$x_{t+1} - x_t = F(x_t, y_t), \qquad y_{t+1} - y_t = H(x_t, y_t).$$

Step 1: the loci. We draw the $\Delta x = 0$ locus $\{(x, y) : F(x, y) = 0\}$ and the $\Delta y = 0$ locus $\{(x, y) : H(x, y) = 0\}$. Their intersections are the steady states.

Step 2: the arrows. On each side of the $\Delta x = 0$ locus, we determine the sign of F. If $\partial F/\partial y < 0$, then $\Delta x < 0$ above the locus (horizontal arrow pointing left) and $\Delta x > 0$ below it. We proceed in the same way for Δy with vertical arrows. The two loci divide the plane into (typically four) regions, and in each region the combination of a horizontal and a vertical arrow indicates the direction of motion.

Step 3: the trajectories. Trajectories follow the arrows. If, in two opposite regions, the arrows point towards the steady state and in the other two away from it, the steady state is a saddle, and the saddle path runs through the regions where the arrows point towards the steady state.

Step 4: the economics. We identify the predetermined variable and the jump variable. Given the initial value of the predetermined variable, the jump variable must place the economy on the saddle path, and other choices violate feasibility or the transversality condition.

Cautions specific to discrete time. (i) Paths are sequences of points, not continuous curves, and a step can jump across a locus, so arrows indicate directions only locally. (ii) We must be explicit about the dating of the variables on the axes. For example, the Ramsey $\Delta c = 0$ locus is the vertical line $k = k^*$ when the axis shows the capital that earns the return, k_{t+1}. Expressed in terms of beginning-of-period capital k_t, it becomes a steep upward-sloping curve through the same steady state (Remark 4.3). (iii)

Local stability should be confirmed by computing the eigenvalues of the linearized system.

Example A.2 (A linear saddle)**.** Consider, around the steady state $(1, 1)$,

$$\Delta x_t = 0.1(x_t - 1) - 0.1(y_t - 1), \qquad \Delta y_t = -0.05(x_t - 1),$$

with x predetermined and y a jump variable. The $\Delta x = 0$ locus is the line $y = x$, and $\Delta x < 0$ above it. The $\Delta y = 0$ locus is the vertical line $x = 1$, and $\Delta y < 0$ to its right. In deviations, the system matrix is

$$M = \begin{pmatrix} 1.1 & -0.1 \\ -0.05 & 1 \end{pmatrix},$$

with $\operatorname{tr} M = 2.1$, $\det M = 1.095$ and $P(1) = 1 - 2.1 + 1.095 = -0.005 < 0$, so the steady state is a saddle. The eigenvalues are $\mu_{1,2} = (2.1 \mp \sqrt{0.03})/2 \approx 0.963, 1.137$. The stable eigenvector satisfies $(1.1 - \mu_1)v_x = 0.1v_y$, so the saddle path is $y - 1 \approx 1.37\,(x - 1)$, steeper than the $\Delta x = 0$ locus. Given x_0, the jump variable must start at $y_0 = 1 + 1.37(x_0 - 1)$.

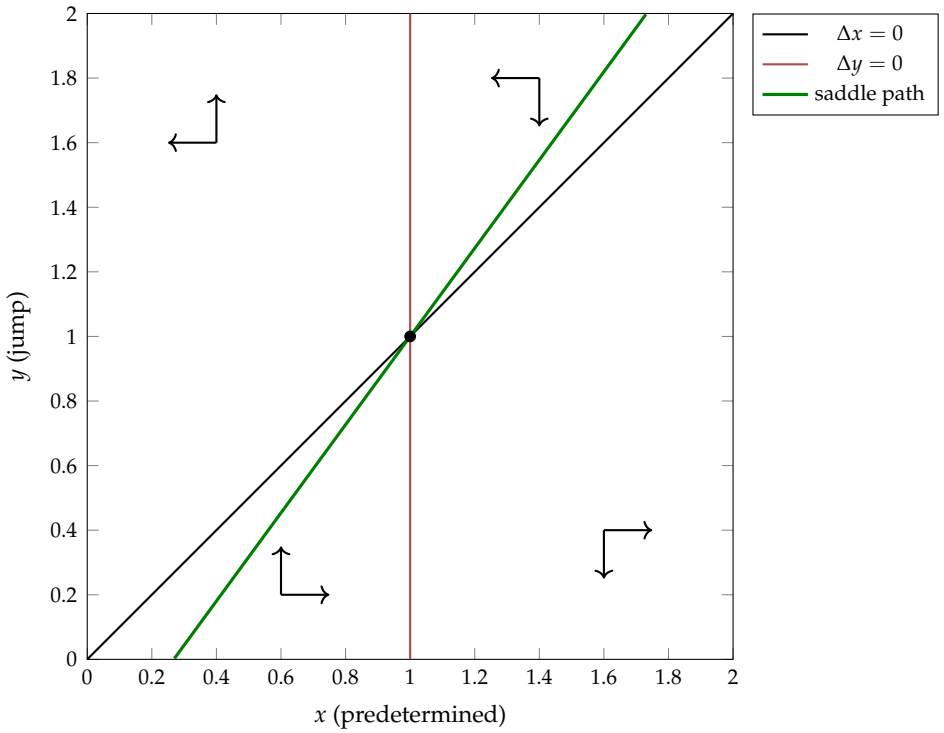


Figure A.1: Phase diagram of the linear saddle example. Arrows show the direction of motion in each region, and only paths starting on the saddle path converge to the steady state $(1, 1)$.

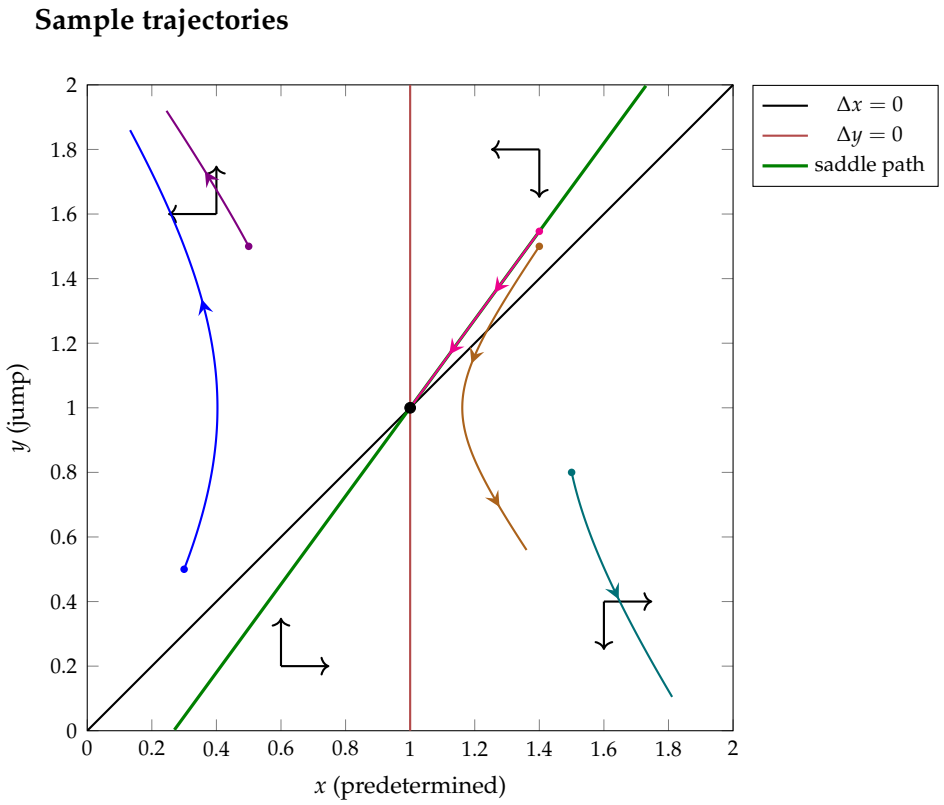


Figure A.2: Phase diagram of the linear saddle example. Direction arrows show the motion in each region, and colored curves are sample trajectories (starting points marked with dots). Only trajectories on the saddle path, such as the magenta curve that starts at $(1.4, 1.55)$ and converges to $(1, 1)$, reach the steady state. Trajectories off the saddle path diverge, and even the orange one, which passes very close to $(1, 1)$, eventually veers away.

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